

Elementary Ordinary Differential Equations

Methods and Applications

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Chapter 1

Introduction and Preliminaries

Something goes here

Chapter 2

First Order Differential Equations

Intro goes here.

Chapter 3

Second and Higher Order Equations and Applications

Intro goes here.

3.1 Introduction

Objectives

- Understand basics of n^{th} order linear differential equations

The n^{th} order linear differential equations are of the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x) \quad (3.1.1)$$

where the functions $a_0(x), \dots, a_{n-1}(x), a_n(x)$ and $f(x)$ are continuous on some interval I . We say that (3.1.1) is **homogeneous** on I if $f(x) = 0$ for all x in I and **nonhomogeneous** on I if $f(x) \neq 0$ for some x in I . We will usually have $a_n(x) \neq 0$ on I . If $a_n(x) = 0$ for some point x of I , we will simply restrict our attention to a smaller interval, denoted again by I , on which $a_n(x) \neq 0$. With these consideration, (3.1.1) can be rewritten in the form

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x), \quad (3.1.2)$$

$a_0(x), \dots, a_{n-1}(x)$ and $f(x)$ are continuous on some interval I . This is achieved by dividing both sides of (3.1.1) by $a_n(x)$ to obtain

$$y^{(n)} + \frac{a_{n-1}(x)}{a_n(x)}y^{(n-1)} + \cdots + \frac{a_1(x)}{a_n(x)}y' + \frac{a_0(x)}{a_n(x)}y = \frac{f(x)}{a_n(x)}$$

and redefining the functions

$$\frac{a_{n-1}(x)}{a_n(x)}, \dots, \frac{a_1(x)}{a_n(x)}, \frac{a_0(x)}{a_n(x)}, \frac{f(x)}{a_n(x)}$$

as

$$a_{n-1}(x), \dots, a_1(x), a_0(x), f(x),$$

respectively.

Definition 3.1.1 A function $y = \phi(x)$ defined on I is said to be a **solution**

of (3.1.2) if

$$\phi^{(n)}(x) + a_{n-1}(x)\phi^{(n-1)}(x) + \cdots + a_1(x)\phi'(x) + a_0(x)\phi(x) = f(x)$$

for all x in I , that is, if (3.1.2) is satisfied for all x in I when we substitute $\phi(x)$ for y . $\blacklozenge$

In Definition 3.1.1, we note the following.

- i. Derivatives of $\phi(x)$ at the endpoints (if any) of I are understood as one-sided.
- ii. Solutions $y = \phi(x)$ may be complex-valued, though real solutions are preferred in applications. For $\phi(x) = \phi_1(x) + i\phi_2(x)$ with real ϕ_1 and ϕ_2 ,

$$\phi'(x) = \phi_1'(x) + i\phi_2'(x),$$

where $i = \sqrt{-1}$, and higher derivatives are computed similarly.

Definition 3.1.2 Let $a_0(x), \dots, a_{n-1}(x), f(x)$ be functions defined on some interval I , x_0 be a point in I , and $y_1, \dots, y_{n-1}$ be fixed numbers. The n^{th} order linear differential equation

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x)$$

subject to the conditions (called initial conditions)

$$y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1},$$

is called an **initial value problem (IVP)**. $\blacklozenge$

It is essential to know when an initial value problem has a solution if $a_0(x), \dots, a_{n-1}(x), f(x)$ are continuous on some interval I containing x_0 . This is addressed by the existence and uniqueness theorem stated below, whose proof requires advanced calculus and is beyond the scope of this book.

Theorem 3.1.3 (Existence and Uniqueness). *Let $a_0(x), \dots, a_{n-1}(x), f(x)$ be continuous functions on some interval I , x_0 be a point in I , and $y_1, \dots, y_{n-1}$ be fixed numbers. Then the IVP*

$$\begin{cases} y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x) \\ y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}, \end{cases}$$

has a unique solution $y = \phi(x)$ defined on I .

Remark 3.1.4 When $f(x) = 0$ for all x in I and $y_1 = y_2 = \cdots = y_{n-1} = 0$, it follows from Theorem 3.1.3 that the function $\phi(x) = 0$ for all x in I is the unique solution of the IVP. By the uniqueness, if at least one of $y_1, \dots, y_{n-1}$ is nonzero, then the unique solution of the IVP in Theorem 3.1.3 must be a nonzero function.

The superposition principle discussed below provides a general procedure for constructing new solutions from known solutions.

Theorem 3.1.5 (Superposition Principle). *Suppose that $y_1, \dots, y_k$ are solutions of the homogeneous linear equation*

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = 0,$$

where $a_0(x), \dots, a_{n-1}(x)$ are continuous on some interval I . Then, for any

constants $c_1, \dots, c_k$ (possibly complex), the function

$$y(x) = c_1 y_1(x) + \dots + c_k y_k(x) = \sum_{j=1}^k c_j y_j(x)$$

is a solution on I . The expression $\sum_{j=1}^k c_j y_j(x)$ is called a **linear combination** (or **superposition**) of $y_1, \dots, y_k$, with **weights** $c_1, \dots, c_k$.

Proof. We treat the special case $n = 2$ here for clarity of this proof and leave the general case as an exercise. Let y_1 and y_2 be solutions on I of

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.1.3)$$

and let $y = c_1 y_1 + c_2 y_2$ for constants c_1 and c_2 . Then

$$y' = c_1 y_1' + c_2 y_2' \quad \text{and} \quad y'' = c_1 y_1'' + c_2 y_2''$$

on I . We note that the differentiation rules also apply when c_1 and c_2 are complex. Substituting these into the left side of the differential equation gives

$$\begin{aligned} & y'' + a_1(x)y' + a_0(x)y \\ &= c_1 (y_1'' + a_1(x)y_1' + a_0(x)y_1) + c_2 (y_2'' + a_1(x)y_2' + a_0(x)y_2) \end{aligned}$$

on I . Since y_j satisfies

$$y_j'' + a_1(x)y_j' + a_0(x)y_j = 0$$

on I for $j = 1, 2$, we obtain

$$y'' + a_1(x)y' + a_0(x)y = 0.$$

Thus, $y = c_1 y_1 + c_2 y_2$ is a solution of (3.1.3) on I . ■

Remark 3.1.6 The superposition principle provides the following basic operations for constructing new solutions from known solutions.

- i. If y_1 is a solution, then $c y_1$ is also a solution for any constant c .
- ii. If y_1 and y_2 are solutions, then so is $y_1 + y_2$.

Example 3.1.7 We can verify that $y_1 = \cos x$ and $y_2 = \sin x$ are solutions of $y'' + y = 0$, which is a homogeneous linear equation. Moreover, $y = 7 \cos x$ and $y = 2 \cos x - 3 \sin x$ are also solutions. In fact, $y = c_1 \cos x$ and $y = c_1 \cos x + c_2 \sin x$ are solutions for any constants c_1 and c_2 . ▲

When building new solutions of a homogeneous linear differential equation from a collection of known solutions, it is desirable to use a minimal set of solutions as building blocks. Even if several solutions are available, we may not actually need many of them, and some solutions can be discarded based on how they are related to the remaining solutions. This important technique is addressed in the next subsection.

3.1.1 Linear Independence

Two functions y_1 and y_2 defined on an interval I are said to be linearly independent on I if neither function is a constant multiple of the other. For instance, the functions e^x and e^{-x} are linearly independent on $(-\infty, \infty)$, as are $\sin x$ and $\cos x$ on $(-\infty, \infty)$. The concept of linear independence of functions extends

naturally to any finite collection of functions, as formalized in the following definition.

Definition 3.1.8 The functions $y_1, \dots, y_k$ defined on an interval I , where k is a positive integer, are said to be **linearly independent on I** if the relation

$$c_1 y_1(x) + \dots + c_k y_k(x) = 0$$

holds for all x in I *only when* the constants $c_1, \dots, c_k$ satisfy $c_1 = \dots = c_k = 0$. If this condition does not hold, that is, if there exist constants $c_1, \dots, c_k$, *not all zero*, satisfying the above relation for all x in I , then the functions $y_1, \dots, y_k$ are said to be **linearly dependent on I** . ♦

We say that the set $\{y_1, \dots, y_k\}$ is linearly independent on I if the functions $y_1, \dots, y_k$ are linearly independent on I . Similarly, we say that the set $\{y_1, \dots, y_k\}$ is linearly dependent on I if the functions $y_1, \dots, y_k$ are linearly dependent on I .

The linear dependence or independence of a set of functions on an interval may change when the interval is changed or when functions are added or removed. It follows directly from the definition that if the set $\{y_1, \dots, y_k\}$ is linearly independent on I , then it remains linearly independent on any larger interval J containing I , provided the functions are defined on J . Similarly, if the set $\{y_1, \dots, y_k\}$ is linearly dependent on I , it will also be linearly dependent on every subinterval J of I .

Moreover, using the definition above, any subset of a linearly independent set of functions is itself linearly independent, whereas a set that contains a linearly independent set can be linearly dependent. Similarly, a set containing a linearly dependent set of functions is necessarily linearly dependent, whereas a subset of a linearly dependent set can be linearly independent.

Suppose the functions $y_1, \dots, y_k$ are linearly dependent on an interval I . Then there exist constants $c_1, \dots, c_k$, not all zero, such that

$$c_1 y_1(x) + \dots + c_k y_k(x) = 0 \quad (3.1.4)$$

for all x in I . Since the constants are not all zero, at least one of them must be nonzero. Suppose, for instance, that $c_1 \neq 0$. Then, from (3.1.4), we can express

$$y_1(x) = \left(-\frac{c_2}{c_1}\right) y_2(x) + \left(-\frac{c_3}{c_1}\right) y_3(x) + \dots + \left(-\frac{c_k}{c_1}\right) y_k(x)$$

on I . This shows that y_1 is a linear combination of the remaining functions. Similarly, if $c_2 \neq 0$, then

$$y_2(x) = \left(-\frac{c_1}{c_2}\right) y_1(x) + \left(-\frac{c_3}{c_2}\right) y_3(x) + \dots + \left(-\frac{c_k}{c_2}\right) y_k(x)$$

on I , and so y_2 is a linear combination of the remaining functions. In general, for any integer k_0 with $2 \leq k_0 \leq k$ and $c_{k_0} \neq 0$, (3.1.4) implies that y_{k_0} is a linear combination of the remaining functions. Thus, if $\{y_1, \dots, y_k\}$ is a linearly dependent set of functions $y_1, \dots, y_k$ defined on I , then at least one of the functions can be expressed as a linear combination of the others. In particular, this means that any set $\{y_1, \dots, y_k\}$ containing the zero function is necessarily linearly dependent.

From a linearly dependent set, it is often desirable to remove any function that can be expressed as a linear combination of the remaining functions. By repeatedly removing such functions, we eventually obtain a linearly independent set. The functions in this linearly independent set can then be used to construct new functions by forming linear combinations.

Example 3.1.9 Determine whether $1, \cos^2 x$, and $\sin^2 x$ are linearly independent on $(-\infty, \infty)$.

Solution. Suppose there are constants c_1, c_2 , and c_3 that satisfy

$$c_1 + c_2 \cos^2 x + c_3 \sin^2 x = 0$$

for all x in $(-\infty, \infty)$. The main question here is: can we have at least one of c_1, c_2, c_3 nonzero? In view of $\cos^2 x + \sin^2 x = 1$, we can take $c_1 = -1, c_2 = c_3 = 1$. Therefore $1, \cos^2 x$, and $\sin^2 x$ are linearly dependent on $(-\infty, \infty)$. $\blacktriangle$

Example 3.1.10 Determine whether $1, e^x$ and xe^x are linearly independent on $(-\infty, \infty)$.

Solution. Suppose there are constants c_1, c_2 , and c_3 that satisfy

$$c_1 + c_2 e^x + c_3 x e^x = 0 \quad (3.1.5)$$

for all x in $(-\infty, \infty)$. Again, can we have at least one of c_1, c_2, c_3 nonzero? Differentiating (3.1.5) both sides with respect to x gives

$$c_2 e^x + c_3 e^x + c_3 x e^x = 0$$

for all x in $(-\infty, \infty)$. Since e^x is never zero, we have

$$c_2 + c_3 + c_3 x = 0 \quad (3.1.6)$$

for all x in $(-\infty, \infty)$. Differentiating (3.1.6) both sides with respect to x yields $c_3 = 0$. Using $c_3 = 0$ in (3.1.6) yields $c_2 = 0$. Using $c_2 = c_3 = 0$ in (3.1.5), we get $c_1 = 0$. Therefore $1, e^x$ and xe^x are linearly independent on $(-\infty, \infty)$. $\blacktriangle$

3.1.2 The Wronskian

The methods used in Example 3.1.9 and Example 3.1.10 to test the linear independence of functions can become cumbersome, especially when the functions involved are more complicated than those in the examples. In this book, we will typically need to determine whether a set of solutions of homogeneous linear differential equations is linearly independent. For such sets of solutions, the linear independence of solutions can be determined by using the **Wronskian**, a determinant involving solutions and their derivatives, introduced in 1812 by the Polish mathematician Józef Wronski.

Definition 3.1.11 (Wronskian). Let $y_1, y_2, \dots, y_n$ be functions that are $n - 1$ times differentiable on an interval I . Their **Wronskian**, denoted by $W(y_1, \dots, y_{n-1})(x)$ (or simply by $W(x)$), is a function on I defined by

$$W(y_1, \dots, y_n)(x) = \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix}$$

on I . $\blacklozenge$

Example 3.1.12 Compute the Wronskian of $y_1 = e^x$ and $y_2 = xe^{-x}$ on $(-\infty, \infty)$.

Solution. The Wronskian $W(y_1, y_2)(x)$ of $y_1 = e^x$ and $y_2 = xe^{-x}$ is

$$W(e^x, xe^{-x}) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = \begin{vmatrix} e^x & xe^{-x} \\ e^x & e^{-x} - xe^{-x} \end{vmatrix} = 1 - 2x$$

on $(-\infty, \infty)$. ▲

The next theorem provides a Wronskian test for linear independence of differentiable functions.

Theorem 3.1.13 (Wronskian Test for Linear Independence). *Let $y_1, \dots, y_k$ be $k-1$ times differentiable functions on an interval I with*

$$W(y_1, \dots, y_k)(x) \neq 0 \text{ on } I.$$

Then $y_1, \dots, y_k$ are linearly independent on I .

Proof. For simplicity, we treat the case $k = 2$; the general case is analogous. Suppose that y_1 and y_2 are differentiable functions on an interval I such that $W(y_1, y_2)(x) \neq 0$ on I . Let c_1 and c_2 be constants such that

$$c_1 y_1(x) + c_2 y_2(x) = 0 \tag{3.1.7}$$

for all x in I . Differentiating gives

$$c_1 y_1'(x) + c_2 y_2'(x) = 0 \tag{3.1.8}$$

for all x in I . Since $W(y_1, y_2)(x) \neq 0$ for every x in I , the linear system consisting of the (3.1.7) and (3.1.8) has the unique solution $c_1 = c_2 = 0$ for every x in I . Hence y_1 and y_2 are linearly independent on I . ■

Remark 3.1.14

1. The conclusion of [Theorem 3.1.13](#) holds even if $W(y_1, \dots, y_k)(x_0) \neq 0$ at just one point x_0 in I . The case $k = 2$ appears as (i) in [Exercises 3.1.3](#).
2. The converse of [Theorem 3.1.13](#) is not true. For example, take the differentiable functions $y_1 = x^2$ and $y_2 = x|x|$ on $(-\infty, \infty)$. We know from Example that y_1, y_2 are linearly independent on $(-\infty, \infty)$; however, their Wronskian

$$W(x^2, x|x|) = \begin{vmatrix} x^2 & x|x| \\ 2x & 2|x| \end{vmatrix} = 0$$

on $(-\infty, \infty)$.

3. We will see in [Theorem 3.2.1](#) for the case $k = 2$ and in [Theorem 3.4.1](#) for the general case that the converse of [Theorem 3.1.13](#) also holds if $y_1, \dots, y_k$ are solutions of a k^{th} order homogeneous linear differential equation.
4. A useful formula for the Wronskian of two solutions of a second order linear homogeneous equation is derived in [Section 3.2](#). Problem [Exercise 3.2.3.7](#) in [Exercises 3.2.3](#) shows how the Wronskian can be used to determine a second linearly independent solution when one solution is known.

3.1.3 Exercises

1. Determine whether each set of functions below is linearly independent on the specified interval using [Definition 3.1.8](#) or the discussion that follows it.

i. $\{1, x, x^2\}, \quad (-\infty, \infty)$	viii. $\{\sin x, \cos x\}, \quad (-\infty, \infty)$
ii. $\{1, x - 1, 2x\}, \quad (-\infty, \infty)$	ix. $\{1, \sin x, \cos x\}, \quad (-\infty, \infty)$
iii. $\{x, x \}, \quad (0, \infty)$	x. $\{0, \sin x, \cos x\}, \quad (-\infty, \infty)$
iv. $\{x, x \}, \quad (-\infty, 0)$	xi. $\{\sin^2 x, \cos^2 x\}, \quad (-\infty, \infty)$
v. $\{x, x \}, \quad (-\infty, \infty)$	xii. $\{1, \sin^2 x, \cos^2 x\}, \quad (-\infty, \infty)$
vi. $\{e^{2x}, e^{-2x}\}, \quad (-\infty, \infty)$	xiii. $\{\cos(\ln x), \sin(\ln x)\}, \quad (0, \infty)$
vii. $\{e^{2x}, xe^{2x}\}, \quad (-\infty, \infty)$	xiv. $\left\{\frac{1}{x}, x^3\right\}, \quad (0, \infty)$
2. Suppose that y_1, y_2 are differentiable functions on an interval I .
 - i. Let x_0 be point in I such that $W(y_1, y_2)(x_0) \neq 0$. Show that y_1, y_2 are linearly independent on I .
 - ii. Determine whether the functions $y_1 = e^x$ and $y_2 = xe^{-x}$ considered in [Example 3.1.12](#) are linearly independent on $(-\infty, \infty)$.
 - iii. Find examples of y_1, y_2 that are linearly independent on I and satisfy $W(y_1, y_2)(x) = 0$ for all x in I .

3.2 Second Order Homogeneous Linear Differential Equations

Objectives

- Understand methods of solutions for second order homogeneous linear differential equations

We now discuss the Wronskian of solutions of second order homogeneous linear equations.

Let y_1 and y_2 be solutions of the homogeneous linear differential equation

$$y'' + a_1(x)y' + a_0(x)y = 0,$$

where $a_1(x)$ and $a_0(x)$ are continuous functions on an interval I . The **Wronskian** of y_1 and y_2 , denoted by $W(y_1, y_2)(x)$, $W(y_1(x), y_2(x))$, or simply $W(x)$, is defined as

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_1'(x)y_2(x).$$

Let us examine the derivative of the Wronskian $W(x)$. We have

$$\begin{aligned} W'(x) &= y_1(x)y_2''(x) + y_1'(x)y_2'(x) - y_1'(x)y_2'(x) - y_2(x)y_1''(x) \\ &= y_1(x)y_2''(x) - y_2(x)y_1''(x) \\ &= y_1(x)(-a_1(x)y_2'(x) - a_0(x)y_2(x)) \\ &\quad - y_2(x)(-a_1(x)y_1'(x) - a_0(x)y_1(x)) \\ &= -a_1(x)W(x), \end{aligned}$$

where we used $y_1''(x) = -a_1(x)y_1'(x) - a_0(x)y_1(x)$ and $y_2''(x) = -a_1(x)y_2'(x) - a_0(x)y_2(x)$. It then follows that

$$W(x) = W(x_0) e^{-\int_{x_0}^x a_1(t) dt}, \quad (3.2.1)$$

where x_0 is a fixed number in I . The formula (3.2.1) is known as **Abel's formula**. It follows from this formula that $W(x)$ is a constant when $a_1(x) = 0$ on I . Also, if $W(x_0) = 0$, then $W(x) = 0$ for all x in I . On the other hand, since

$$e^{-\int_{x_0}^x a_1(t) dt} > 0 \quad \text{for all } x \text{ in } I,$$

it follows that $W(x) < 0$ for all x in I if $W(x_0) < 0$, and $W(x) > 0$ for all x in I if $W(x_0) > 0$. Thus, either $W(x)$ is identically zero or it does not change its sign in I .

The analysis of the Wronskian together with the formula (3.2.1) above also hold for higher order homogeneous linear equations (see Section?? Higher Order).

Theorem 3.2.1 (Criterion for Linearly Independent Solutions). *Two solutions y_1 and y_2 of*

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.2.2)$$

where $a_1(x)$ and $a_0(x)$ are continuous functions on an interval I , are linearly independent on I if and only if $W(y_1, y_2)(x) \neq 0$ for all x in I .

Proof. Let y_1 and y_2 be linearly independent solutions of (3.2.2) on I . Suppose, for a contradiction, that there is a point x_0 in I such that $W(y_1, y_2)(x_0) = 0$. Let c_1 and c_2 be constants. Then the linear system

$$\begin{cases} c_1 y_1(x_0) + c_2 y_2(x_0) = 0 \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) = 0 \end{cases} \quad (3.2.3)$$

has a solution c_1, c_2 , not both zero because the determinant of the coefficient matrix is zero. Fix c_1, c_2 , not both zero, satisfying (3.2.3), and define the function

$$\phi(x) = c_1 y_1(x) + c_2 y_2(x)$$

on I . By (3.2.3) and the superposition principle, $y = \phi(x)$ is a solution on I of the initial value problem

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad y(x_0) = 0, \quad y'(x_0) = 0.$$

The uniqueness part of Theorem Theorem 3.1.3 implies $\phi(x) = 0$ on I , that is,

$$c_1 y_1(x) + c_2 y_2(x) = 0$$

on I . Thus, y_1 and y_2 are linearly dependent on I , contradicting our assumption. Hence $W(y_1, y_2)(x) \neq 0$ for every x in I .

Suppose that $W(y_1, y_2)(x) \neq 0$ on I . By [Theorem 3.2.1](#), y_1 and y_2 are linearly independent on I . Note that y_1 and y_2 need not be solutions [\(3.2.2\)](#) for this part of the theorem to hold. ■

3.2.1 Existence of Fundamental Sets of Solutions

Definition 3.2.2 Consider the second order homogeneous linear differential equation

$$y'' + a_1(x)y' + a_0(x)y = 0,$$

where $a_1(x)$ and $a_0(x)$ are continuous functions on an interval I . Two solutions y_1 and y_2 that are linearly independent on I are said to form a **fundamental set of solutions** of the differential equation. Each of these solutions, y_1 and y_2 , is referred to as a fundamental solution. ◆

An important question arises: does a fundamental set of solutions always exist? The following theorem ensures the existence of a fundamental set of solutions for second order homogeneous linear equations when $a_1(x)$ and $a_0(x)$ are continuous. These concepts extend naturally to higher order equations.

Theorem 3.2.3 Consider the second order homogeneous linear differential equation

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.2.4)$$

where $a_1(x)$ and $a_0(x)$ are continuous functions on an interval I . Then there exist two functions y_1 and y_2 on I that form a fundamental set of solutions for [\(3.2.4\)](#).

Proof. By [Theorem 3.1.3](#), there exists a unique solutions of y_1 the initial value problem

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad y(x_0) = 1, \quad y'(x_0) = 0,$$

and a unique y_2 of the initial value problem

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad y(x_0) = 0, \quad y'(x_0) = 1.$$

The two solutions y_1 and y_2 are linearly independent on I . In fact, let c_1, c_2 be constants such that $c_1y_1(x) + c_2y_2(x) = 0$ for all x in I . Since x_0 is in I , we have $c_1y_1(x_0) + c_2y_2(x_0) = 0$, which implies $c_1 = 0$. We then only have $c_2y_2(x) = 0$ for all x in I . Differentiating $c_2y_2(x) = 0$ yields $c_2y_2'(x) = 0$ for all x in I . Since $y_2'(x_0) = 1$, we must have $c_2 = 0$. Thus, $\{y_1, y_2\}$ forms a fundamental set of solutions for [\(3.2.4\)](#) on I .

Let $\{y_1, y_2\}$ be a fundamental set of solutions of [\(3.2.4\)](#) on the interval I , where both $a_0(x)$ and $a_1(x)$ are continuous. Denote by $W(x)$ the Wronskian of y_1 and y_2 . Let x_0 be a point in I . Since y_1 and y_2 are linearly independent on I , we have $W(x_0) \neq 0$. Then the Wronskian $\widetilde{W}(x)$ of $\frac{1}{W(x_0)}y_1$ and y_2 is

$$\begin{aligned} \widetilde{W}(x) &= \frac{y_1(x)}{W(x_0)}y_2'(x) - \frac{y_1'(x)}{W(x_0)}y_2(x) \\ &= \frac{W(x)}{W(x_0)}. \end{aligned}$$

Obviously, $\widetilde{W}(x_0) = 1$. Hence we have

$$\widetilde{W}(x) = \widetilde{W}(x_0)e^{-\int_{x_0}^x a_1(t) dt} = e^{-\int_{x_0}^x a_1(t) dt}.$$

Thus, the solutions in a fundamental set can be scaled so that the Wronskian of the scaled solutions is

$$e^{-\int_{x_0}^x a_1(t) dt},$$

where x_0 is a fixed point in I . ■

Example 3.2.4 Find the Wronskian of a fundamental set of solutions of each of the following differential equations.

i. $y'' + \frac{2}{x}y' + \frac{10}{x^2}y = 0$

ii. $y'' + \ln x y' + 10y = 0$

iii. $y'' - 2xy' + 5y = 0$

Solution. (i) Take $x_0 = 1$ in $(0, \infty)$. There is a fundamental set of solutions whose Wronskian is

$$e^{-\int_1^x a_1(t) dt} = e^{-\int_1^x \frac{2}{t} dt} = e^{-2 \ln x} = \frac{1}{x^2}.$$

(ii) Take $x_0 = 1$ in $(0, \infty)$. There is a fundamental set of solutions whose Wronskian is

$$e^{-\int_1^x a_1(t) dt} = e^{-\int_1^x \ln t dt} = e^{x - x \ln x - 1}.$$

(iii) Take $x_0 = 0$ in $(-\infty, \infty)$. There is a fundamental set of solutions whose Wronskian is

$$e^{-\int_0^x a_1(t) dt} = e^{-\int_0^x (-2t) dt} = e^{x^2}.$$

The next theorem provides a method for finding the general solutions of second order homogeneous linear differential equations. ▲

Theorem 3.2.5 Let $\{y_1, y_2\}$ form a fundamental set of solutions of the second order homogeneous linear differential equation

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.2.5)$$

where $a_1(x)$ and $a_0(x)$ are continuous functions on an interval I . Then the functions of the form

$$y = c_1 y_1(x) + c_2 y_2(x) \quad (3.2.6)$$

are solutions on I for any constants c_1 and c_2 . Moreover, every solution of (3.2.5) is of form (3.2.6) for some constants c_1 and c_2 .

Proof. For any c_1 and c_2 constants, the Theorem 3.1.5 implies that

$$y = c_1 y_1(x) + c_2 y_2(x) \quad (3.2.7)$$

is a solution of (3.2.5) on I .

To show that every solution of (3.2.5) is of this form (3.2.6), let $\phi(x)$ be a solution of (3.2.5) on I and x_0 be in I . Denote by a and b the constants $\phi(x_0)$ and $\phi'(x_0)$, respectively. Then $y = \phi(x)$ solves the initial value problem

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad y(x_0) = a, y'(x_0) = b. \quad (3.2.8)$$

On the other hand, let c_1 and c_2 be constants to be determined so that $y(x) = c_1 y_1(x) + c_2 y_2(x)$ satisfies $y(x_0) = a, y'(x_0) = b$. Then

$$\begin{cases} c_1 y_1(x_0) + c_2 y_2(x_0) = a, \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) = b. \end{cases}$$

Since $\{y_1, y_2\}$ is a linearly independent set of solutions, Theorem 3.2.1 implies $W(y_1, y_2)(x_0) \neq 0$, and therefore, by the Cramer's rule, we find the unique

values of c_1 and c_2 given by

$$c_1 = \frac{\begin{vmatrix} a & y_2(x_0) \\ b & y_2'(x_0) \end{vmatrix}}{W(y_1, y_2)(x_0)} \text{ and } c_2 = \frac{\begin{vmatrix} y_1(x_0) & a \\ y_1'(x_0) & b \end{vmatrix}}{W(y_1, y_2)(x_0)}.$$

With these values of c_1 and c_2 , the function $y = c_1 y_1 + c_2 y_2$ solves the same initial value problem (3.2.8) as $\phi(x)$. By the uniqueness part of Theorem Theorem 3.1.3, we conclude that $\phi(x) = c_1 y_1(x) + c_2 y_2(x)$ on I . ■

3.2.2 Constructing Fundamental Sets of Solutions

We now discuss methods for determining fundamental sets of solutions for second order homogeneous linear differential equations with constant coefficients, specifically

$$ay'' + by' + cy = 0, \quad (3.2.9)$$

where a, b, c are constant coefficients and $a \neq 0$. This equation can equivalently be written in the standard form

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = 0. \quad (3.2.10)$$

Because the coefficient functions are constants and defined on $(-\infty, \infty)$, the interval I in the existence and uniqueness Theorem 3.1.3 for (3.2.9) is the entire set of real numbers $(-\infty, \infty)$. Consequently, the solutions in any fundamental set are defined on $(-\infty, \infty)$.

Motivated from first order linear equations, we expect that solutions appear as exponential functions. So, we let $y = e^{\lambda x}$ be a *trial* solution of (3.2.9) for some number (possibly complex) λ . Differentiating $y = e^{\lambda x}$ twice with respect to x and substituting y, y', y'' into (3.2.9) yields

$$(a\lambda^2 + b\lambda + c)e^{\lambda x} = 0$$

for all x in $(-\infty, \infty)$. In particular, when $x = 0$, we obtain

$$a\lambda^2 + b\lambda + c = 0, \quad (3.2.11)$$

which is called the *auxiliary* (or *characteristic*)¹ equation for (3.2.9). The roots² of (3.2.11) are given by

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad (3.2.12)$$

which are *distinct real*, *complex*, or *repeated* according as $b^2 - 4ac > 0$, $b^2 - 4ac < 0$, or $b^2 - 4ac = 0$. We now construct a fundamental set of solutions for each case.

Case I. $b^2 - 4ac > 0$ (Distinct Real Roots). When $b^2 - 4ac > 0$, the auxiliary equation (3.2.11) possesses two distinct real roots λ_1 and λ_2 given by (3.2.12). Take

$$\lambda_1 = -\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a} \text{ and } \lambda_2 = -\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}. \quad (3.2.13)$$

¹The polynomial $p(\lambda) = a\lambda^2 + b\lambda + c$ is called the *characteristic polynomial* for the differential equation $ay'' + by' + cy = 0$.

²*Solutions* would be a more precise term, but we use *roots* here to avoid confusion with solutions of differential equations.

Since $\lambda_1 \neq \lambda_2$, it follows that $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$ are linearly independent solutions of (3.2.9). We confirm the linear independence of y_1 and y_2 by evaluating their Wronskian:

$$W(y_1, y_2)(x) = \begin{vmatrix} e^{\lambda_1 x} & e^{\lambda_2 x} \\ \lambda_1 e^{\lambda_1 x} & \lambda_2 e^{\lambda_2 x} \end{vmatrix} = (\lambda_2 - \lambda_1) e^{\lambda_1 x} e^{\lambda_2 x} \neq 0$$

on $(-\infty, \infty)$. Therefore, $\{y_1, y_2\}$ is a fundamental set of solutions. By Theorem 3.2.5, the general solution of (3.2.9) is

$$\boxed{y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}}, \quad (3.2.14)$$

where c_1 and c_2 are arbitrary constants.

Case II. $b^2 - 4ac < 0$ (Complex Roots). When $b^2 - 4ac < 0$, we have $\sqrt{b^2 - 4ac} = i\sqrt{4ac - b^2}$, where $i = \sqrt{-1}$, the imaginary unit in the complex number system. Then the auxiliary equation (3.2.11) possesses two distinct complex roots λ_1 and λ_2 given by (3.2.12). Take

$$\lambda_1 = -\frac{b}{2a} + i\frac{\sqrt{4ac - b^2}}{2a} \quad \text{and} \quad \lambda_2 = -\frac{b}{2a} - i\frac{\sqrt{4ac - b^2}}{2a}. \quad (3.2.15)$$

Let

$$\alpha = -\frac{b}{2a} \quad \text{and} \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}.$$

It is evident that $\beta \neq 0$. Then the roots $\lambda_1 = \alpha + i\beta$ and $\lambda_2 = \alpha - i\beta$ of the auxiliary equation (3.2.11) give the complex-valued solutions $z_1 = e^{\lambda_1 x}$ and $z_2 = e^{\lambda_2 x}$ of (3.2.9). To extract real-valued solutions, we first use the Euler formula

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad \theta \text{ in radian,}$$

and write

$$\begin{aligned} z_1 &= e^{\lambda_1 x} = e^{(\alpha + i\beta)x} = e^{\alpha x} e^{i\beta x} = e^{\alpha x} \cos(\beta x) + i e^{\alpha x} \sin(\beta x), \text{ and} \\ z_2 &= e^{\lambda_2 x} = e^{(\alpha - i\beta)x} = e^{\alpha x} e^{-i\beta x} = e^{\alpha x} \cos(\beta x) - i e^{\alpha x} \sin(\beta x). \end{aligned}$$

It is evident that

$$z_1 + z_2 = 2e^{\alpha x} \cos(\beta x) \quad \text{and} \quad z_1 - z_2 = 2ie^{\alpha x} \sin(\beta x),$$

which imply

$$\begin{aligned} e^{\alpha x} \cos(\beta x) &= \frac{1}{2} (z_1 + z_2) = \frac{1}{2} z_1 + \frac{1}{2} z_2 \\ e^{\alpha x} \sin(\beta x) &= \frac{1}{2i} (z_1 - z_2) = \frac{1}{2i} z_1 - \frac{1}{2i} z_2. \end{aligned}$$

Since z_1 and z_2 are solutions of (3.2.10) (and equivalently of (3.2.9)), it follows from the superposition principle (Theorem 3.1.5) that $e^{\alpha x} \cos(\beta x)$ and $e^{\alpha x} \sin(\beta x)$ are solutions of the homogeneous equation (3.2.9). Denote $y_1 = e^{\alpha x} \cos(\beta x)$ and $y_2 = e^{\alpha x} \sin(\beta x)$. Then y_1 and y_2 are linearly independent solutions because their Wronskian satisfies

$$\begin{aligned} W(y_1, y_2)(x) &= \begin{vmatrix} e^{\alpha x} \cos(\beta x) & e^{\alpha x} \sin(\beta x) \\ -\beta e^{\alpha x} \sin(\beta x) + \alpha e^{\alpha x} \cos(\beta x) & \beta e^{\alpha x} \cos(\beta x) + \alpha e^{\alpha x} \sin(\beta x) \end{vmatrix} \\ &= e^{2\alpha x} \beta \\ &\neq 0 \end{aligned}$$

for all x in $(-\infty, \infty)$. Thus, $\{y_1, y_2\}$ is a fundamental set of solutions. By Theorem 3.2.5, the general solution of (3.2.9) is

$$y = c_1 e^{\alpha x} \cos(\beta x) + c_2 e^{\alpha x} \sin(\beta x) = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x)), \quad (3.2.16)$$

where c_1 and c_2 are arbitrary constants.

Case III. $b^2 - 4ac = 0$ (**Repeated Root**). In this case, the auxiliary equation (3.2.11) has a repeated real root

$$\lambda_1 = \lambda_2 = -\frac{b}{2a},$$

as given in (3.2.12). Consequently, we obtain only a single solution:

$$y_1 = e^{\lambda_1 x} = e^{-\frac{b}{2a}x}.$$

Since a fundamental set of solutions for (3.2.9) requires two linearly independent solutions, we must determine a second solution y_2 so that $\{y_1, y_2\}$ forms a fundamental set of solutions.

We next discuss a method known as **reduction of order**, which allows us to find a second linearly independent solution of a second order homogeneous linear differential equation, even when the equation has variable coefficients, as in (3.2.17), given that one nontrivial solution is already known.

Let $y_1(x)$ be a known solution of

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.2.17)$$

where $a_0(x)$ and $a_1(x)$ are continuous functions on an interval I , satisfying either $y_1(x) > 0$ for all x in I or $y_1(x) < 0$ for all x in I . Theorem 3.1.3 guarantees the existence of such a nonzero solution on I for the initial condition $y(x_0) = y_0 \neq 0$. We assume $y_1(x) > 0$ on I ; the other case can be treated similarly.

Our goal is to find a second solution y_2 such that $\{y_1, y_2\}$ is linearly independent on I . To this end, let $u(x)$ be a function on I such that $y_2(x) = u(x)y_1(x)$ is a solution of (3.2.17). If we find a non-constant function u , then it will follow that $\{y_1, y_2\}$ is a fundamental set of solutions. Then, suppressing x from the arguments for simplicity, we have

$$\begin{aligned} y_2' &= u'y_1 + uy_1' \quad \text{and} \\ y_2'' &= u''y_1 + 2u'y_1' + uy_1''. \end{aligned}$$

Substituting these into (3.2.17) gives

$$u''y_1 + 2u'y_1' + uy_1'' + a_1u'y_1 + a_1uy_1' + a_0uy_1 = 0.$$

which becomes

$$u''y_1 + u'(2y_1' + a_1y_1) + u(y_1'' + a_1y_1' + a_0y_1) = 0.$$

Since y_1 is a solution of (3.2.17), we have $y_1'' + a_1y_1' + a_0y_1 = 0$, and therefore we obtain

$$u''y_1 + u'(2y_1' + a_1y_1) = 0.$$

We observe that there is no u in this equation. Let $v = u'$. Then the equation becomes

$$v'y_1 + v(2y_1' + a_1y_1) = 0,$$

which is of the first order. Since y_1 never vanishes on I , we can rewrite the above equation as

$$v' + \left(\frac{2y_1' + a_1 y_1}{y_1} \right) v = 0. \quad (3.2.18)$$

A solution of (3.2.18) is given by

$$v(x) = e^{-\int \left(2 \frac{y_1'(x)}{y_1(x)} + a_1(x) \right) dx} = e^{-2 \ln |y_1(x)|} e^{-\int a_1(x) dx} = \frac{e^{-\int a_1(x) dx}}{[y_1(x)]^2}.$$

Since $v(x) = u'(x)$, we obtain

$$u(x) = \int \frac{e^{-\int a_1(x) dx}}{[y_1(x)]^2} dx$$

on I , and consequently, the solution $y_2(x)$ is given by

$$y_2(x) = y_1(x) \int \frac{e^{-\int a_1(x) dx}}{[y_1(x)]^2} dx. \quad (3.2.19)$$

on I .³ Thus, $\{y_1, y_2\}$ is a fundamental set of solutions for (3.2.17). We now return to finding a second linearly independent solution y_2 for **Case III**. Since

$$y_1 = e^{\lambda_1 x} = e^{-\frac{b}{2a}x}$$

and $a_1(x) = \frac{b}{a}$ is a constant, the reduction of the order formula (3.2.19) gives

$$y_2(x) = e^{-\frac{b}{2a}x} \int \frac{e^{-\int (b/a) dx}}{e^{-\frac{b}{a}x}} dx = e^{-\frac{b}{2a}x} \int 1 dx = x e^{-\frac{b}{2a}x}$$

on I . Since we only need a second solution y_2 , the constants of integration are set to zero for convenience. Thus, $\{y_1, y_2\}$ forms a fundamental set of solutions. By Theorem 3.2.5, the general solution of (3.2.9) is

$$y = c_1 e^{-\frac{b}{2a}x} + c_2 x e^{-\frac{b}{2a}x} = (c_1 + c_2 x) e^{-\frac{b}{2a}x}, \quad (3.2.20)$$

where c_1 and c_2 are arbitrary constants.

An alternative method for finding a second linearly independent solution when the characteristic polynomial has a repeated root is outlined in Exercises.

Remark 3.2.6

1. The method discussed above for finding $u(x)$ is known as the **reduction of order** because the resulting differential equation (3.2.18) for $v(x)$ is of first order; one order less than that of the original equation (3.2.17).
2. If the known solution y_1 vanishes at some points in the interval I , we can first apply the reduction of the order on a subinterval J of I where y_1 does not change sign, and then extend the solution to all of I using a theorem on continuation of solutions (which is beyond the scope of this book). In practice, we may formally compute $y_2(x)$ using (3.2.19) and then verify that y_2 is actually well-defined on the entire interval I . For this, see Example 3.2.11.

³This formula is due the French mathematician Joseph Liouville

Example 3.2.7 Find the general solution of

$$y'' - 16y = 0.$$

Solution. Assume a solution of the form $y = e^{\lambda x}$. Then $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substituting into the differential equation gives

$$\lambda^2 e^{\lambda x} - 16e^{\lambda x} = 0,$$

which gives

$$(\lambda^2 - 16)e^{\lambda x} = 0.$$

Since $e^{\lambda x} \neq 0$, the auxiliary equation that we obtain is

$$\lambda^2 - 16 = 0,$$

whose distinct real roots are

$$\lambda_1 = 4 \quad \text{and} \quad \lambda_2 = -4.$$

By (3.2.14), the general solution is

$$y(x) = c_1 e^{4x} + c_2 e^{-4x},$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.2.8 Find the general solution of

$$y'' - y' - 6y = 0.$$

Solution. Assume a solution of the form $y = e^{\lambda x}$. Then $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substituting into the differential equation gives

$$\lambda^2 e^{\lambda x} - \lambda e^{\lambda x} - 6e^{\lambda x} = 0,$$

which gives

$$(\lambda^2 - \lambda - 6)e^{\lambda x} = 0.$$

Since $e^{\lambda x} \neq 0$, the auxiliary equation that we obtain is

$$\lambda^2 - \lambda - 6 = 0,$$

whose distinct real roots are

$$\lambda_1 = 3 \quad \text{and} \quad \lambda_2 = -2.$$

By (3.2.14), the general solution is

$$y(x) = c_1 e^{3x} + c_2 e^{-2x},$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.2.9 Find the general solution of

$$3y'' + 2y' + y = 0.$$

Solution. Assume a solution of the form $y = e^{\lambda x}$. Then $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substituting into the differential equation gives

$$(3\lambda^2 + 2\lambda + 1)e^{\lambda x} = 0.$$

Then the auxiliary equation is

$$3\lambda^2 + 2\lambda + 1 = 0,$$

whose complex roots are

$$\lambda = \frac{-2 \pm \sqrt{-8}}{6} = \frac{-1 \pm \sqrt{2}i}{3} = -\frac{1}{3} \pm i\frac{\sqrt{2}}{3}.$$

Substituting $\alpha = -\frac{1}{3}$ and $\beta = \frac{\sqrt{2}}{3}$ in (3.2.16), the general solution is

$$y = e^{-x/3} \left(c_1 \cos \left(\frac{\sqrt{2}}{3}x \right) + c_2 \sin \left(\frac{\sqrt{2}}{3}x \right) \right),$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.2.10 Find the general solution of

$$y'' + 8y' + 16y = 0.$$

Solution. Assume a solution of the form $y = e^{\lambda x}$. Then $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substituting into the differential equation gives

$$\lambda^2 e^{\lambda x} + 8\lambda e^{\lambda x} + 16e^{\lambda x} = 0,$$

which gives

$$(\lambda^2 + 8\lambda + 16)e^{\lambda x} = 0.$$

Since $e^{\lambda x} \neq 0$, the auxiliary equation that we obtain is

$$\lambda^2 + 8\lambda + 16 = 0,$$

whose repeated real root is

$$\lambda = -4.$$

By (3.2.20), the general solution is

$$y(x) = (c_1 + c_2 x)e^{-4x},$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.2.11 Given that $y_1 = \sin x$ solves $y'' + y = 0$ on $(-\infty, \infty)$, find a second linearly independent solution using the formula (3.2.19) for the reduction of the order.

Solution. For $y'' + y = 0$, we have $a_1 = 0$. Also, the given solution $y_1 = \sin x$ vanishes at the points $x = n\pi$, $n = 0, \pm 1, \pm 2, \dots$. Therefore, we restrict y_1 to an interval J not containing these points. For example, take $J = (0, \pi)$. Applying the formula (3.2.19), a second solution y_2 is given by

$$\begin{aligned} y_2(x) &= \sin x \int \frac{1}{\sin^2 x} dx \\ &= \sin x \int \csc^2 x dx = -\sin x \cot x \\ &= -\cos x \end{aligned}$$

on J . By scaling, we redefine $y_2 = \cos x$ on $J = (0, \pi)$. Since $y_2 = \cos x$ can be extended by the same formula to all of $(-\infty, \infty)$ and

$$W(\cos x, \sin x) = 1 \neq 0$$

for all x in $(-\infty, \infty)$, it follows [Theorem 3.2.1](#) that $y_1 = \sin x$ and $y_2 = \cos x$ are linearly independent on $(-\infty, \infty)$. ▲

Techniques for finding fundamental sets of solutions for homogeneous linear equations with variable coefficients are discussed in [Section 3.6](#). In this section, we will only discuss the **Cauchy–Euler** equations of the form

$$x^2 y'' + bxy' + cy = 0, \quad x > 0, \quad (3.2.21)$$

where b and c are real constants. The method uses $y = x^\lambda$ as a trial solution, with λ to be determined from the auxiliary equation

$$\lambda^2 + (b-1)\lambda + c = 0,$$

whose roots are real and distinct, complex, or repeated according as

$$(b-1)^2 - 4c > 0, \quad (b-1)^2 - 4c < 0, \quad \text{or} \quad (b-1)^2 - 4c = 0.$$

We next discuss three examples, one for each case.

Example 3.2.12 Find the general solution of the Cauchy-Euler equation

$$x^2 y'' + 4xy' + y = 0, \quad x > 0.$$

Solution. Take $y = x^\lambda$ as a trial solution, where λ is a number to be determined. Differentiating $y = x^\lambda$ twice with respect to x and substituting y, y', y'' into the differential equation, we obtain

$$(\lambda^2 + 3\lambda + 1)x^\lambda = 0$$

for all x in an interval I , so the auxiliary equation is

$$\lambda^2 + 3\lambda + 1 = 0$$

with roots

$$\lambda = \frac{-3 \pm \sqrt{5}}{2}.$$

Let

$$\lambda_1 = \frac{-3 + \sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{-3 - \sqrt{5}}{2}.$$

Obviously, $\lambda_1 < 0$ and $\lambda_2 < 0$. From the two options $(-\infty, 0)$ and $(0, \infty)$, let us take $I = (0, \infty)$. Then $y_1 = x^{\lambda_1}$ and $y_2 = x^{\lambda_2}$ are solutions on I , and their Wronskian

$$\begin{aligned} W(y_1, y_2)(x) &= \begin{vmatrix} x^{\lambda_1} & x^{\lambda_2} \\ \lambda_1 x^{\lambda_1-1} & \lambda_2 x^{\lambda_2-1} \end{vmatrix} \\ &= (\lambda_2 - \lambda_1)x^{(\lambda_1+\lambda_2-2)} \\ &= (\lambda_2 - \lambda_1)x^{-5} \\ &\neq 0 \end{aligned}$$

for all x in I since $\lambda_1 + \lambda_2 = -3$. Thus $\{y_1, y_2\}$ is a fundamental set of solutions, and therefore the general solution of the Cauchy-Euler equation is

$$y = c_1 x^{\lambda_1} + c_2 x^{\lambda_2} = c_1 x^{\frac{-3+\sqrt{5}}{2}} + c_2 x^{\frac{-3-\sqrt{5}}{2}}$$

on I , where c_1 and c_2 are arbitrary constants. ▲

Example 3.2.13 Solve the Cauchy–Euler equation

$$x^2 y'' - y' + 5y = 0, \quad x > 0.$$

Solution. Let $y = x^\lambda$ be a trial solution. Then

$$y' = \lambda x^{\lambda-1}, \quad y'' = \lambda(\lambda-1)x^{\lambda-2}.$$

Substituting into the differential equation gives

$$(\lambda^2 - 2\lambda + 5)x^\lambda = 0.$$

Then the auxiliary equation is

$$\lambda^2 - 2\lambda + 5 = 0$$

whose solutions are

$$\lambda = \frac{2 \pm \sqrt{-16}}{2} = 1 \pm 2i.$$

For complex numbers $\lambda = 1 \pm 2i$, we have

$$x^{1 \pm 2i} = x x^{\pm 2i} = x e^{\pm 2i \ln x} = x (\cos(2 \ln x) \pm i \sin(2 \ln x)).$$

Let $y_1 = x \cos(2 \ln x)$ and $y_2 = x \sin(2 \ln x)$. It is routine to verify that y_1 and y_2 are linearly independent solutions on $(0, \infty)$. Then the general solution of the Cauchy–Euler equation is

$$y(x) = x(c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)), \quad x > 0,$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.2.14 Solve the Cauchy–Euler equation

$$x^2 y'' - y' + y = 0, \quad x > 0.$$

Solution. Let $y = x^\lambda$ be a trial solution. Then

$$y' = \lambda x^{\lambda-1}, \quad y'' = \lambda(\lambda-1)x^{\lambda-2}.$$

Substituting into the differential equation yields

$$(\lambda^2 - 2\lambda + 1)x^\lambda = 0.$$

Then the auxiliary equation is

$$\lambda^2 - 2\lambda + 1 = 0, \quad \text{i.e., } (\lambda - 1)^2 = 0.$$

Its solutions are

$$\lambda = 1, 1 \text{ (repeated)}$$

We thus have one solution as $y_1 = x$. By the reduction of the order formula (3.2.19), a second linearly independent solution y_2 is given by

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int a_1(x) dx}}{y_1^2} dx \quad \text{with } a_1(x) = -\frac{1}{x} \\ &= x \int \frac{e^{-\int (-\frac{1}{x}) dx}}{x^2} dx \\ &= x \int \frac{1}{x} dx \\ &= x \ln x. \end{aligned}$$

Hence the general solution of the Cauchy-Euler equation is

$$y = c_1x + c_2x \ln x, \quad x > 0,$$

where c_1 and c_2 are arbitrary constants. ▲

3.2.3 Exercises

1. Verify that the indicated solution set for each differential equation below is linearly independent on the specified interval I by computing Wronskian of the solutions (see [Theorem 3.2.1](#)).

- i. $y'' = 0$, $\{7, x\}$, $I = (-\infty, \infty)$
- ii. $y'' + y = 0$, $\{\sin x, \cos x\}$, $I = (-\infty, \infty)$
- iii. $y'' + y = 0$, $\{\sin x, \sin x + \cos x\}$, $I = (-\infty, \infty)$
- iv. $y'' - 4y = 0$, $\{e^{2x}, e^{-2x}\}$, $I = (-\infty, \infty)$
- v. $y'' - 4y = 0$, $\{e^{2x}, e^{-2x}\}$, $I = (-\infty, \infty)$
- vi. $y'' - 4y' + 4y = 0$, $\{e^{2x}, xe^{2x}\}$, $I = (-\infty, \infty)$
- vii. $y'' - 4y' + 4y = 0$, $\{e^{2x}, (x+1)e^{2x}\}$, $I = (-\infty, \infty)$
- viii. $y'' - 4y' + 13y = 0$, $\{e^{2x} \cos(3x), e^{2x} \sin(3x)\}$, $I = (-\infty, \infty)$
- ix. $y'' - \frac{2}{x}y' = 0$, $\{1, x^3\}$, $I = (0, \infty)$
- x. $x^2y'' + xy' + y = 0$, $\{\cos(\ln x), \sin(\ln x)\}$, $I = (0, \infty)$
- xi. $x^2y'' + 4xy' + y = 0$, $\{x^{\frac{-3+\sqrt{5}}{2}}, x^{\frac{-3-\sqrt{5}}{2}}\}$, $I = (0, \infty)$

2. Find the general solution of each differential equation below (see [Theorem 3.2.5](#)).

- i. $y'' - 5y' + 6y = 0$
- ii. $y'' + 11y' + 30y = 0$
- iii. $2y'' + 5y' + 3y = 0$
- iv. $y'' - 7y' - 30y = 0$
- v. $y'' + 4y' = 0$
- vi. $y'' + 4y' + 13y = 0$
- vii. $y'' - 5y' + y = 0$
- viii. $2y'' + 5y' + y = 0$

3. Find the general solution of each of the following Cauchy-Euler differential equations (see [Example 3.2.12](#)).

- i. $x^2y'' + xy' - y = 0$
- ii. $xy'' - 2y' = 0$
- iii. $x^2y'' - xy' - 3y = 0$
- iv. $x^2y'' - 2y = 0$
- v. $x^2y'' + xy' + y = 0$
- vi. $x^2y'' + 3xy' + 5y = 0$

4. Find the solution of each of the following initial value problems:

- i. $y'' - 16y = 0$, $y(0) = 0$, $y'(1) = 1$
(Hint: See [Example 3.2.7](#) for the general solution.)
- ii. $y'' - y' - 6y = 0$, $y(0) = -1$, $y'(0) = 1$
(Hint: See [Example 3.2.8](#) for the general solution.)
- iii. $3y'' + 2y' + y = 0$, $y(0) = 1$, $y'(0) = 0$

(Hint: See [Example 3.2.9](#) for the general solution.)

iv. $y'' + 8y' + 16y = 0$, $y(1) = 1$, $y'(1) = -1$

(Hint: See [Example 3.2.10](#) for the general solution.)

5. Solve the following initial value problems by using the general solutions of the related equations obtained in [Exercise 3.2.3.2](#) and [Exercise 3.2.3.3](#).

i. $y'' - 5y' + 6y = 0$, $y(0) = 0$, $y'(0) = 1$

ii. $y'' + 11y' + 30y = 0$, $y(0) = 0$, $y'(0) = 1$

iii. $2y'' + 5y' + 3y = 0$, $y(0) = 1$, $y'(0) = 0$

iv. $y'' - 7y' - 30y = 0$, $y(0) = 1$, $y'(0) = 0$

v. $y'' + 4y' = 0$, $y(0) = 1$, $y'(0) = 1$

vi. $y'' + 4y' + 13y = 0$, $y(0) = 1$, $y'(0) = 0$

vii. $y'' - 5y' + y = 0$, $y(0) = 0$, $y'(0) = 1$

viii. $2y'' + 5y' + y = 0$, $y(0) = 0$, $y'(0) = 1$

ix. $x^2y'' + xy' - y = 0$, $y(1) = 1$, $y'(1) = 0$

x. $x^2y'' + xy' + y = 0$, $y(1) = 1$, $y'(1) = 1$

xi. $x^2y'' - xy' - 3y = 0$, $y(1) = 1$, $y'(1) = 0$

xii. $x^2y'' + 3xy' + 5y = 0$, $y(1) = 1$, $y'(1) = 0$

6. Suppose that the characteristic polynomial

$$p(\lambda) = a\lambda^2 + b\lambda + c$$

of the differential equation

$$ay'' + by' + cy = 0, \quad a \neq 0,$$

where a, b, c are constants, has a repeated root λ_0 . Then $y_1 = e^{\lambda_0 x}$ is a solution of the differential equation. To find a second linearly independent solution, proceed as follows.

- a. Let $y = e^{\lambda x}$ with parameter λ . Show that

$$a \frac{\partial^2}{\partial x^2}(e^{\lambda x}) + b \frac{\partial}{\partial x}(e^{\lambda x}) + c(e^{\lambda x}) = p(\lambda)e^{\lambda x}.$$

- b. Differentiate both sides of the equation in [Item 3.2.3.6.a](#) with respect to λ to obtain

$$a \frac{\partial^2}{\partial x^2}(xe^{\lambda x}) + b \frac{\partial}{\partial x}(xe^{\lambda x}) + c(xe^{\lambda x}) = (p'(\lambda) + \lambda p(\lambda))e^{\lambda x}.$$

- c. Using $p(\lambda_0) = p'(\lambda_0) = 0$, conclude that $y_2 = xe^{\lambda_0 x}$ is also a solution.
- d. Show that $y_1 = e^{\lambda_0 x}$ and $y_2 = xe^{\lambda_0 x}$ are linearly independent on $(-\infty, \infty)$.

7. Let $y_1(x)$ be a known solution of

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.2.22)$$

where $a_0(x)$ and $a_1(x)$ are continuous functions on an interval I , satisfying $y_1(x) > 0$ for all x in I . Use the Wronskian to derive Liouville's formula (3.2.19) as follows:

- a. Suppose that y_2 is a solution of (3.2.22) such that $\{y_1, y_2\}$ is linearly independent on I . Show that

$$\left(\frac{y_2}{y_1} \right)' = \frac{W(x)}{y_1^2} \quad \text{on } I,$$

where $W(x)$ is the Wronskian of y_1 and y_2 given by $W(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$.

- b. Using Item 3.2.3.7.a, show that

$$y_2 = y_1(x) \int \frac{W(x)}{[y_1(x)]^2} dx.$$

- c. Obtain Liouville's formula

$$y_2 = y_1(x) \int \frac{e^{-\int a_1(x) dx}}{[y_1(x)]^2} dx$$

from part Item 3.2.3.7.b.

The same steps also apply when the known solution y_1 satisfies $y_1(x) < 0$ for all x in I . For the general case, see (ii) of Item 2.

3.3 Second Order Nonhomogeneous Linear Differential Equations

Objectives

- Understand the structure of the general solution of second order nonhomogeneous linear differential equations.
- Learn methods for solving second order nonhomogeneous linear differential equations.

In this section, we discuss methods for solving second order nonhomogeneous linear differential equations. The same ideas extend naturally to higher order equation (see Section 3.4). The following is concerning how the general solutions of nonhomogeneous linear differential equations must be formatted.

Theorem 3.3.1 (General Solution). *Consider the nonhomogeneous equation*

$$y'' + a_1(x)y' + a_0(x)y = f(x), \quad (3.3.1)$$

where a_1 , a_0 and f are continuous functions on an interval I and the associated homogeneous equation

$$y'' + a_1(x)y' + a_0(x)y = 0 \quad (3.3.2)$$

on I . Let y_h be the general solution of (3.3.2) and y_p be a particular solution of

the nonhomogeneous equation (3.3.1) on I . Then the general solution of (3.3.1) is of the form

$$\boxed{y = y_h(x) + y_p(x)} \quad (3.3.3)$$

on I .

Proof. Let y_p be a particular solution of (3.3.1). Let y be any other solution of (3.3.1). Then the difference

$$\phi(x) = y(x) - y_p(x) \quad (3.3.4)$$

is a solution of the homogeneous equation (3.3.2). In fact, for all x in I , we have

$$\begin{aligned} & \phi''(x) + a_1(x)\phi'(x) + a_0(x)\phi(x) \\ &= (y''(x) - y_p''(x)) + a_1(x)(y'(x) - y_p'(x)) + a_0(x)(y(x) - y_p(x)) \\ &= (y''(x) + a_1(x)y'(x) + a_0(x)y(x)) \\ &\quad - (y_p''(x) + a_1(x)y_p'(x) + a_0(x)y_p(x)) \\ &= f(x) - f(x) \\ &= 0. \end{aligned}$$

Therefore, $\phi(x)$ takes the form of $y_h(x)$ on I . It follows from (3.3.4) that

$$y = y_h(x) + y_p(x) \quad \text{on } I$$

is the general solution of (3.3.1). ■

Remark 3.3.2 Theorem 3.3.1 provides the fundamental structure for the general solution of nonhomogeneous linear differential equations. The function y_h in (3.3.3) is called the **complementary function** of the general solution, and it is given by

$$y_h(x) = c_1 y_1(x) + c_2 y_2(x),$$

where y_1 and y_2 form a fundamental set of solutions for (3.3.2) and c_1 and c_2 are arbitrary constants.

In practice, solving a nonhomogeneous equation begins with finding the complementary function by determining a fundamental set of solutions to the associated homogeneous equation. We then find a particular solution of the nonhomogeneous equation, and the sum of the two provides the general solution. Finding a particular solution is typically the more difficult step; we outline two procedures for finding it in this section.

3.3.1 Variation of Parameters

The **variation of parameters** is a systematic method for constructing a particular solution of a nonhomogeneous linear differential equation of the form

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x), \quad (3.3.5)$$

where the forcing term g and coefficient functions a_0, a_1 , and a_2 are assumed to be continuous on an interval I with $a_2(x) \neq 0$ on I . We will be working with scenarios in which $a_2(x)$ is either only positive or only negative on I .

To formulate the method, we put (3.3.5) in the standard form

$$y'' + a_1(x)y' + a_0(x)y = f(x), \quad (3.3.6)$$

by first dividing both sides of (3.3.5) by $a_2(x)$ to obtain

$$y'' + \frac{a_1(x)}{a_2(x)}y' + \frac{a_0(x)}{a_2(x)}y = \frac{g(x)}{a_2(x)},$$

and then redefining

$$\frac{a_1(x)}{a_2(x)} \quad \text{and} \quad \frac{a_0(x)}{a_2(x)}$$

as $a_1(x)$ and $a_0(x)$, respectively, and defining $f(x) = \frac{g(x)}{a_2(x)}$.

Assume that $\{y_1, y_2\}$ is a fundamental set of solutions on I for the homogeneous equation

$$y'' + a_1(x)y' + a_0(x)y = 0, \quad (3.3.7)$$

associated to the nonhomogeneous equation (3.3.6). Then the general solution to (3.3.6) is

$$y_h(x) = c_1y_1(x) + c_2y_2(x), \quad (3.3.8)$$

for all x in I , where c_1 and c_2 are arbitrary constants.

To find a particular solution of the nonhomogeneous equation (3.3.6), the method of variation of parameters assumes a solution of the form

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x),$$

for all x in I , where u_1 and u_2 are functions to be determined. *The choice of y_p reflects the name of the method: variation of parameters. The constants c_1 and c_2 in the solution y_h of the homogeneous equation (3.3.7) are replaced by the functions $u_1(x)$ and $u_2(x)$, thereby varying the parameters in order to obtain a particular solution to (3.3.6).* For the remainder of the discussion of the method, we suppress x in all functions to minimize distractions.

Differentiating $y_p = u_1y_1 + u_2y_2$, we have

$$y'_p = u_1y'_1 + u'_1y_1 + u_2y'_2 + u'_2y_2 = u_1y'_1 + u_2y'_2 + (u'_1y_1 + u'_2y_2)$$

and

$$\begin{aligned} y''_p &= u_1y''_1 + u'_1y'_1 + u_2y''_2 + u'_2y'_2 + (u'_1y_1 + u'_2y_2)' \\ &= u_1y''_1 + u_2y''_2 + u'_1y'_1 + u'_2y'_2 + (u'_1y_1 + u'_2y_2)' \end{aligned}$$

We impose the condition

$$u'_1y_1 + u'_2y_2 = 0,$$

which is trivially satisfied when u_1 and u_2 are constant functions. The condition simplifies the expressions for y'_p and y''_p as follows:

$$\begin{aligned} y'_p &= u_1y'_1 + u_2y'_2, \quad \text{and} \\ y''_p &= u_1y''_1 + u_2y''_2 + u'_1y'_1 + u'_2y'_2. \end{aligned}$$

Substituting y_p , y'_p , and y''_p for y , y' , and y'' , respectively, into (3.3.6) gives

$$u'_1y'_1 + u_1y''_1 + u'_2y'_2 + u_2y''_2 + a_1u_1y'_1 + a_1u_2y'_2 + a_0u_1y_1 + a_0u_2y_2 = f,$$

which can be rewritten as

$$u_1(y''_1 + a_1y'_1 + a_0y_1) + u_2(y''_2 + a_1y'_2 + a_0y_2) + u'_1y'_1 + u'_2y'_2 = f.$$

Since y_1 and y_2 satisfy the homogeneous equation (3.3.7), the expressions in parentheses are zero, and therefore we obtain the system

$$\begin{cases} y_1u'_1 + y_2u'_2 = 0, \\ y'_1u'_1 + y'_2u'_2 = f. \end{cases}$$

This system can be solved for u'_1 and u'_2 :

$$u'_1 = -\frac{y_2 f}{W}, \quad u'_2 = \frac{y_1 f}{W},$$

where $W = y_1 y'_2 - y_2 y'_1$ is the Wronskian of y_1 and y_2 . By integrating, we obtain

$$u_1 = -\int \frac{y_2 f}{W} dx, \quad u_2 = \int \frac{y_1 f}{W} dx. \quad (3.3.9)$$

Therefore, y_p is given by

$$y_p = -y_1 \int \frac{y_2 f}{W} dx + y_2 \int \frac{y_1 f}{W} dx. \quad (3.3.10)$$

Consequently, the general solution of (3.3.6) is

$$y = y_h + y_p = c_1 y_1 + c_2 y_2 - y_1 \int \frac{y_2 f}{W} dx + y_2 \int \frac{y_1 f}{W} dx.$$

Remark 3.3.3 The main advantage of the method of *variation of parameters* over the method of *undetermined coefficients* is that it is applicable to all non homogeneous linear differential equations, including the equations with variable coefficients. However, the evaluation of the integrals in (3.3.9) can become quite cumbersome even when $f(x)$ involves only elementary functions, such as the ones treated with the method of undetermined coefficients discussed Subsection 3.3.2.

We illustrate the method of variation of parameters with the following initial value problem which can be readily solved using the method of undetermined coefficients.

Example 3.3.4 Solve the initial value problem

$$y'' + 8y' + 16y = 4x, \quad y(0) = 0, \quad y'(0) = 0$$

by variation of parameters.

Solution. The auxiliary equation for the associated homogeneous equation is $\lambda^2 + 8\lambda + 16 = 0$ which gives $\lambda = -4, -4$, and so the functions

$$y_1(x) = e^{-4x}, \quad y_2(x) = x e^{-4x}$$

form a fundamental set solutions to $y'' + 8y' + 16y = 0$. The Wronskian of y_1 and y_2 is

$$W(x) = \begin{vmatrix} e^{-4x} & x e^{-4x} \\ -4e^{-4x} & (1-4x)e^{-4x} \end{vmatrix} = e^{-8x}(1-4x+4x) = e^{-8x}.$$

We note that the given nonhomogeneous equation is in the standard form with $f(x) = 4x$ and therefore a particular solution by the method of variation of parameters is given by

$$y_p = -y_1 \int \frac{y_2 f}{W} dx + y_2 \int \frac{y_1 f}{W} dx.$$

Then the integration by parts with the Kronecker method (see Figure 3.3.5) gives

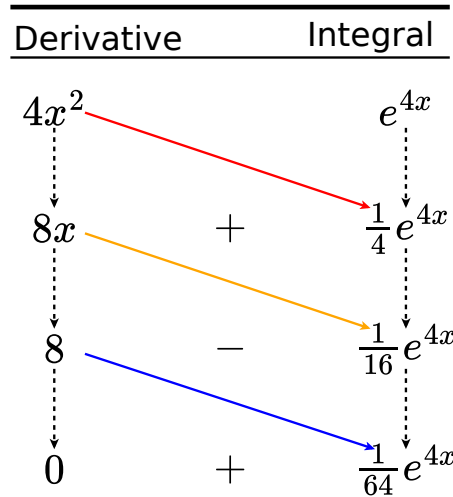


Figure 3.3.5 Kronecker method for integration by parts

$$\int \frac{y_2 f}{W} dx = \int \frac{4x^2 e^{-4x}}{e^{-8x}} dx = x^2 e^{4x} - \frac{1}{2} x e^{4x} + \frac{1}{8} e^{4x}.$$

In a similar manner, we also evaluate

$$\int \frac{y_1 f}{W} dx = \int \frac{4x e^{-4x}}{e^{-8x}} dx = \int 4x e^{4x} dx = x e^{4x} - \frac{1}{4} e^{4x}.$$

Then y_p becomes

$$\begin{aligned} y_p &= -e^{-4x} \left(x^2 e^{4x} - \frac{1}{2} x e^{4x} + \frac{1}{8} e^{4x} \right) + x e^{-4x} \left(x e^{4x} - \frac{1}{4} e^{4x} \right) \\ &= -x^2 + \frac{x}{2} - \frac{1}{8} + x^2 - \frac{x}{4} \\ &= \frac{x}{4} - \frac{1}{8}. \end{aligned}$$

The general solution of the nonhomogeneous differential equation is

$$y(x) = c_1 e^{-4x} + c_2 x e^{-4x} + \frac{x}{4} - \frac{1}{8},$$

where c_1 and c_2 are arbitrary constants. Since $y(0) = 0$, we get $c_1 = \frac{1}{8}$. We find

$$y'(x) = -4c_1 e^{-4x} + c_2 e^{-4x} - 4c_2 x e^{-4x} + \frac{1}{4}.$$

Since $y'(0) = 0$, we have $-4c_1 + c_2 + \frac{1}{4} = 0$. Using $c_1 = \frac{1}{8}$, we get $-\frac{1}{2} + c_2 + \frac{1}{4} = 0$, so that $c_2 = \frac{1}{4}$. Hence the solution to the given initial value problem is

$$y(x) = \frac{1}{8} e^{-4x} + \frac{1}{4} x e^{-4x} + \frac{x}{4} - \frac{1}{8}$$

on $(-\infty, \infty)$. ▲

3.3.2 Method of Undetermined Coefficients

We note that variation of parameters often leads to integrals that can be quite involved and sometimes unexpectedly time-consuming even for simple forcing terms such as $f(x) = \sin x$. For instance, to find a particular solution of

$$y'' - 9y = \sin x,$$

one may instead *guess* a solution of the form $y_p(x) = A \sin x$ (or a slightly more general form, $y_p(x) = A \sin x + B \cos x$, if needed). See [Example 3.3.6](#) and [Example 3.3.8](#) below. This strategy of guessing a trial function for y_p based on type of $f(x)$ and determining the unknown constant coefficients in the trial function is known as the method of ***undetermined coefficients***. However, a demerit of this method is that it becomes quite complicated when differential equations has variable coefficients.

Example 3.3.6 Find a particular solution of

$$y'' - 9y = \sin x \tag{3.3.11}$$

by using the method of undetermined coefficients and write down the general solution.

Solution. Let

$$y_p = A \sin x$$

be a particular solution of the equation, where A is a constant to be determined. Differentiating yields

$$y'_p = A \cos x, \quad y''_p = -A \sin x.$$

Substituting these into (3.3.11) gives

$$-A \sin x + 9A \sin x = \sin x,$$

i.e.,

$$8A \sin x = \sin x,$$

an identity in x . This implies $A = \frac{1}{8}$. Therefore,

$$y_p = \frac{1}{8} \sin x$$

is a particular solution.

To determine the complementary function of the general solution, the auxiliary equation for the associated homogeneous equation

$$y'' - 9y = 0$$

is

$$\lambda^2 - 9 = 0,$$

whose roots are $\lambda_1 = 3$ and $\lambda_2 = -3$. Then the complementary function y_h is given by

$$y_h = c_1 e^{3x} + c_2 e^{-3x},$$

where c_1 and c_2 are arbitrary constants. Finally, the general solution of (3.3.11) is

$$y = y_h(x) + y_p(x) = c_1 e^{3x} + c_2 e^{-3x} + \frac{1}{8} \sin x,$$

where c_1 and c_2 are arbitrary constants. ▲

Remark 3.3.7

- i. The method of undetermined coefficients is most effective when the forcing function $f(x)$ is a polynomial, exponential, sine, cosine, or a linear combination of these functions.
- ii. When variation of parameters may involve complicated integrals, the method of undetermined coefficients may be a quicker alternative technique for finding particular solutions of equations with constant coefficients and simple combination of elementary functions for $f(x)$ remarked in part (i). The method of undetermined coefficients, however, requires a well-informed guessing for y_p , which makes it challenging to apply even when $f(x)$ is a common function, such as $\sec x$ or $\tan x$.
- iii. A particular solution of a differential equation is just one member of the family represented by its general solution, and thus infinitely many particular solutions exist. In [Example 3.3.6](#), the guess for y_p was relatively easy to make by observing that the equation has no first derivative term and its forcing function is $\sin x$.

In the next example, the equation also has a first derivative term, so the trial solution used for y_p in [Example 3.3.6](#) does not work.

Example 3.3.8 Find a particular solution of

$$y'' - 6y' + 9y = \sin x \quad (3.3.12)$$

and then write down the general solution.

Solution. First, we use the same trial solution as in [Example 3.3.6](#) and see what happens. Let

$$y_p = A \sin x$$

be a particular solution, where A is a constant to be determined. Differentiating yields

$$y_p' = A \cos x, \quad y_p'' = -A \sin x.$$

Substituting these into (3.3.12) would give

$$-A \sin x - 6A \cos x + 9A \sin x = \sin x,$$

i.e.,

$$(8A - 1) \sin x - 6A \cos x = 0,$$

which is not an identity in x for any value of A because $\sin x$ and $\cos x$ are linearly independent. Thus, no value of A makes $y_p = A \sin x$ a valid particular solution. This occurs because $\cos x$ is involved in the process. Therefore, we modify our trial function for y_p and take

$$y_p = A \sin x + B \cos x$$

where A and B are constants to be determined. Differentiating twice gives

$$y_p' = A \cos x - B \sin x, \quad y_p'' = -A \sin x - B \cos x.$$

Substituting these into (3.3.12) gives

$$(-A \sin x - B \cos x) - 6(A \cos x - B \sin x) + 9(A \sin x + B \cos x) = \sin x,$$

i.e.,

$$(8A + 6B) \sin x + (8B - 6A) \cos x = \sin x,$$

an identity in x . Since $\sin x$ and $\cos x$ are linearly independent, we obtain

$$\begin{cases} 8A + 6B = 1, \\ 8B - 6A = 0. \end{cases}$$

Solving these equations for A and B yields $A = \frac{2}{25}$ and $B = \frac{3}{50}$. Thus,

$$y_p = \frac{2}{25} \sin x + \frac{3}{50} \cos x$$

is a particular solution.

To determine the complementary function of the general solution of (3.3.12), the auxiliary equation for the associated homogeneous equation

$$y'' - 6y' + 9y = 0$$

is

$$\lambda^2 - 6\lambda + 9 = 0,$$

which has a repeated root $\lambda = 3$. Then the complementary function y_h is given by

$$y_h = c_1 e^{3x} + c_2 x e^{3x},$$

where c_1 and c_2 are arbitrary constants. Hence the general solution of (3.3.12) is

$$y = y_h(x) + y_p(x) = c_1 e^{3x} + c_2 x e^{3x} + \frac{2}{25} \sin x + \frac{3}{50} \cos x,$$

where c_1 and c_2 are arbitrary constants. ▲

In the previous two examples, we chose trial solutions y_p before finding the complementary function without any difficulty. In general, however, a trial solution may include terms that duplicate in the complementary function, which are redundant (even useless) since they satisfy the associated homogeneous equation. Therefore, it is useful to determine the complementary function first and then select a trial function for a particular solution. We will typically follow this procedure from now on.

Example 3.3.9 Find the general solution of

$$y'' - 4y' + 4y = e^{2x}. \quad (3.3.13)$$

Solution. First, we find the complementary function. The auxiliary equation for the associated homogeneous equation

$$y'' - 4y' + 4y = 0$$

is

$$\lambda^2 - 4\lambda + 4 = 0,$$

which has a repeated root $\lambda = 2$. Then the complementary function y_h is given by

$$y_h = c_1 e^{2x} + c_2 x e^{2x},$$

where c_1 and c_2 are arbitrary constants.

Next, we choose a trial solution for the particular solution. Since the right-hand side is e^{2x} , and this term appears in the complementary function, we need to multiply by x^2 . Thus, we take

$$y_p = Ax^2 e^{2x}$$

where A is a constant to be determined. Differentiating twice gives

$$y_p' = Ae^{2x}(2x^2 + 2x), \quad y_p'' = Ae^{2x}(4x^2 + 8x + 4).$$

Substituting these into (3.3.13) gives

$$Ae^{2x}(4x^2 + 8x + 4) - 4Ae^{2x}(2x^2 + 2x) + 4Ax^2e^{2x} = e^{2x}.$$

This yields

$$4Ae^{2x} = e^{2x}.$$

Therefore, $A = 1/4$ and

$$y_p = \frac{1}{4}x^2e^{2x}.$$

Hence the general solution is

$$y = c_1e^{2x} + c_2xe^{2x} + \frac{1}{4}x^2e^{2x},$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.3.10 Find the general solution of

$$y'' + y = \cos x. \quad (3.3.14)$$

Solution. The auxiliary equation of the associated homogeneous equation

$$y'' + y = 0$$

is

$$\lambda^2 + 1 = 0,$$

whose complex roots are

$$\lambda = \pm i.$$

Therefore, the complementary function of the general solution of the nonhomogeneous equation is

$$y_h = c_1 \cos x + c_2 \sin x,$$

where c_1 and c_2 arbitrary constants. If y_h was not known in advance, we might initially guess

$$y_p = A \cos x + B \sin x,$$

with A and B as undetermined coefficients. However, this form of y_p duplicates y_h and it cannot satisfy the nonhomogeneous equation. A point of action is to modify y_p to

$$y_p = x(A \cos x + B \sin x).$$

This will work! Let us proceed to determine A and B . Differentiating y_p gives

$$\begin{aligned} y_p' &= A \cos x + B \sin x + x(-A \sin x + B \cos x), \\ y_p'' &= -2A \sin x + 2B \cos x + x(-A \cos x - B \sin x). \end{aligned}$$

Substituting these into the nonhomogeneous equation (3.3.14) yields

$$-2A \sin x + 2B \cos x = \cos x.$$

For this to be an identity in x , we must have

$$A = 0, \quad B = \frac{1}{2},$$

so

$$y_p = \frac{1}{2}x \sin x$$

is a particular solution of (3.3.14), and therefore its general solution is

$$y = c_1 \cos x + c_2 \sin x + \frac{1}{2}x \sin x,$$

where c_1 and c_2 arbitrary constants. ▲

When the forcing function can be decomposed into a sum of several forcing functions, then the sum of particular solutions with these individual forcing functions yields a particular solution of the original equation. The following theorem addresses this property.

Theorem 3.3.11 *Let $a_0(x), a_1(x), \dots, a_{n-1}(x)$ and $f_i(x)$, $i = 1, \dots, k$, be continuous on an interval I . Suppose y_{p_i} , $i = 1, \dots, k$, is a solution of*

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = f_i(x), \quad (3.3.15)$$

for each $i = 1, \dots, k$. Then the function

$$y_p = \sum_{j=1}^k y_{p_j}(x)$$

is a solution of

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = \sum_{j=1}^k f_j(x) \quad (3.3.16)$$

on I

Proof. Define the linear differential operator L on the class of n -times differentiable functions by

$$L[y](x) = y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y.$$

Since each y_{p_j} satisfies (3.3.15), we have

$$L[y_{p_j}](x) = f_j(x) \quad \text{on } I, \quad j = 1, \dots, k.$$

Then the linearity of the differential operator L implies

$$L \left[\sum_{j=1}^k y_{p_j} \right] (x) = \sum_{j=1}^k L[y_{p_j}](x) = \sum_{j=1}^k f_j(x).$$

This shows that the function

$$y_p = \sum_{j=1}^k y_{p_j}$$

is a solution of (3.3.15) on I . ■

Example 3.3.12 Find first a particular solution and then the general solution of

$$y'' + 4y = 4 \sin(2x) + 8 \sin x + 2x^2 - 4x + 3e^x. \quad (3.3.17)$$

Solution. The auxiliary equation of the associated homogeneous equation

$$y'' + 4y = 0$$

is

$$\lambda^2 + 4 = 0$$

whose complex roots are $\pm 2i$. Therefore, the complementary function of the general solution of (3.3.17) is

$$y_h = c_1 \cos x + c_2 \sin x,$$

where c_1 and c_2 are arbitrary constants.

To find a particular solution of (3.3.17), we split the forcing term into four simpler terms:

$$\begin{aligned} f_1(x) &= 4 \sin(2x), & f_2(x) &= 8 \sin x, \\ f_3(x) &= 2x^2 - 4x, & f_4(x) &= 3e^x. \end{aligned}$$

In view of Theorem 3.3.11, we find particular solutions $y_{p_1}, y_{p_2}, y_{p_3}, y_{p_4}$ and then add them together.

Since $\sin(2x)$ in $f_1(x) = 4 \sin(2x)$ is a solution of the homogeneous equation, we take

$$y_{p_1}(x) = x(A \cos(2x) + B \sin(2x)),$$

and find $A = -1$, $B = 0$. Thus,

$$\boxed{y_{p_1}(x) = -x \cos(2x).}$$

For $f_2(x) = 8 \sin x$, we take

$$y_{p_2}(x) = C \cos x + D \sin x$$

and find $C = 0$, $D = \frac{8}{3}$. Thus,

$$\boxed{y_{p_2}(x) = \frac{8}{3} \sin x.}$$

For $f_3(x) = 2x^2 - 4x$, we try the quadratic

$$y_{p_3}(x) = Ex^2 + Fx + G.$$

and find $E = \frac{1}{2}$, $F = -1$, $G = -\frac{1}{4}$. Thus,

$$\boxed{y_{p_3}(x) = \frac{1}{2}x^2 - x - \frac{1}{4}.}$$

For $f_4(x) = 3e^x$, we take

$$y_{p_4}(x) = He^x.$$

and find $H = \frac{3}{5}$. Thus,

$$\boxed{y_{p_4}(x) = \frac{3}{5}e^x.}$$

By Theorem 3.3.11, a particular solution of (3.3.17) is given by

$$y_p(x) = y_{p_1}(x) + y_{p_2}(x) + y_{p_3}(x) + y_{p_4}(x)$$

$$= -x \cos(2x) + \frac{8}{3} \sin x + \frac{1}{2}x^2 - x - \frac{1}{4} + \frac{3}{5}e^x.$$

Hence the general solution of (3.3.17) is

$$\begin{aligned} y(x) &= y_h(x) + y_p(x) \\ &= c_1 \cos x + c_2 \sin x - x \cos(2x) + \frac{8}{3} \sin x \\ &\quad + \frac{1}{2}x^2 - x - \frac{1}{4} + \frac{3}{5}e^x, \end{aligned}$$

where c_1 and c_2 are arbitrary constants. ▲

Some Basic Trial Particular Solutions

In principle, the method of undetermined coefficients can always be employed; however, it requires an educated guess for particular trial solution y_p . Table 3.3.13 summarizes the basic form of a trial particular solution $y_p(x)$ for a given forcing function $f(x)$ when the forcing function is a polynomial, exponential, sine, cosine, or a linear combination of these functions, provided no term in a trial $y_p(x)$ duplicates any term in the complementary function $y_h(x)$. If a term in a trial $y_p(x)$ duplicates a term in $y_h(x)$, then we modify $y_p(x)$ by multiplying it by x^k , where k is the smallest integer such that no term in the modified $y_p(x)$ duplicates a term in $y_h(x)$. These ideas also apply to higher order linear equations.

Table 3.3.13 Basic trial particular solutions

$g(x)$	Basic trial particular solution $y_p(x)$
$p_n(x) = a_n x^n + \cdots + a_0$	$q_n(x) = A_n x^n + \cdots + A_0$
$Ce^{\alpha x}$	$Ae^{\alpha x}$
$p_n(x)e^{\alpha x}$	$q_n(x)e^{\alpha x}$
$C \cos(\beta x)$	$A \cos(\beta x) + B \sin(\beta x)$
$C \sin(\beta x)$	$A \cos(\beta x) + B \sin(\beta x)$
$p_n(x) \cos(\beta x)$	$q_n(x) \cos(\beta x) + r_n(x) \sin(\beta x)$
$p_n(x) \sin(\beta x)$	$q_n(x) \cos(\beta x) + r_n(x) \sin(\beta x)$
$Ce^{\alpha x} \cos(\beta x)$	$Ae^{\alpha x} \cos(\beta x) + Be^{\alpha x} \sin(\beta x)$
$Ce^{\alpha x} \sin(\beta x)$	$Ae^{\alpha x} \cos(\beta x) + Be^{\alpha x} \sin(\beta x)$

In Table 3.3.13, the general polynomial $p_n(x)$ is of degree n with known coefficients and $q_n(x)$, $r_n(x)$ are general polynomials of degree n with undetermined coefficients. Also, C , α , β are known constants, and A and B are undetermined constants.

3.3.3 Method of Operator Factorization

In this subsection, we discuss yet another method for finding particular solutions of a second order nonhomogeneous linear equations of the form

$$y'' + a_1(x)y' + a_0(x)y = f(x), \quad (3.3.18)$$

where $a_1(x)$, $a_0(x)$, $f(x)$ are continuous on some interval I . For simplicity, we will restrict our attention to equations with constant coefficients and include only a few examples involving variable coefficients. The method extends naturally to higher-order linear equations, which we will briefly discuss in Section 3.4.

We let D denote the differential operator $\frac{d}{dx}$ defined by

$$Dy = \frac{dy}{dx}.$$

Higher order differential operators are denoted by integer powers of D . For example,

$$D^2y = \frac{d^2y}{dx^2}, \quad D^3y = \frac{d^3y}{dx^3},$$

and so forth. The differential operator of order 0 is the identity operator denoted by D^0 and defined by

$$D^0y = y.$$

With these notations, the second order equation (3.3.18) can be written as

$$D^2y + a_1(x)Dy + a_0(x)y = f(x),$$

or more concisely,

$$p(D)y = f(x),$$

where

$$p(D) = D^2 + a_1(x)D + a_0(x)$$

denotes the corresponding second order differential operator.

To introduce the operator method, we begin by considering the first order linear equation

$$y' + a_1(x)y = f(x), \quad (3.3.19)$$

where $a_1(x)$ and $f(x)$ are continuous on an interval I . The equation in operator notation can be written as

$$(D + a_1(x))y = f(x).$$

Formally, a solution may be expressed as

$$y = \frac{1}{D + a_1(x)}f(x).$$

Since a solution of (3.3.19) is given by

$$\begin{aligned} y &= \frac{1}{D + a_1(x)}f(x) \\ &= c_1 e^{-\int a_1(x) dx} + e^{-\int a_1(x) dx} \int \left(f(x) e^{\int a_1(x) dx} \right) dx, \end{aligned} \quad (3.3.20)$$

where c_1 is an arbitrary constant. A particular solution y_p (with $c_1 = 0$) is given by

$$\boxed{y_p = e^{-\int a_1(x) dx} \int f(x) e^{\int a_1(x) dx} dx,} \quad (3.3.21)$$

In particular, if $a_1(x) = -k$, a constant, then the formula (3.3.21) takes the form

$$\boxed{y_p = \frac{1}{D - k}f(x) = e^{kx} \int f(x) e^{-kx} dx.} \quad (3.3.22)$$

We next study (3.3.18) in the case where a_0 and a_1 are constants. When formally substituting D with λ in

$$p(D) = D^2 + a_1(x)D + a_0(x),$$

we get

$$p(\lambda) = \lambda^2 + a_1\lambda + a_0,$$

which is the characteristic polynomial for the homogeneous equation

$$y'' + a_1y' + a_0y = 0.$$

If the polynomial $p(\lambda)$ factors as

$$p(\lambda) = (p - \lambda_1)(p - \lambda_2),$$

where λ_1, λ_2 may be real or complex, then the operator $p(D)$ factors correspondingly as

$$p(D) = (D - \lambda_1)(D - \lambda_2).$$

Thus, $p(D)$ can be written a composition of two first order differential operators. Moreover, we find

$$\begin{aligned} (D - \lambda_1)(D - \lambda_2)y &= (D - \lambda_1)(Dy - \lambda_2y) \\ &= D(Dy - \lambda_2y) - \lambda_1(Dy - \lambda_2y) \\ &= D^2y - \lambda_2Dy - \lambda_1Dy + \lambda_1\lambda_2y \\ &= D^2y - (\lambda_1 + \lambda_2)Dy + \lambda_1\lambda_2y. \end{aligned}$$

Since $\lambda_1 + \lambda_2 = \lambda_2 + \lambda_1$ and $\lambda_1\lambda_2 = \lambda_2\lambda_1$, it follows that

$$p(D) = (D - \lambda_1)(D - \lambda_2)y = (D - \lambda_2)(D - \lambda_1)y,$$

and hence the factors commute. A particular solution of

$$p(D)y = f(x)$$

is expressed formally by

$$y_p = \frac{1}{p(D)}f(x) = \frac{1}{D - \lambda_1} \frac{1}{D - \lambda_2}f(x).$$

In this situation, a solution of

$$p(D)y = f(x)$$

can be written formally as

$$\boxed{y = \frac{1}{p(D)}f(x) = \frac{1}{D - \lambda_1} \frac{1}{D - \lambda_2}f(x),} \quad (3.3.23)$$

and it can be obtained by applying (3.3.22) successively to these two first order operators.

Thus, the method of operators is based on the following idea: if we can factor the differential operator $p(D)$ into a product of two first order operators, say $p(D) = (D - \alpha)(D - \beta)$, then we can solve the equation (3.3.18) by solving two first order equations in succession. We will illustrate this method with several examples. These techniques extend naturally to higher order linear equations.

The following example illustrates these techniques.

Example 3.3.14 Find the general solution of

$$y'' - 5y' + 6y = 4xe^x$$

Solution. The equation in the operator form is

$$p(D)y = 4xe^x,$$

where

$$p(D) = D^2 - 5D + 6 = (D - 2)(D - 3).$$

By (3.3.23), a particular solution is formally given by

$$y_p = \frac{1}{D-2} \frac{1}{D-3} 4xe^x.$$

Using (3.3.22) and integrating by parts, we get

$$\frac{1}{D-3} xe^x = e^{3x} \int 4xe^x e^{-3x} dx = 4e^{3x} \int xe^{-2x} dx = -(2x+1)e^x.$$

Consequently, again by (3.3.22) and integrating by parts, we obtain

$$\begin{aligned} y_p &= -\frac{1}{D-2} (2x+1)e^x \\ &= -e^{2x} \int (2x+1)e^x e^{-2x} dx \\ &= e^{2x} \int (2x+1)e^{-x} dx \\ &= (2x+3)e^x. \end{aligned}$$

Hence the general solution of the nonhomogeneous equation is

$$y = c_1 e^{2x} + c_2 e^{3x} + (2x+3)e^x,$$

where c_1 and c_2 are arbitrary constants. ▲

The formula (3.3.23) can sometimes be more conveniently used using partial fraction decomposition:

$$\boxed{\begin{aligned} y_p &= \frac{1}{D-\lambda_1} \frac{1}{D-\lambda_2} f(x) \\ &= \left(\frac{A_1}{D-\lambda_1} + \frac{A_2}{D-\lambda_2} \right) f(x), \end{aligned}} \quad (3.3.24)$$

with the constants A_1 and A_2 to be evaluated. We can then evaluate the individual terms by using (3.3.22). Let us revisit Example 3.3.14 with this technique.

Example 3.3.15 Find a particular solution y_p of

$$y'' - 5y' + 6y = 4xe^x.$$

Solution. In Example 3.3.14, we have

$$y_p = \frac{1}{D-2} \frac{1}{D-3} 4xe^x.$$

Suppose A_1 and A_2 are constants for which

$$\frac{1}{D-2} \frac{1}{D-3} = \frac{A_1}{D-2} + \frac{A_2}{D-3}.$$

Then we find $A_1 = -1$ and $A_2 = 1$. In view of (3.3.24), we have

$$y_p = -\frac{1}{D-2} 4xe^x + \frac{1}{D-3} 4xe^x,$$

which, by (3.3.22) and integration by parts, yields

$$\begin{aligned}
 y_p &= -4e^{2x} \int x e^x e^{-2x} dx + 4e^{3x} \int x e^x e^{-3x} dx \\
 &= -4e^{2x} \int x e^{-x} dx + 4e^{3x} \int x e^{-2x} dx \\
 &= e^x(4x + 4) + e^x(-2x - 1) \\
 &= (2x + 3)e^x.
 \end{aligned}$$

The partial fraction decomposition method employed in Example 3.3.15 works naturally for higher order equations as well. ▲

The next example shows how we deal with the case when certain factor $D - \lambda$ is repeated.

Example 3.3.16 Find a particular solution y_p of

$$y'' - 4y' + 4y = 4e^{2x}.$$

Solution. The equation in the operator form is

$$p(D)y = 4e^{2x},$$

where

$$p(D) = D^2 - 4D + 4 = (D - 2)^2.$$

By (3.3.23), a particular solution is formally given by

$$y_p = \frac{1}{D - 2} \frac{1}{D - 2} 4e^{2x}.$$

By (3.3.22), we have

$$\frac{1}{D - 2} 4e^{2x} = e^{2x} \int 4e^{2x} e^{-2x} dx = 4xe^{2x}.$$

Then, by (3.3.22), we obtain

$$y_p = \frac{1}{D - 2} 4xe^{2x} = e^{2x} \int 4xe^{2x} e^{-2x} dx = e^{2x} \int 4x dx = 2x^2 e^{2x}.$$

Whenever $f(x)$ contains a factor of the form $e^{\alpha x}$, a more convenient method exists. To discuss this, suppose $f(x) = e^{\alpha x} g(x)$ for some continuous function $g(x)$. Suppose ▲

$$p(D) = (D - \lambda_1)(D - \lambda_2),$$

λ_1 and λ_2 real or complex constants. Then

$$\begin{aligned}
 (D - \lambda_2)(e^{\alpha x} g(x)) &= e^{\alpha x} Dg(x) + \alpha e^{\alpha x} g(x) - \lambda_2 e^{\alpha x} g(x) \\
 &= e^{\alpha x} (D + \alpha - \lambda_2)g(x).
 \end{aligned}$$

and

$$\begin{aligned}
 p(D)(e^{\alpha x} g(x)) &= (D - \lambda_1)(D - \lambda_2)(e^{\alpha x} g(x)) \\
 &= (D - \lambda_1)(e^{\alpha x} (D + \alpha - \lambda_2)g(x)) \\
 &= e^{\alpha x} (D + \alpha - \lambda_1)(D + \alpha - \lambda_2)g(x) \\
 &= e^{\alpha x} p(D + \alpha)g(x).
 \end{aligned}$$

Thus, we have

$$p(D)(e^{\alpha x}g(x)) = e^{\alpha x}p(D + \alpha)g(x). \quad (3.3.25)$$

We also have a similar property of the inverse operator; namely,

$$\frac{1}{p(D)}e^{\alpha x}g(x) = e^{\alpha x}\frac{1}{p(D + \alpha)}g(x). \quad (3.3.26)$$

To see this, using (3.3.25), we have

$$p(D)\left(e^{\alpha x}\frac{1}{p(D + \alpha)}g(x)\right) = e^{\alpha x}p(D + \alpha)\frac{1}{p(D + \alpha)}g(x) = e^{\alpha x}g(x),$$

which yields (3.3.26).

The formulas (3.3.25) and (3.3.26) are called the **exponential shifting properties** of $p(D)$ and its inverse operator, respectively. The next example provides an application of the exponential shifting properties.

Example 3.3.17 Find a particular solution of

$$y'' - 5y' + 6y = 4xe^x$$

by using the exponential shifting properties.

Solution. We can write

$$4xe^x = e^x(4x).$$

By the exponential shifting property of the inverse operator, we have

$$y_p = \frac{1}{D^2 - 5D + 6}(4xe^x) = e^x \frac{1}{(D + 1)^2 - 5(D + 1) + 6}(4x).$$

Since

$$(D + 1)^2 - 5(D + 1) + 6 = D^2 - 3D + 2,$$

we have

$$y_p = e^x \frac{1}{D^2 - 3D + 2}(4x).$$

By using partial fraction decomposition and then applying (3.3.22) and integration by parts, we get

$$\begin{aligned} y_p &= e^x \left(\frac{A_1}{D - 1} + \frac{A_2}{D - 2} \right) 4x \\ &= e^x \left(-\frac{4}{D - 1} + \frac{4}{D - 2} \right) x \\ &= e^x \left(-4e^x \int xe^{-x} dx + 4e^{2x} \int xe^{-2x} dx \right) \\ &= (2x + 3)e^x. \end{aligned}$$



When the coefficient functions $a_1(x)$ and $a_0(x)$ are variable, the factors of $p(D)$ generally do not commute, as illustrated in Example 3.3.18.

Example 3.3.18 Find a particular solution of

$$y'' + (x + 1)y' + xy = e^{-x^2/2}.$$

Solution. The equation in the operator form is

$$p(D)y = e^{-x^2/2}, \quad (3.3.27)$$

where

$$p(D) = D^2 + (x+1)D + x.$$

By inspection, $p(D)$ may be factored as $(D+x)(D+1)$ or $(D+1)(D+x)$, and so which one is it? Direct computation shows

$$\begin{aligned} (D+x)(D+1)y &= (D+x)(Dy+y) \\ &= D(Dy+y) + x(Dy+y) \\ &= D^2y + Dy + xDy + xy \\ &= (D^2 + (x+1)D + x)y \\ &= p(D)y, \end{aligned}$$

and

$$\begin{aligned} (D+1)(D+x)y &= (D+1)(Dy+xy) \\ &= D(Dy+xy) + (Dy+xy) \\ &= D^2y + xDy + y + Dy + xy \\ &= (D^2 + (x+1)D + (x+1))y \\ &\neq p(D)y. \end{aligned}$$

Thus, we only have

$$p(D)y = (D+x)(D+1)y.$$

Then (3.3.27) turns into

$$(D+x)(D+1)y = e^{-\frac{1}{2}x^2}.$$

Put $(D+1)y = z$. Then we have

$$(D+x)z = e^{-x^2/2},$$

which is a first order linear equation in z . In view of (3.3.21), a particular solution of this equation is given by

$$z_p = e^{-x^2/2} \int e^{-x^2/2} e^{x^2/2} dx = (x-1)e^{-x^2/2}.$$

The constant of integration is chosen so as to simplify the integral below. Using this z_p for z in $(D+1)y = z$, we obtain

$$(D+1)y = (x-1)e^{-x^2/2}.$$

By (3.3.21), a particular solution of this equation is given by

$$\begin{aligned} y_p &= e^{-x} \int (x-1)e^x e^{-x^2/2} dx \\ &= e^{-x+\frac{1}{2}} \int (x-1)e^{-\frac{1}{2}(x-1)^2} dx \\ &= e^{-x+\frac{1}{2}} \int e^{-t} dt \quad \left(\text{using } t = \frac{1}{2}(x-1)^2 \text{ so that } (x-1)dx = dt \right) \\ &= -e^{-(x-\frac{1}{2})} e^{-\frac{1}{2}(x-1)^2} \\ &= -e^{-\frac{1}{2}x^2}. \end{aligned}$$



We note that a fundamental set of solutions for the equation in [Example 3.3.18](#) involves an integral that cannot be evaluated in terms of elementary functions. We can construct the general solution by leaving these integrals unevaluated. A sketch of this construction is outlined in [Exercises 3.3.4](#).

Example 3.3.19 Find the general solution of the Cauchy-Euler equation

$$x^2 y'' + 4xy' + 2y = e^x, \quad x > 0,$$

using the method of operator factorization.

Solution. In the operator notation, the equation reads

$$(x^2 D^2 + 4xD + 2)y = e^x.$$

By inspection, the differential operator can be factored as

$$(x^2 D^2 + 4xD + 2) = (xD + 1)(xD + 2),$$

and this product is commutative. For the general conditions under which the product of two binomial differential operators with variable coefficients commutes, see [Exercises 3.3.4](#). Then the differential equation reads

$$(xD + 1)(xD + 2)y = e^x.$$

Let $z = (xD + 2)y$. Then the equation in z reads

$$(xD + 1)z = e^x,$$

which is linear in z , that is,

$$\left(D + \frac{1}{x}\right)z = \frac{e^x}{x}.$$

The solution of this equation is given formally by

$$z = \frac{1}{D + \frac{1}{x}} \left(\frac{e^x}{x} \right).$$

By [\(3.3.20\)](#), we obtain

$$\begin{aligned} z &= c_1 e^{-\int \frac{1}{x} dx} + e^{-\int \frac{1}{x} dx} \int \left(\frac{e^x}{x} e^{\int \frac{1}{x} dx} \right) dx \\ &= \frac{c_1}{x} + \frac{1}{x} \int e^x dx \\ &= \frac{c_1}{x} + \frac{e^x}{x}, \end{aligned}$$

where c_1 is an arbitrary constant. Since $z = (xD + 2)y$, we have

$$(xD + 2)y = \frac{c_1}{x} + \frac{e^x}{x},$$

that is,

$$\left(D + \frac{2}{x}\right)y = \frac{c_1}{x^2} + \frac{e^x}{x^2},$$

the solution of which is given formally by

$$y = \frac{1}{D + \frac{2}{x}} \left(\frac{c_1}{x^2} + \frac{e^x}{x^2} \right).$$

In view of (3.3.20), the general solution is given by

$$\begin{aligned}
 y &= c_2 e^{-\int \frac{2}{x} dx} + e^{-\int \frac{2}{x} dx} \int \left(\left(\frac{c_1}{x^2} + \frac{e^x}{x^2} \right) e^{\int \frac{2}{x} dx} \right) dx \\
 &= \frac{c_2}{x^2} + \frac{1}{x^2} \int (c_1 + e^x) dx \\
 &= \frac{c_2}{x^2} + \frac{1}{x^2} (c_1 x + e^x) \\
 &= \frac{c_1}{x} + \frac{c_2}{x^2} + \frac{e^x}{x^2},
 \end{aligned}$$

where c_2 is an arbitrary constant. ▲

3.3.4 Exercises

Subsection 3.3.1 Variation of Parameters.

1. Solve the following differential equations using the method of variation of parameters to find particular solutions.

- | | |
|---------------------------------|---|
| i. $y'' + y = 2e^x$ | ix. $y'' + y = \sec^2 x$ |
| ii. $y'' - y = xe^{-2x}$ | x. $y'' + 4y' + 4y = e^{-2x}$ |
| iii. $y'' + y = \sin x$ | xi. $y'' + 3y' + 2y = \frac{1}{1 + e^x}$ |
| iv. $y'' + y = x \sin x$ | xii. $y'' + 3y' + 2y = \cos(e^x)$ |
| v. $y'' + y = \sec x$ | xiii. $y'' + 4y' + 4y = 2 \cos(2x)$ |
| vi. $y'' + y = \tan x$ | xiv. $y'' + 4y' + 4y = 2e^{-2x} \cos(2x)$ |
| vii. $y'' + y = \sin^2 x$ | |
| viii. $y'' + y = \sec x \tan x$ | |

2. Find the general solution of each differential (Cauchy-Euler) equation below using the variation of parameters to find particular solutions.

- | | |
|--------------------------------|------------------------------|
| i. $x^2 y'' + xy' - y = 5$ | iii. $x^2 y'' - xy' + y = x$ |
| ii. $x^2 y'' + xy' - y = 2x^4$ | iv. $xy'' - y' = x$ |

3. Find the general solution of each differential equation below. One non-trivial solution to the associated homogeneous equation is provided.

- | | |
|--|--|
| i. $x^2 y'' + xy' - 4y = \frac{1}{x^2}; \quad u(x) = x^2$ | iv. $(x-1)y'' - xy' + y = (x-1)^2; \quad u(x) = x,$ |
| ii. $x^2 y'' + xy' - 4y = \frac{1}{x} + 1; \quad u(x) = x^2$ | v. $(\tan^2 x)y'' - (2 \tan x)y' + (\sec^2 x + 1)y = \sin^2 x \tan x; \quad u(x) = \sin x$ |
| iii. $x^2 y'' + xy' + 4y = \sin x$ | |

4. Show that y_p in (3.3.10) may be written as

$$y_p(x) = \int_{x_0}^x G(x, t) f(t) dt,$$

where x, x_0 are in I and

$$G(x, t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W(t)} = \frac{\begin{vmatrix} y_1(t) & y_2(t) \\ y_1(x) & y_2(x) \end{vmatrix}}{\begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}}$$

with $W(t) = W(y_1(t), y_2(t))$, the Wronskian of y_1 and y_2 . The function $G(x, t)$ is called the **Green's function** for (3.3.7).

5. Use the Green's function derived in Exercise 3.3.4.4 to find a particular solution of

$$y'' + y = \frac{1}{\sin x}, \quad 0 < x < \pi.$$

6. Use Exercise 3.3.4.4 to express a particular solution $y_p(x)$ of $y'' + k^2 y = f(x)$, where $k > 0$ is in the form

$$y_p(x) = \frac{1}{k} \int_{x_0}^x f(t) \sin(k(x-t)) dt.$$

The integral on the right with $x_0 = 0$ is called the *convolution* of $f(x)$ and $\sin(kx)$, in the context of the Laplace transform (see Definition 5.5.1 in Chapter 5).

7. Show that the solution of (3.3.6) with the initial conditions $y(x_0) = 0$ and $y'(x_0) = 0$ is

$$y(x) = \int_{x_0}^x G(x, t) f(t) dt.$$

Hint. Use the following rule for differentiating under the integral sign: if

$$F(x) = \int_{x_0}^x g(x, t) dt,$$

then

$$F'(x) = g(x, x) + \int_{x_0}^x \frac{\partial g}{\partial x}(x, t) dt.$$

8. Use the result in Exercise 3.3.4.6 to obtain the solution of the differential equation in Example 3.3.4 satisfying $y(0) = 0, y'(0) = 0$.

Subsection 3.3.2 Method of Undetermined Coefficients.

9. Find the general solution of each of the following nonhomogeneous linear differential equations.

i. $y'' - 5y' + 4y = 5x + 1$

ii. $y'' - y' + 2y = x^2 + x + 1$

iii. $y'' - 5y' + 4y = 5e^{2x}$

iv. $y'' + 4y = 5 \cos x$

v. $y'' + 4y = 5 \sin(2x)$

vi. $y'' + 4y' + 5y = 2e^{-x} \sin x$

vii. $y'' + 4y' + 4y = 3e^{2x}$

viii. $y'' - 4y' + 4y = 3xe^{2x}$

ix. $y'' - 4y' + 5y = 2e^{2x} \sin x$

x. $y'' + 16y = 2 \sin(4x) + 3 \cos(4x)$

xi. $y'' - 5y' + 4y = 5e^x + 3e^{4x}$

xii. $y'' + 16y = x^3 + 1 + 2e^{4x} + \sin x$

xiii. $y'' - y' = 1 + 7e^x$

xiv. $y'' + y' = 2 + \sin x - 7e^{-x}$

10. Let ω be a fixed positive constant. Find the solution of the initial value problem

$$y'' + \omega^2 y = 5 \cos(\omega x), \quad y(0) = 0, y'(0) = 0$$

by following the steps below.

- i. Find the solution of the initial value problem

$$y'' + \omega^2 y = 5 \cos(\gamma x), \quad y(0) = 0, y'(0) = 0$$

for $\gamma \neq \omega$.

- ii. Denote by $y_\gamma(x)$ the solution obtained in [Item 3.3.4.10.i](#). For each x in $(-\infty, \infty)$, evaluate the limit

$$\lim_{\gamma \rightarrow \omega} y_\gamma(x).$$

- iii. Verify that the function defined by

$$y = \lim_{\gamma \rightarrow \omega} y_\gamma(x)$$

is the solution of

$$y'' + \omega^2 y = 5 \cos(\omega x), \quad y(0) = 0, y'(0) = 0.$$

- iv. Obtain the solution found in part [Item 3.3.4.10.iii](#) by using the method of undetermined coefficients to find a particular solution.
11. Solve the following initial value problems and plot the corresponding solutions using a graphing utility. Describe any differences you observe in the behavior of the two solutions.

i. $y'' + y = \sin(2x), \quad y(0) = 1, y'(0) = 0$

ii. $y'' + y = \sin x, \quad y(0) = 1, y'(0) = 0$

Subsection 3.3.3 Method of Operator Factorization.

12. Find a particular solution of

$$y'' - 9y = e^{3x}$$

by using the method used in [Example 3.3.14](#).

13. Find a particular solution of

$$y'' - 9y = e^{-3x}$$

by using the method used in [Example 3.3.15](#).

14. Find a particular solution of each of the following equations by using the method used in [Example 3.3.17](#) (the exponential shifting properties).

i. $y'' - 6y' + 9y = xe^{3x}$

ii. $y'' - 2y' - 3y = 16e^{5x}$

15. Use the exponential shifting properties to show that the general solution of the homogeneous equation

$$(D - \lambda)^2 y = 0,$$

where λ a real constant, has

$$y = (c_1 + c_2 x)e^{\lambda x},$$

with c_1 and c_2 arbitrary constants.

16. Find a particular solution of

$$y'' + (x+1)y' + (x+1)y = e^{-x}, \quad x > 0,$$

by using the method used in [Example 3.3.18](#).

17. Determine a fundamental set of solutions of

$$y'' + (x+1)y' + xy = 0 \tag{3.3.28}$$

by following the steps below.

- i. Express [\(3.3.28\)](#) in operator form as

$$(D+x)(D+1)y = 0.$$

- ii. Solve $(D+1)y = 0$ to obtain a nonzero solution y_1 , and verify that y_1 is indeed a solution of [\(3.3.28\)](#).

- iii. Find a function y_2 by using the Liouville reduction formula [\(3.2.19\)](#) so that $\{y_1, y_2\}$ forms a fundamental set of solutions for [\(3.3.28\)](#).

18. Let $a_0(x), a_1(x), b_0(x), b_1(x)$ be differentiable functions on an interval I . Prove that

$$(a_1 D + a_0)(b_1 D + b_0) = (b_1 D + b_0)(a_1 D + a_0)$$

if and only if

$$a_1 b_1' = a_1' b_1 \quad \text{and} \quad a_1 b_0' = a_0' b_1.$$

19. Let $a_1(x)$ be a continuous function on some interval I . Let $\alpha(x)$ and $g(x)$ be differentiable functions on I . Prove the following general exponential shifting properties.

(a) $(D + a_1(x))(e^{\alpha(x)} g(x)) = e^{\alpha(x)}(D + \alpha'(x) + a_1(x))g(x).$

(b) $\frac{1}{D + a_1(x)}(e^{\alpha(x)} g(x)) = e^{\alpha(x)} \frac{1}{D + \alpha'(x) + a_1(x)} g(x).$

20. Use the shifting properties established in [Exercise 3.3.4.19](#) to find a particular solution of each of the following equations.

$$(a) \quad y'' + (x+1)y' + (x+1)y = e^{-x}, \quad x > 0,$$

$$(b) \quad y'' + (x+1)y' + xy = e^{-x^2/2}, \quad x > 0,$$

3.4 Higher Order Linear Differential Equations

Objectives

- Learn solution techniques for higher order linear differential equations

In this section we discuss methods for solving the n^{th} order linear differential equations of the form

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x), \quad (3.4.1)$$

where the coefficient functions $a_0(x), \dots, a_{n-1}(x)$ and the forcing term $f(x)$ are continuous on some interval I .

3.4.1 Homogeneous Equations

To find the general solution of (3.4.1), we first find the general solution y_h of the associated homogeneous equation

$$y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = 0 \quad (3.4.2)$$

and a particular solution y_p of (3.4.1). Then the general solution of the non-homogeneous equation (3.4.1) is given by

$$\boxed{\begin{aligned} y &= y_h(x) + y_p(x) \\ &= c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x) + y_p(x) \end{aligned}} \quad (3.4.3)$$

on I , where $\{y_1, y_2, \dots, y_n\}$ is a fundamental set of solutions of (3.4.2) and $c_1, \dots, c_n$ are arbitrary constants. The function $y_h(x)$, also denoted by $y_c(x)$, is called the **complementary function** of the general solution of (3.4.2). The existence of a fundamental set of solutions for (3.4.2) follows by arguments analogous to those used in Theorem 3.2.3. In fact, let y_i be the unique solution of the initial value problem consisting of the equation (3.4.2) subject to the initial conditions

$$y(x_0) = 0, \dots, y^{(i-1)}(x_0) = 1, \dots, y^{(n-1)}(x_0) = 0, \quad (3.4.4)$$

where $i = 1, \dots, n$. It follows that $\{y_1, \dots, y_n\}$ is a fundamental solutions of (3.4.2) (see Exercise 3.4.3.4 in Exercises 3.4.3).

Let $y_1, \dots, y_n$ be solutions of (3.4.2). In Exercise 3.4.3.5 in Exercises 3.4.3, you will derive the Abel formula for the Wronskian of $y_1, \dots, y_n$ as

$$W(y_1, \dots, y_n)(x) = W(y_1, \dots, y_n)(x_0)e^{\int_{x_0}^x a_{n-1}(t) dt}, \quad x \in I.$$

The linear independence of solutions $y_1, y_2, \dots, y_n$ of (3.4.2) can be determined using their Wronskian (see Definition 3.1.11), as stated in the following theorem.

Theorem 3.4.1 *Let $y_1, y_2, \dots, y_n$ be solutions of (3.4.2) on I . Then they are*

linearly independent on I if and only if

$$W(y_1, \dots, y_n)(x) \neq 0 \text{ for all } x \text{ in } I.$$

Proof. The proof is similar to the proof of [Theorem 3.2.1](#) and is omitted. ■

Methods for constructing a fundamental set of solutions for (3.4.2) vary. When we have means to find a single nonzero solution, the reduction of order discussion in [Section 3.2](#) can be employed to obtain additional linearly independent solutions. This method applies not only to second order homogeneous linear equations but also extends to higher order equations, as discussed in the next theorem.

Theorem 3.4.2 *Let y_1 be a solution of (3.4.2) such that $y_1(x) \neq 0$ on I , and let $y = uy_1$ be a solution of (3.4.2) for a nonconstant function u on I . If*

$$\{v_1, \dots, v_{n-1}\}$$

is a fundamental set of solutions for the equation of order $n - 1$ in $v = u'$ obtained by substituting $y = uy_1$ into (3.4.2) and if $u'_k = v_k$ for $k = 1, \dots, n - 1$, then

$$\{y_1, u_1y_1, u_2y_1, \dots, u_{n-1}y_1\}$$

forms a fundamental set of solutions for (3.4.2) on I .

Proof. We present a proof for the case $n = 3$. The general case proceeds in a similar way, and a careful reader is encouraged to verify it, noting that it involves the use of binomial coefficients that are produced for higher order derivatives of the product function $y = uy_1$.

For $y = uy_1$ to be a solution of (3.4.2), we must have

$$\begin{aligned} & u'''y_1 + 3u''y_1' + 3u'y_1'' + uy_1''' \\ & + a_2u''y_1 + 2a_2u'y_1' + a_2uy_1'' \\ & + a_1u'y_1 + a_1uy_1' \\ & + a_0uy_1 = 0, \end{aligned}$$

which gives

$$\begin{aligned} & u'''y_1 + (3y_1' + a_2y_1)u'' + (3y_1'' + 2a_2y_1' + a_1y_1)u' \\ & + (y_1''' + a_2y_1'' + a_1y_1' + a_0y_1)u = 0 \end{aligned}$$

Since y_1 is a solution of (3.4.2), we have $y_1''' + a_2y_1'' + a_1y_1' + a_0y_1 = 0$. Consequently, we obtain

$$u'''y_1 + (3y_1' + a_2y_1)u'' + (3y_1'' + 2a_2y_1' + a_1y_1)u' = 0.$$

Letting $v = u'$ gives

$$v''y_1 + (3y_1' + a_2y_1)v' + (3y_1'' + 2a_2y_1' + a_1y_1)v = 0,$$

which is a second order homogeneous linear equation in v . Suppose that $\{v_1, v_2\}$ is a fundamental set of solution of the second order equation and let u_k be functions satisfying $u'_k = v_k$, $k = 1, 2$. To show

$$\{y_1, u_1y_1, u_2y_1\}$$

is a fundamental set of solutions for (3.4.2), let c_0, c_1, c_2 be real numbers such that

$$c_0y_1(x) + c_1u_1(x)y_1(x) + c_2u_2(x)y_1(x) = 0$$

for all x in I . Since $y_1(x) \neq 0$ on I , we must have

$$c_0 + c_1 u_1(x) + c_2 u_2(x) = 0$$

for all x in I . Differentiating yields

$$c_1 u_1'(x) + c_2 u_2'(x) = 0$$

for all x in I , that is,

$$c_1 v_1(x) + c_2 v_2(x) = 0$$

for all x in I . The linear independence of $\{v_1, v_2\}$ on I implies that $c_1 = c_2 = 0$. Consequently, $c_0 = 0$ as well. Therefore,

$$\{y_1, u_1 y_1, u_2 y_1\}$$

forms a linearly independent set of solutions of (3.4.2) for $n = 3$, and hence constitutes a fundamental set of solutions. ■

Example 3.4.3 Find a fundamental set of solutions for

$$y^{(4)} - 4y''' + 6y'' - 4y' + y = 0, \quad (3.4.5)$$

given that $y = e^x$ is a solution. Also, write down the general solution.

Solution. Let $y = ue^x$ a solution of (3.4.5), where u is a nonconstant function on $(-\infty, \infty)$. Differentiating four times and substituting into the equation gives

$$\begin{aligned} & u^{(4)}e^x + 4u'''e^x + 6u''e^x + 4u'e^x + ue^x \\ & - 4u'''e^x - 12u''e^x - 12u'e^x - 4ue^x \\ & + 6u''e^x + 12u'e^x + 6ue^x \\ & - 4u'e^x - 4ue^x \\ & + ue^x = 0. \end{aligned}$$

After canceling terms, we obtain $u^{(4)}e^x = 0$, which yields $u^{(4)}(x) = 0$ in $(-\infty, \infty)$. Let $v = u'$. Then we find $v'''(x) = 0$ on I . Integrating, we obtain

$$v(x) = Ax^2 + Bx + C,$$

where A, B, C are arbitrary constants.

Take $v_1(x) = 1, v_2(x) = 2x$, and $v_3(x) = 3x^2$. Then $\{v_1, v_2, v_3\}$ forms a fundamental set of solutions for $v'''(x) = 0$ on I . Since $u' = v$, we take $u_1(x) = x, u_2(x) = x^2$, and $u_3(x) = x^3$, so that

$$\{e^x, xe^x, x^2e^x, x^3e^x\}$$

forms a fundamental set of solutions for (3.4.5). The general solution of the equation is

$$y = c_1 e^x + c_2 x e^x + c_3 x^2 e^x + c_4 x^3 e^x = e^x (c_1 + c_2 x + c_3 x^2 + c_4 x^3),$$

where c_1, c_2, c_3 , and c_4 are arbitrary constants. ▲

3.4.2 Nonhomogeneous Equations

To find a particular solution $y_p(x)$ of (3.4.1) by using the method of variation of parameters, we take $y_p(x)$ as

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x) + \cdots + u_n(x)y_n(x), \quad (3.4.6)$$

where $u_1(x), u_2(x), \dots, u_n(x)$ are to be determined.

Differentiating (3.4.6) successively, and to simplify calculations, imposing the conditions

$$\begin{aligned} u'_1 y_1 + u'_2 y_2 + \cdots + u'_n y_n &= 0, \\ u'_1 y'_1 + u'_2 y'_2 + \cdots + u'_n y'_n &= 0, \\ &\vdots \\ u'_1 y_1^{(n-2)} + u'_2 y_2^{(n-2)} + \cdots + u'_n y_n^{(n-2)} &= 0, \end{aligned}$$

we obtain

$$u'_1 y_1^{(n-1)} + u'_2 y_2^{(n-1)} + \cdots + u'_n y_n^{(n-1)} = f(x).$$

These n equations can be solved for $u'_1, u'_2, \dots, u'_n$ by using Cramer's rule as follows. We first write down the system of n equations in the matrix form

$$\begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} \begin{bmatrix} u'_1 \\ u'_2 \\ \vdots \\ u'_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ f(x) \end{bmatrix}, \quad (3.4.7)$$

which can be solved for $u'_1, u'_2, \dots, u'_n$ by using **Cramer's rule**. In fact, we have

$$u'_j = \frac{W_j}{W},$$

where W is the Wronskian of $y_1, \dots, y_n$ and W_k is the determinant obtained by replacing the j -th column of W by the column on the right-hand side of (3.4.7). We find

$$u_j(x) = \int \frac{W_j(x)}{W(x)} dx$$

for each $j = 1, \dots, n$ and hence $y_p(x)$ is then fully determined by (3.4.6).

As discussed in Remark 3.3.3, the method of variation of parameters is quite general and works for any linear nonhomogeneous differential equation (even with variable coefficients), so long as the required integrals for $u_1, u_2, \dots, u_n$ make sense.

Example 3.4.4 Find the general solution of

$$y''' + y' = \tan x$$

by using the variation of parameters.

Solution. The homogeneous equation associated to the nonhomogeneous equation $y''' + y' = \tan x$ is

$$y''' + y' = 0$$

with the auxiliary equation

$$r^3 + r = 0 \quad \text{so that} \quad r(r^2 + 1) = 0.$$

Then we have

$$r = 0, \quad r = i, \quad r = -i,$$

and therefore the functions

$$y_1(x) = 1, \quad y_2(x) = \sin x, \quad y_3(x) = \cos x$$

form a fundamental set of solutions for the homogeneous equation. The general solution to $y''' + y = \tan x$ is then given by

$$y(x) = c_1 + c_2 \sin x + c_3 \cos x + y_p(x),$$

where $y_p(x)$ is a particular solution of the nonhomogeneous equation.

By the method of variation of parameters, we assume that $y_p(x)$ is of the form

$$y_p(x) = u_1(x) + u_2(x) \sin x + u_3(x) \cos x,$$

where the functions u_1, u_2, u_3 are to be determined so that u'_1, u'_2, u'_3 satisfy

$$\begin{cases} u'_1 + u'_2 \sin x + u'_3 \cos x = 0, \\ u'_2 \cos x - u'_3 \sin x = 0, \\ u'_2 \sin x - u'_3 \cos x = \tan x. \end{cases}$$

The system in the matrix form is

$$\begin{bmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 0 & -\sin x & -\cos x \end{bmatrix} \begin{bmatrix} u'_1 \\ u'_2 \\ u'_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \tan x \end{bmatrix}.$$

The Wronskian $W(x)$ of $1, \sin x, \cos x$ is

$$W(x) = \begin{vmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 0 & -\sin x & -\cos x \end{vmatrix}.$$

Using the co-factor expansion along the first column, we have

$$W(x) = \begin{vmatrix} \cos x & -\sin x \\ -\sin x & -\cos x \end{vmatrix} = -\cos^2 x - \sin^2 x = -1.$$

We compute

$$W_1(x) = \begin{vmatrix} 0 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ \tan x & -\sin x & -\cos x \end{vmatrix}$$

by expanding along the first column and obtain

$$W_1(x) = \tan x \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = \tan x (-\sin^2 x - \cos^2 x) = -\tan x.$$

Then we have

$$u'_1 = \frac{W_1}{W} = \frac{-\tan x}{-1} = \tan x,$$

and consequently, we can take

$$u_1(x) = -\ln|\cos x|.$$

We next compute

$$W_2(x) = \begin{vmatrix} 1 & 0 & \cos x \\ 0 & 0 & -\sin x \\ 0 & \tan x & -\cos x \end{vmatrix}$$

by expanding along the first column and obtain

$$W_2(x) = \begin{vmatrix} 0 & -\sin x \\ \tan x & -\cos x \end{vmatrix} = 0 - (-\sin x) \tan x = \sin x \tan x.$$

Then we have

$$u_2' = \frac{W_2}{W} = \frac{\sin x \tan x}{-1} = -\sin x \tan x = -\frac{\sin^2 x}{\cos x} = \cos x - \sec x,$$

and so

$$u_2(x) = \int (\cos x - \sec x) dx.$$

We can take

$$u_2(x) = \sin x - \ln|\sec x + \tan x|.$$

We also compute

$$W_3(x) = \begin{vmatrix} 1 & \sin x & 0 \\ 0 & \cos x & 0 \\ 0 & -\sin x & \tan x \end{vmatrix}$$

by expanding along the first column and obtain

$$W_3(x) = \begin{vmatrix} \cos x & 0 \\ -\sin x & \tan x \end{vmatrix} = \cos x \tan x = \sin x,$$

and so

$$u_3' = \frac{W_3}{W} = \frac{\cos x \tan x}{-1} = -\sin x.$$

We can take

$$u_3(x) = \cos x.$$

Consequently, y_p is given by

$$\begin{aligned} y_p(x) &= u_1(x) + u_2(x) \sin x + u_3(x) \cos x \\ &= \ln|\sec x| + \sin x (\sin x - \ln|\sec x + \tan x|) + \cos^2(x) \\ &= 1 + \ln|\sec x| - \sin x \ln|\sec x + \tan x|. \end{aligned}$$

Hence the general solution to $y''' + y' = \tan x$ is

$$y(x) = c_1 + c_2 \sin x + c_3 \cos x + \ln|\sec x| - \sin x \ln|\sec x + \tan x|,$$

where c_1 , c_2 and c_3 are arbitrary constants. ▲

3.4.3 Exercises

- Find the general solution of each differential equation below by using variation of parameters to find a particular solution.

i. $y''' + y' = x + \sin x$

iii. $y''' + y' = \cot x$

ii. $y''' + 4y' = \tan(2x)$

iv. $y''' + 9y' = \csc(3x)$

- Solve each differential equation below by using the provided fundamental set of solutions for the associated homogeneous equation to find a particular solution by variation of parameters.

i. $x^3 y''' - x^2 y'' + 2xy' + 2y = -\frac{1}{x}, \quad \{x, x^2, x \ln x\}$

ii. $x^3 y''' + x^2 y'' - 2xy' + 2y = -\frac{1}{x^2}, \quad \{x, x^2, 1/x\}$

iii. $xy''' - y'' - xy' + y = 4x^2 e^x, \quad \{x, e^x, e^{-x}\}$

iv. $x^3 y''' - (x+3)x^2 y'' + 2(x+3)xy' - 2(x+3)y = x^5, \quad \{x, x^2, x \ln x\}$

v. $x^4 y^{(4)} + 4x^3 y''' + x^2 y'' + xy' - y = \ln x, \quad \{1/x, x, x \ln x, x(\ln x)^2\}$

- Find a fundamental set of solutions for each differential equation below by using the given solution for the reduction of order.

i. $y^{(4)} - 6y''' + 13y'' - 12y' + 4y = 0, \quad y_1(x) = e^x$

ii. $x^3 y''' - x^2 y'' + 2xy' + 2y = 0, \quad y_1(x) = x \text{ for } x > 0$

iii. $x^3 y''' + x^2 y'' - 2xy' + 2y = 0, \quad y_1(x) = x \text{ for } x > 0$

iv. $xy''' - y'' - xy' + y = 0, \quad y_1(x) = x \text{ for } x > 0$

- Following the arguments similar to the proof of [Theorem 3.2.3](#), prove that the solutions $y_1, \dots, y_n$ of the initial value problems consisting of the equation [\(3.4.2\)](#) and n initial conditions [\(3.4.4\)](#) are linearly independent on I .
- Let $y_1, \dots, y_n$ be solutions of [\(3.4.2\)](#). Following the arguments used to derive [\(3.2.1\)](#), prove that the Wronskian $W(y_1, \dots, y_n)$ is given by

$$W(y_1, \dots, y_n)(x) = W(y_1, \dots, y_n)(x_0) e^{\int_{x_0}^x a_{n-1}(t) dt}, \quad x \in I.$$

3.5 Applications

Objectives

- Be able to apply methods of solutions from previous sections to some elementary problems in applications.

In this section we will study three elementary examples of applications where the methods from previous sections can be effectively used.

3.5.1 Spring-Mass System

A common example of a second order linear differential equation with constant coefficients arises in modeling the motion of a mass attached to a spring whose other end is fixed to a support. Let us consider a spring suspended vertically

from a fixed ceiling (rigid support) with the mass attached to its lower end, as illustrated in Figure 3.5.1.

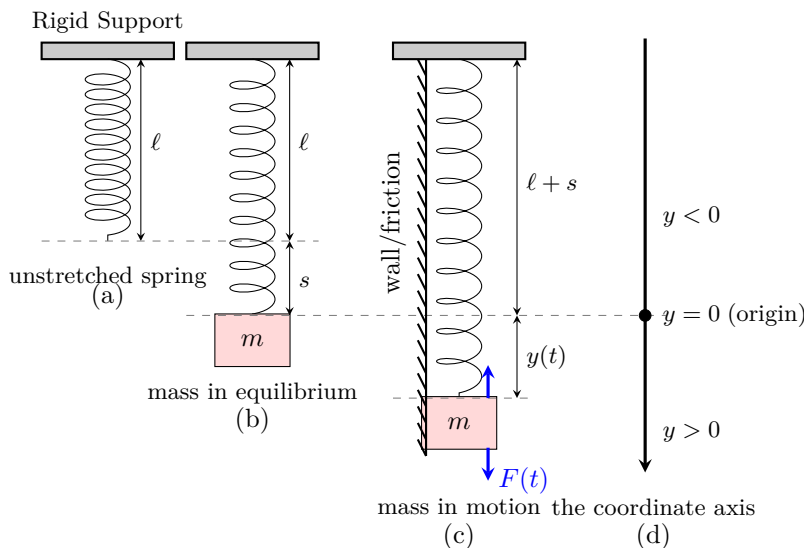


Figure 3.5.1 A mass-spring system showing unstretched, equilibrium, motion, and coordinate-axis.

Let ℓ denote the natural (unstretched) length of the spring, and let s be the amount by which the spring stretches when the mass m is attached and allowed to hang in equilibrium under gravity. We assume that the motion of the mass is restricted to the vertical line through the point of the equilibrium of the mass. To take this vertical line (see Figure 3.5.1 (d)) as the coordinate axis, we take the equilibrium point as the origin and assign the vertically downward direction as the positive direction.

Denote by $y(t)$ the displacement of the mass at time t from its equilibrium position (the origin). In addition to the restoring force $F_s(t)$ of the spring and gravitational force mg , assume that the mass is acted upon by a damping force $F_d(t)$ as well as an external force $F(t)$. See Figure 3.5.1 (c) for wall friction as an example of a damping force. Thus, we have

$$\text{the net force on the mass} = F_s(t) + mg + F_d(t) + F(t).$$

According to Newton's second law, we then have

$$my''(t) = F_s(t) + mg + F_d(t) + F(t). \quad (3.5.1)$$

- How do we model the spring force $F_s(t)$?

Since a spring that is stretched or compressed tends to return to its natural length, the spring force wants to pull the mass toward its equilibrium position. For an ideal spring—one made of a homogeneous material with uniform coiling—Robert Hooke, a 17th-century English physicist, established that the restoring force required to stretch or compress the spring (without damaging it) is directly proportional to the resulting displacement.¹ This means that

$$F_s(t) = -k(s + y(t)), \quad (3.5.2)$$

¹For a spring made up of a heterogeneous material and with nonuniform coiling, the restoring force may be modeled by a nonlinear function of the displacement. For example, $F_s(t) = -k|y(t)|^{p-2}y(t)$ with a fixed number p in $(1, \infty)$ and $k > 0$ a fixed real number.

where k is a fixed positive constant, called the spring constant which measures the stiffness of the spring. The negative $(-)$ means that the restoring force is directed toward the origin.

One may ask: *how do we practically calculate k of a spring?* The simplest way to calculate k is to simply suspend the spring from a support with the mass attached to its other end and observe the stretch s the spring experiences for its equilibrium under gravity. Since the mass is at rest, the net force acting on the mass is zero, that is,

$$\text{the net force on the mass} = 0 = -ks + mg,$$

which yields

$$mg = ks, \quad (3.5.3)$$

and so

$$k = \frac{mg}{s}.$$

- How do we model $F_d(t)$?

It is intuitive that the damping force (or drag) increases as the speed of the mass increases. Hence, the simplest way to model the damping force is to assume that it is directly proportional to the velocity.² This means that

$$F_d(t) = -by'(t), \quad (3.5.4)$$

where b is the constant of proportionality, called the damping coefficient of the medium offering the resistance to the motion of the mass.

Using (3.5.2) and (3.5.4) in (3.5.1), we obtain

$$my''(t) = -ks - ky(t) + mg - by'(t) + F(t),$$

which, in view of (3.5.3), yields

$$my''(t) + by'(t) + ky(t) = F(t). \quad (3.5.5)$$

This is a second order linear differential equation with constant coefficient, and it can be solved using the methods from the previous sections. For example, the method of undetermined coefficients or variation of parameters can be used.

We now examine several interesting and special cases of (3.5.5) before discussing the general form.

I. Free Undamped Motion ($b = 0$, $F(t) = 0$). When $b = 0$ (no damping) and $F(t) = 0$ (no external forces), the mass-spring system is said to have a free undamped motion and the system is called a *simple harmonic oscillator*. The resulting differential equation for the displacement of the mass from its equilibrium position is

$$my''(t) + ky(t) = 0,$$

which, with $w^2 = k/m$, becomes

$$y''(t) + \omega^2 y(t) = 0, \quad (3.5.6)$$

also referred to as a simple harmonic oscillator. Then the general solution of the simple harmonic oscillator is

$$y(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t), \quad (3.5.7)$$

²For a medium that resists the motion of the mass depending on a power of the speed, the damping force may be modeled by a nonlinear function of the velocity. For example, $F_s(t) = -b|y'(t)|^{q-2}y'(t)$ with a fixed number q in $(1, \infty)$ and $b > 0$ a fixed real number.

where c_1 and c_2 are arbitrary constants which can be determined when the initial conditions $y(t_0) = y_0, y'(t_0) = y_1$ are provided.

The displacement $y(t)$ described by (3.5.7) can be interpreted most effectively when it is written in the form of a single sine or cosine function. To express it in the form of a single sine function, we seek to find two constants A and ϕ so that

$$y(t) = A \sin(\omega t - \phi). \quad (3.5.8)$$

Then

$$y(t) = A \sin(\omega t) \cos(\phi) - A \cos(\omega t) \sin(\phi). \quad (3.5.9)$$

Since the terms in (3.5.7) and (3.5.9) must match, we obtain

$$\begin{cases} A \sin(\phi) = -c_1, \\ A \cos(\phi) = c_2. \end{cases}$$

Squaring and adding these equations gives

$$A^2 = c_1^2 + c_2^2.$$

We take

$$A = \sqrt{c_1^2 + c_2^2}.$$

It is possible that $A = 0$, in which case, we have $y(t) = 0$ for all t , the trivial solution, and this means that the mass is at rest in the equilibrium position for all t . Therefore the only interesting case is when $A > 0$. If $c_2 \neq 0$, then

$$\arctan\left(\frac{-c_1}{c_2}\right) \text{ lies in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

We have

$$\phi = \begin{cases} \arctan\left(\frac{-c_1}{c_2}\right) & \text{if } c_2 > 0, \\ \arctan\left(\frac{-c_1}{c_2}\right) + \pi & \text{if } c_2 < 0, \\ \frac{\pi}{2} & \text{if } c_1 < 0 \text{ and } c_2 = 0, \\ -\frac{\pi}{2} & \text{if } c_1 > 0 \text{ and } c_2 = 0. \end{cases}$$

Also, note that (3.5.8) can be written as

$$y(t) = A \cos(\omega t - \phi - \frac{\pi}{2}).$$

It is now clear from (3.5.8) that the mass oscillates about the origin with the amplitude A , which is the maximum displacement from the equilibrium position. The time period for one complete oscillation is $T = 2\pi/\omega$, and therefore the number ω is the circular frequency of the oscillations, that is, the number of cycles per unit 2π units of time.

Example 3.5.2 A 2-pound mass stretches a spring 6 inches. At time $t = 0$, the mass is released from a position 8 inches below equilibrium with an initial upward velocity of $\frac{4}{3}$ ft/s.

- i. Find the equation of motion in both the forms (3.5.7) and (3.5.8).
- ii. Graph the velocity versus the displacement.

Solution.

- i. Let $y(t)$ denote in feet the displacement of the mass from the equilibrium position measured positively in the vertically downward direction. We

have $mg = 2$ lb, spring stretch, $s = 6$ in $= 0.5$ ft. We have $m = \frac{mg}{g} = \frac{2}{32} = \frac{1}{16}$ slug, and the spring constant $k = \frac{mg}{s} = \frac{2}{0.5} = 4$ lb/ft. The differential equation of motion is

$$my'' + ky = 0,$$

which becomes

$$\frac{1}{16}y'' + 4y = 0,$$

and therefore

$$y'' + 64y = 0$$

with the initial condition $y(0) = -\frac{2}{3}$, $y'(0) = \frac{4}{3}$. The general solution of the differential equation is

$$y(t) = c_1 \cos(8t) + c_2 \sin(8t).$$

Applying the initial conditions gives

$$y(0) = -\frac{2}{3} = c_1.$$

Also, since

$$y'(t) = -8c_1 \sin(8t) + 8c_2 \cos(8t)$$

and $y'(0) = \frac{4}{3}$, we obtain $8c_2 = \frac{4}{3}$, that is, $c_2 = \frac{1}{6}$. Hence the equation of the motion is

$$y(t) = -\frac{2}{3} \cos(8t) + \frac{1}{6} \sin(8t),$$

which is of the form (3.5.7). Using the method discussed for (3.5.8) to find the equation of the motion in the amplitude-phase, we compute

$$A = \sqrt{c_1^2 + c_2^2} = \sqrt{\frac{4}{9} + \frac{1}{36}} = \frac{\sqrt{17}}{6}$$

and

$$\phi = \arctan\left(\frac{-c_1}{c_2}\right) = \arctan\left(\frac{2/3}{1/6}\right) = \arctan(4) \approx 1.3258 \text{ rad},$$

and therefore

$$y(t) = \frac{\sqrt{17}}{6} \sin(8t - \phi) \approx \frac{\sqrt{17}}{6} \sin(8t - 1.3258).$$

Its graph is shown in Figure 3.5.3.

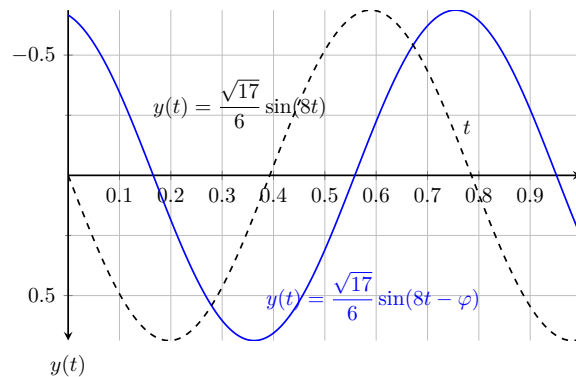


Figure 3.5.3 The graph of $y(t) \approx \frac{\sqrt{17}}{6} \sin(8t - 1.3258)$

ii. We find

$$y'(t) = 8 \frac{\sqrt{17}}{6} \cos(8t - \varphi).$$

Then

$$\left(\frac{y(t)}{\frac{\sqrt{17}}{6}} \right)^2 + \left(\frac{y'(t)}{\frac{8\sqrt{17}}{6}} \right)^2 = 1$$

for all t . This takes the form

$$272(y(t))^2 + 17(y'(t))^2 = 272$$

for all t and is an ellipse in the $y'y$ -plane as shown in Figure 3.5.4. The graph is called the orbit of the simple harmonic oscillator and visualizes the oscillatory motion.

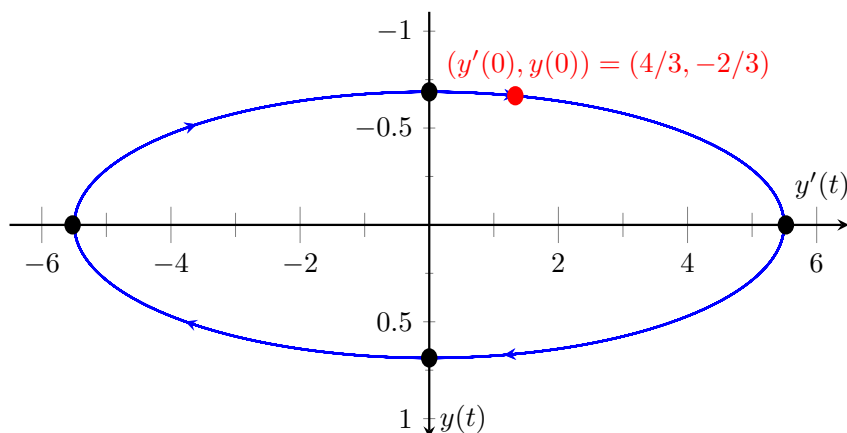


Figure 3.5.4 The graph of $y(t) \approx \frac{\sqrt{17}}{6} \sin(8t - 1.3258)$



II. Free Damped Motion ($b > 0$, $F(t) = 0$). When $b > 0$ (with damping) and $F(t) = 0$ (no external forces), the mass-spring system is said to have a *free damped motion*. The resulting differential equation for the motion of the mass is

$$my''(t) + by'(t) + ky(t) = 0,$$

which, with $2\beta = \frac{b}{m}$ and $\omega^2 = \frac{k}{m}$, becomes

$$y'' + 2\beta y' + \omega^2 y = 0, \quad (3.5.10)$$

and the differential equation (3.5.10) is known as a *free damped spring-mass system*. Its general solution depends on the relative strengths of the damping and spring forces. When the spring force dominates, the motion is oscillatory with amplitude decaying to zero as $t \rightarrow \infty$ and when the damping force dominates, the motion is non-oscillatory. There exists a threshold value of the damping coefficient—called the critical damping—above which no oscillations occurs, and below which the system exhibits damped oscillations whose amplitude decays to zero as $t \rightarrow \infty$. To analyze this, we start with the auxiliary equation for (3.5.10); namely,

$$r^2 + 2\beta r + \omega^2 = 0,$$

with $y(t) = e^{rt}$ is a trial solution. It then follows that

$$r = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega^2}}{2} = -\beta \pm \sqrt{\beta^2 - \omega^2}. \quad (3.5.11)$$

There are three significantly different cases of a free damped motion.

1. **Underdamped Motion:** When $\beta < \omega$, (3.5.11) yields

$$r = -\beta \pm i\sqrt{\omega^2 - \beta^2},$$

so that the general solution of (3.5.10) is

$$y(t) = e^{-\beta t} \left(c_1 \cos(\sqrt{\omega^2 - \beta^2} t) + c_2 \sin(\sqrt{\omega^2 - \beta^2} t) \right),$$

where c_1 and c_2 are arbitrary constants are determined when the initial conditions $y(t_0) = y_0, y'(t_0) = y_1$ are provided. The displacement $y(t)$ is periodic with the period $2\pi/\sqrt{\omega^2 - \beta^2}$. Moreover, since $\beta > 0$, we observe that

$$\lim_{t \rightarrow \infty} e^{-\beta t} \cos(\sqrt{\omega^2 - \beta^2} t) = 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} e^{-\beta t} \sin(\sqrt{\omega^2 - \beta^2} t) = 0.$$

It then follows that

$$\lim_{t \rightarrow \infty} y(t) = 0.$$

With regard to the velocity of the mass, we find

$$\begin{aligned} y'(t) &= -\beta e^{-\beta t} \left(c_1 \cos(\sqrt{\omega^2 - \beta^2} t) + c_2 \sin(\sqrt{\omega^2 - \beta^2} t) \right) \\ &\quad + \sqrt{\omega^2 - \beta^2} e^{-\beta t} \left(-c_1 \sin(\sqrt{\omega^2 - \beta^2} t) + c_2 \cos(\sqrt{\omega^2 - \beta^2} t) \right). \end{aligned}$$

Again, we observe that

$$\lim_{t \rightarrow \infty} y'(t) = 0.$$

Hence, the mass approaches the equilibrium position while oscillating with the circular frequency $\sqrt{\omega^2 - \beta^2}$ and the velocity approaches zero as t goes to ∞ .

Since $\beta = b/(2m)$ and $\omega^2 = k/m$, $\beta < \omega$ is equivalent to $b < 2\sqrt{mk}$. We conclude that when the damping force is too weak to counteract a strong spring force, the mass undergoes damped oscillations: the amplitude gradually decreases to zero. In this case, the mass-spring system is said to be **underdamped**.

2. **Overdamped Motion:** When $\beta > \omega$, (3.5.11) yields $r = -\beta \pm \sqrt{\beta^2 - \omega^2}$, so that the general solution of (3.5.10) is

$$y(t) = c_1 e^{(-\beta + \sqrt{\beta^2 - \omega^2})t} + c_2 e^{(-\beta - \sqrt{\beta^2 - \omega^2})t},$$

where c_1 and c_2 are arbitrary constants are determined when the initial conditions $y(t_0) = y_0, y'(t_0) = y_1$ are provided. Therefore $y(t)$ is not periodic. Moreover, since $\beta > 0$ and $0 < \sqrt{\beta^2 - \omega^2} < \beta$, it follows that both $-\beta + \sqrt{\beta^2 - \omega^2}$ and $-\beta - \sqrt{\beta^2 - \omega^2}$ are negative. Consequently, we have

$$\lim_{t \rightarrow \infty} e^{(-\beta + \sqrt{\beta^2 - \omega^2})t} = 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} e^{(-\beta - \sqrt{\beta^2 - \omega^2})t} = 0.$$

It then follows that

$$\lim_{t \rightarrow \infty} y(t) = 0.$$

We can show that the mass passes through the equilibrium position at most once by setting $y(t) = 0$ and finding the time t at which this happens.

Regarding the velocity of the mass, we find

$$y'(t) = c_1(-\beta + \sqrt{\beta^2 - \omega^2})e^{(-\beta + \sqrt{\beta^2 - \omega^2})t} \\ + c_2(-\beta - \sqrt{\beta^2 - \omega^2})e^{(-\beta - \sqrt{\beta^2 - \omega^2})t},$$

Again, we observe that

$$\lim_{t \rightarrow \infty} y'(t) = 0.$$

We can also show that the mass reaches an extreme position at most once by setting $y'(t) = 0$ and finding the time t at which this happens. Since $\beta = b/(2m)$ and $\omega^2 = k/m$, $\beta > \omega$ is equivalent to $b > 2\sqrt{mk}$. We conclude that when the spring force is too weak to counteract a strong damping force, the mass approaches the equilibrium position with no oscillation and its velocity approaches zero, by changing its sign at most once, as t goes to ∞ . In this case, the mass-spring system is said to be **overdamped**.

3. **Critically Damped Motion:** When $\beta = \omega$, (3.5.11) yields $r = -\beta, -\beta$, so that the general solution of (3.5.10) is

$$y(t) = c_1e^{-\beta t} + c_2te^{-\beta t} = (c_1 + c_2t)e^{-\beta t},$$

where c_1 and c_2 are arbitrary constants which can be determined when the initial conditions $y(t_0) = y_0, y'(t_0) = y_1$ are known. The velocity of the mass is given by

$$y'(t) = -\beta(c_1 + tc_2)e^{-\beta t} + c_2e^{-\beta t} = (c_2 - \beta c_1 - \beta c_2t)e^{-\beta t}.$$

Since $\beta > 0$, we find $e^{-\beta t} \rightarrow 0$ and $te^{-\beta t} \rightarrow 0$ as $t \rightarrow \infty$. These imply that

$$\lim_{t \rightarrow \infty} y(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} y'(t) = 0.$$

As noted earlier in the case of an overdamped motion, the mass in this case can also pass through the equilibrium at most once and can reach an extreme position at most once while its velocity changes its sign. As t goes to ∞ , the mass approaches its equilibrium position and its velocity also tends to zero.

Since $\beta = b/(2m)$ and $\omega^2 = k/m$, $\beta = \omega$ is equivalent to $b = 2\sqrt{mk}$. It follows from the case of underdamped motion that when the damping coefficient b is even slightly less than $2\sqrt{mk}$, the mass undergoes an oscillatory motion. The spring-mass system with the smallest damping coefficient b that maintains a nonoscillatory motion is to be **critically damped**.

III. Forced Undamped Motion ($b = 0, F(t) \neq 0$). When $b = 0$ (no damping) and $F(t) \neq 0$ (with external force), the spring-mass system is governed by the differential equation

$$my''(t) + ky(t) = F(t).$$

which, with $w^2 = k/m$ and $f(t) = F(t)/m$, becomes

$$y''(t) + \omega^2 y(t) = f(t). \quad (3.5.12)$$

The nonhomogeneous equation (3.5.12) can be solved by using the methods of previous sections. The general solution is of the form

$$y(t) = y_h(t) + y_p(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t) + y_p(t), \quad (3.5.13)$$

where c_1 and c_2 are arbitrary constants and $y_p(t)$ is a particular solution of (3.5.12). Recall that

$$y_h(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

is the general solution of the simple harmonic oscillator

$$y''(t) + \omega^2 y(t) = 0,$$

associated with (3.5.12).

Example 3.5.5 A 10-kilogram mass attached to the end of a vertically hanging spring causes the spring to stretch 39.2 centimeters. At time $t = 0$, the spring-mass system is set into motion by an external force $F(t) = 180 \sin(10t)$, where $F(t)$ is measured in newtons and is positive in the downward direction, and time is measured in seconds. Determine the equation of motion and estimate the maximum excursion of the mass from its equilibrium position.

Solution. We let $y(t)$ denote the displacement of the mass from the equilibrium position measured positively in the vertically downward direction.

We have $m = 10$ kg, $s = 39.2$ cm = 0.392 m, $F(t) = 180 \sin(10t)$ N. For the mass in equilibrium, we have $mg = ks$, and so

$$k = \frac{mg}{s} = \frac{10 \times 9.8}{0.392} = 250 \text{ N/m}.$$

The natural frequency of the system is

$$\omega = \sqrt{\frac{k}{m}} = 5 \text{ cycles in } 2\pi \text{ seconds} = 5 \text{ rad/s}$$

The differential equation of motion is therefore

$$10y'' + 250y' = 180 \sin(10t),$$

or equivalently,

$$y'' + 25y = 18 \sin(10t). \quad (3.5.14)$$

The general solution of $y'' + 25y = 0$ is

$$y_h(t) = c_1 \cos(5t) + c_2 \sin(5t),$$

where c_1 and c_2 are arbitrary constants. We find a particular solution $y_p(t)$ of (3.5.14) by variation of parameters. Let

$$y_p(t) = A \cos(10t) + B \sin(10t)$$

be a particular solution of (3.5.14), where A and B are to be determined. We find that $y_p''(t) = -100y_p(t)$. Substituting y_p for y and y_p'' for y'' into (3.5.14) yields

$$-100y_p(t) + 25y_p(t) = 18 \sin(10t),$$

that is,

$$-75A \cos(5t) - 75B \sin(5t) = 18 \sin(10t).$$

Equating the coefficients of the like terms, we have $-75A = 0$ and $-75B = 18$, that is, $A = 0$ and $B = -6/25$. Thus, we have

$$y_p(t) = -\frac{6}{25} \sin(5t).$$

The general solution of (3.5.14) is

$$y(t) = c_1 \cos(5t) + c_2 \sin(5t) - \frac{6}{25} \sin(10t),$$

where c_1 and c_2 are arbitrary constants. The initial conditions for $y(t)$ are $y(0) = 0, y'(0) = 0$. Substituting these gives $c_1 = 0$ and $5c_2 - \frac{12}{5} = 0$, i.e., $c_1 = 0$ and $c_2 = \frac{12}{25}$. Thus, we have

$$y(t) = \frac{12}{25} \sin(5t) - \frac{6}{25} \sin(10t).$$

The graph of the solution is shown in Figure 3.5.6. Using the critical numbers and second derivative test, we can verify that the maximum occurs at $t = \frac{2\pi}{15}$. The maximum value is

$$y\left(\frac{2\pi}{15}\right) = \frac{9\sqrt{3}}{25} \approx 0.6235.$$

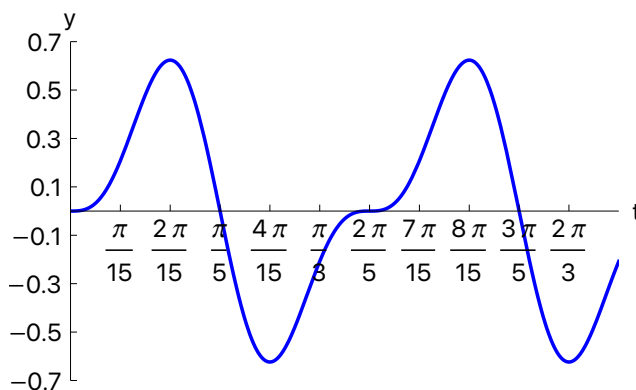


Figure 3.5.6 The graph of $y(t) = \frac{12}{25} \sin(5t) - \frac{6}{25} \sin(10t)$

In Example 3.5.5, the frequency of the external source is $\gamma = 10$ which is not equal to or near the natural frequency $\omega = 5$.

Resonance: An important aspect of analyzing undamped forced spring-mass systems is understanding how large (in absolute values) the values of $y_p(t)$ can become when $F(t)$ is sinusoidal. If the amplitude of $y_p(t)$ grows without bound, the system is said to be in **resonance**. In practical applications, resonance may be undesirable because it can lead to excessive energy buildup and potential system failure. Since the system (3.5.12) is undamped, the only way energy can dissipate from the system is by the nature of the source term $F(t)$. However, it is possible that the energy buildup can happen in the system even when $F(t)$ is sign-changing. The following example demonstrates how resonance occurs.

Example 3.5.7 With the external force in Example 3.5.5 replaced with $F(t) = 180 \sin(5t)$, determine the equation of motion, analyze the amplitude of the resulting motion, and verify that the system exhibits resonance.

Solution. Similarly to the solution in Example 3.5.5, the initial value problem for the current example is

$$y'' + 25y = 18 \sin(5t), \quad y(0) = 0, \quad y'(0) = 0. \quad (3.5.15)$$

Recall from Example 3.5.5 that the natural frequency of the system is $\omega = 10$. We note here that the frequency of the external force $\gamma = 10$ matches the

natural frequency $\omega = 10$. The general solution of $y'' + 25y = 0$ is

$$y_h(t) = c_1 \cos(5t) + c_2 \sin(5t),$$

where c_1 and c_2 are arbitrary constants. We find a particular solution $y_p(t)$ of (3.5.15) by variation of parameters. Let

$$y_p(t) = u_1(t) \cos(5t) + u_2(t) \sin(5t),$$

where $u_1'(t)$ and $u_2'(t)$ satisfy

$$\begin{cases} u_1'(t) \cos(5t) + u_2'(t) \sin(5t) = 0, \\ -5u_1'(t) \sin(5t) + 5u_2'(t) \cos(5t) = 18 \sin(5t). \end{cases}$$

Solving this system gives

$$u_1'(t) = -\frac{18}{5} \sin^2(5t) = -\frac{9}{5} (1 - \cos(10t)),$$

and

$$u_2'(t) = \frac{18}{5} \sin(5t) \cos(5t) = \frac{9}{5} \sin(10t).$$

Integrating yields

$$u_1(t) = -\frac{9}{5}t + \frac{9}{50} \sin(10t), \quad u_2(t) = -\frac{9}{50} \cos(10t).$$

Hence,

$$\begin{aligned} y_p(t) &= u_1 \cos(5t) + u_2 \sin(5t) \\ &= -\frac{9}{5}t \cos(5t) + \frac{9}{50} \sin(10t) \cos(5t) - \frac{9}{50} \cos(10t) \sin(5t) \\ &= -\frac{9}{5}t \cos(5t) + \frac{9}{50} \sin(5t). \end{aligned}$$

The general solution is

$$y(t) = c_1 \cos(5t) + c_2 \sin(5t) - \frac{9}{5}t \cos(5t),$$

c_1 and c_2 are constants to be determined so that $y(0) = 0, y'(0) = 0$. Substituting these gives $c_1 = 0$ and $5c_2 - \frac{9}{5} = 0$, i.e., $c_2 = \frac{9}{25}$. Thus, we have

$$y(t) = -\frac{9}{5}t \cos(5t) + \frac{9}{25} \sin(5t).$$

It is evident that the amplitude $a(t)$ of $y(t)$ is given by

$$a(t) = \frac{9}{5} \sqrt{t^2 + \frac{1}{25}}$$

which shows that $|y(t)|$ grows without bound as $t \rightarrow \infty$. Hence, the system exhibits **resonance** in this case, where the forcing frequency coincides the natural frequency—illustrating the classical example of resonance in an undamped forced spring-mass system. The plots of the solution $y(t)$ and amplitude $a(t)$ are shown in Figure 3.5.8.

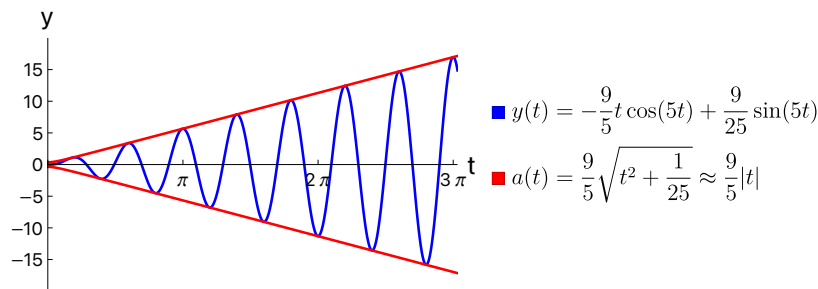


Figure 3.5.8 The graph of $y(t) = -\frac{9}{5}t \cos(5t) + \frac{9}{25} \sin(5t)$ and its amplitude $a(t) = \frac{9}{5} \sqrt{t^2 + \frac{1}{25}}$



IV. Forced Damped Motion ($b > 0$, $F(t) \neq 0$). When $b > 0$ and $F(t) \neq 0$, the spring-mass system is governed by the differential equation (3.5.16); namely,

$$my''(t) + by'(t) + ky(t) = F(t). \quad (3.5.16)$$

The nonhomogeneous equation (3.5.16) can be solved by using the methods of previous sections. The general solution is of the form

$$y(t) = y_h(t) + y_p(t), \quad (3.5.17)$$

where $y_h(t)$ is the general solution of the homogeneous equation

$$my''(t) + by'(t) + ky(t) = 0,$$

which describes the free damped motion corresponding to $F(t) = 0$ and $y_p(t)$ is a particular solution of (3.5.16). We recall from all three cases (underdamped, critically damped, and overdamped) of a free damped motion that

$$\lim_{t \rightarrow \infty} y_h(t) = 0, \quad (3.5.18)$$

and therefore $y_h(t)$ is a **transient term** in (3.5.17) and it is either oscillatory or nonoscillatory according as $b < 2\sqrt{mk}$ or $b \geq \sqrt{mk}$. We conclude from (3.5.17) and (3.5.18) that

$$y(t) \approx y_p(t) \quad \text{for large } t.$$

The particular solution $y_p(t)$ is called the **steady-state** solution of (3.5.16). For short times, $y_h(t)$ must be included in the solution $y(t) = y_h(t) + y_p(t)$, which then becomes the **transient solution**.

Example 3.5.9 For the resonant system in Example 3.5.7, determine the damping coefficient that should be introduced at $t = 3\pi$ so that the system transitions immediately into a simple harmonic oscillator with the amplitude $27\pi/5$.

Solution. We recall that the solution of the resonant system in Example 3.5.7

$$\begin{cases} y'' + 25y = 18 \sin(5t), \\ y(0) = 0, \quad y'(0) = 0, \end{cases} \quad (3.5.19)$$

is

$$y(t) = -\frac{9}{5}t \cos(5t) + \frac{9}{25} \sin(5t).$$

We observe that $y(3\pi) = 27\pi/5$ and $y'(3\pi) = 0$.

We now proceed to find the damping coefficient $b > 0$ required to turn the resonant system (3.5.19) into a simple harmonic oscillator starting at $t = 3\pi$ with the frequency of oscillation same as the natural frequency of resonant system and the amplitude $27\pi/5$. We consider the initial value problem

$$\begin{cases} 10y'' + by' + 250y = 180 \sin(5t), \\ y(3\pi) = 27\pi/5, \quad y'(3\pi) = 0. \end{cases} \quad (3.5.20)$$

It is evident that for small $b > 0$ the complementary function should be oscillatory. Suppose then that $b^2 < 40000$ and put

$$d = \frac{1}{20} \sqrt{10000 - b^2}.$$

Then the general solution of the differential equation in (3.5.20) can be written in the form

$$y(t) = e^{-\frac{b}{20}(t-3\pi)} \left(c_1 \cos(d(t-3\pi)) + c_2 \sin(d(t-3\pi)) \right) - \frac{36}{b} \cos(5t)$$

where c_1 and c_2 are arbitrary constants. The only way this represents the solution to a harmonic oscillator is when its transient term is zero which happens only when $c_1 = c_2 = 0$ because of the linear independence of $\cos(d(t-3\pi))$ and $\sin(d(t-3\pi))$. Since $y(3\pi) = 27\pi/5$, we have

$$\frac{27\pi}{5} = c_1 + \frac{36}{b}.$$

In order that $c_1 = 0$, we must have

$$b = \frac{20}{3\pi}.$$

The condition $y'(3\pi) = 0$ ensures that $c_2 = 0$ because

$$0 = y'(3\pi) = -\frac{b}{20}c_1 + dc_2,$$

which gives

$$c_2 = \frac{b}{20d}c_1 = 0.$$

Thus, the damped system introduced at $t = 3\pi$ is

$$\begin{cases} 10y'' + \frac{20}{3\pi}y' + 250y = 180 \sin(5t), \\ y(3\pi) = 27\pi/5, \quad y'(3\pi) = 0, \end{cases} \quad (3.5.21)$$

with the solution

$$z(t) = -\frac{27\pi}{5} \cos(5t)$$

for all $t \geq 3\pi$. The problem (3.5.21) is equivalent to the simple harmonic oscillator described by

$$\begin{cases} y'' + 25y = 0, \\ y(3\pi) = 27\pi/5, \quad y'(3\pi) = 0. \end{cases} \quad (3.5.22)$$

This harmonic oscillator has frequency same as the natural frequency of the resonant system (3.5.19) and amplitude equal to $27\pi/5$. The solution of the

resonant system for $0 \leq t \leq 3\pi$ and its continuous transition to a damped forced system that exhibits the simple harmonic oscillator (3.5.22) for $t \geq 3\pi$ is

$$y(t) = \begin{cases} -\frac{9}{5}t \cos(5t) + \frac{9}{25} \sin(5t) & \text{for } 0 \leq t \leq 3\pi \\ -\frac{27\pi}{5} \cos(5t) & \text{for } t \geq 3\pi. \end{cases}$$

The graph of this solution is shown in Figure 3.5.10.

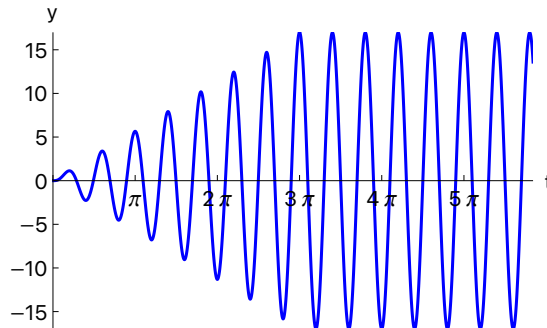


Figure 3.5.10 The graph of $y = y(t)$.

When $b = 0$ and the natural frequency of the system matches that of an external sinusoidal force, as illustrated in Example 3.5.7, the system exhibits resonance. In contrast, we recall that a free damped system approaches its equilibrium as $t \rightarrow \infty$. In most practical applications, the damping constant b is relatively small. Interestingly, when $b > 0$ but small, although the amplitude of the particular solution y_p is not unbounded as $t \rightarrow \infty$, the system displays a behavior approaching resonance because the damping is insufficient to suppress large oscillations. Therefore $\gamma = \omega$ should be avoided even for small damping. The following example illustrates this phenomenon.

Example 3.5.11 In the resonant system discussed in Example 3.5.7, suppose that the surrounding medium offers a damping force that is numerically equal to 0.01 times the instantaneous velocity. Find the equation of motion and analyze the danger of having a low value of the damping coefficient.

Solution. The initial value problem is

$$\begin{cases} y'' + 0.001y' + 25y = 18 \sin(5t), \\ y(0) = 0, \quad y'(0) = 0. \end{cases} \quad (3.5.23)$$

The auxiliary equation for $y'' + 0.001y' + 25y = 0$ is

$$r^2 + 0.001r + 25 = 0$$

with solutions $r = -a \pm ib$, where $a = 0.0005$ and $b = \frac{1}{2}\sqrt{100 - 10^{-6}}$. Then the general solution of the homogeneous differential equation

$$y_h(t) = e^{-at} (c_1 \cos(bt) + c_2 \sin(bt)),$$

where c_1 and c_2 are arbitrary constants. Let $y_p(t) = A \cos(5t) + B \sin(5t)$ be particular solution of the nonhomogeneous equation in (3.5.23). Then

$$y_p'(t) = -5A \sin(5t) + 5B \cos(5t)$$

and

$$y_p''(t) = -25A \cos(5t) - 25B \sin(5t) = -25y_p(t).$$

Substituting y_p for y in the nonhomogeneous equation yields

$$-25y_p(t) - 0.005A \sin(5t) + 0.005B \cos(5t) + 25y_p(t) = 18 \sin(5t),$$

which gives $A = -3600$ and $B = 0$. Then $y_p(t) = -3600 \cos(5t)$, and therefore the general solution of the nonhomogeneous equation is

$$y(t) = e^{-at} (c_1 \cos(bt) + c_2 \sin(bt)) - 3600 \cos(5t).$$

Since $y(0) = 0$, we have $0 = c_1 - 3600$, i.e., $c_1 = 3600$. Also, since $y'(0) = 0$, we have $0 = -ac_1 + bc_2$, which gives $c_2 = ac_1/b = 3600a/b$. Thus, the solution to the initial value problem is

$$y(t) = 3600e^{-at} \left(\cos(bt) + \frac{a}{b} \sin(bt) \right) - 3600 \cos(5t).$$

The graph of the solution is shown in Figure 3.5.12 over $[0, 20]$ shows that the system appears to have resonance-like behavior as in Figure 3.5.8. However, it is evident that $y(t) \approx -3600 \cos(5t)$ for large values of t . In particular, $y(7001\pi) \approx 3600$.

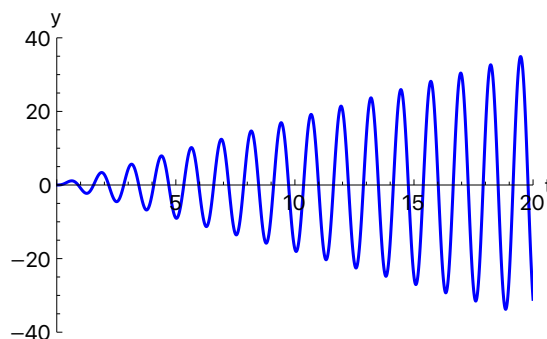


Figure 3.5.12 The graph of $y(t) = 3600e^{-at} \left(\cos(bt) + \frac{a}{b} \sin(bt) \right) - 3600 \cos(5t)$.

▲
In the following example, we examine an undamped forced system in which the frequency of the external force is slightly different from the natural frequency.

Example 3.5.13 In the setting of the resonant problem in Example 3.5.7, let us take the external force

$$F(t) = 180 \cos(5t + 2\epsilon t),$$

where ϵ is very small. Find the equation of motion.

Solution. The initial value problem is

$$\begin{cases} y'' + 25y = 18 \cos(\gamma t), \\ y(0) = 0, \quad y'(0) = 0, \end{cases} \quad (3.5.24)$$

where $\gamma = 5 + 2\epsilon$ with ϵ a small nonzero number. We have thus considered the frequency of the external force close to the natural frequency. The general solution of $y'' + 25y = 0$ is

$$y_h(t) = c_1 \cos(5t) + c_2 \sin(5t),$$

where c_1 and c_2 are arbitrary constants.

Let $y_p(t) = A \cos(\gamma t) + B \sin(\gamma t)$ be a particular solution of the nonhomogeneous equation in (3.5.24). Substituting y_p for y and y_p'' for y'' into the nonhomogeneous equation in (3.5.24) gives

$$(25 - \gamma^2)A \cos(\gamma t) + (25 - \gamma^2)B \sin(\gamma t) = 18 \cos(\gamma t).$$

This gives

$$A = -\frac{18}{\gamma^2 - 25} \quad \text{and} \quad B = 0.$$

Thus,

$$y(t) = -\frac{18}{\gamma^2 - 25} \cos(\gamma t),$$

and therefore the general solution of the nonhomogeneous equation is

$$y_p(t) = c_1 \cos(5t) + c_2 \sin(5t) - \frac{18}{\gamma^2 - 25} \cos(\gamma t).$$

Since $y(0) = 0$ and $y'(0) = 0$, we have

$$0 = c_1 - \frac{18}{\gamma^2 - 25} \quad \text{and} \quad 0 = 5c_2,$$

that is,

$$c_1 = \frac{18}{\gamma^2 - 25} \quad \text{and} \quad c_2 = 0.$$

Thus, the solution of the initial value problem is

$$\begin{aligned} y(t) &= \frac{18}{\gamma^2 - 25} (\cos(5t) - \cos(\gamma t)) \\ &= -\frac{36}{(\gamma + 5)(\gamma - 5)} \sin\left(\frac{1}{2}(\gamma - 5)t\right) \sin\left(\frac{1}{2}(\gamma + 5)t\right) \\ &= -\frac{9}{(5 + \epsilon)\epsilon} \sin(\epsilon t) \sin((5 + \epsilon)t) \\ &= A(t) \sin((5 + \epsilon)t), \end{aligned}$$

where

$$A(t) = -\frac{9}{(5 + \epsilon)\epsilon} \sin(\epsilon t).$$

We observe that

$$|y(t)| \leq |A(t)|$$

for all t . This means that the graph of $y(t)$ will be enveloped in the graphs of $\pm A(t)$, bounding the amplitude of $\sin((5 + \epsilon)t)$ as shown in Figure 3.5.14 for $\epsilon = 0.5$.

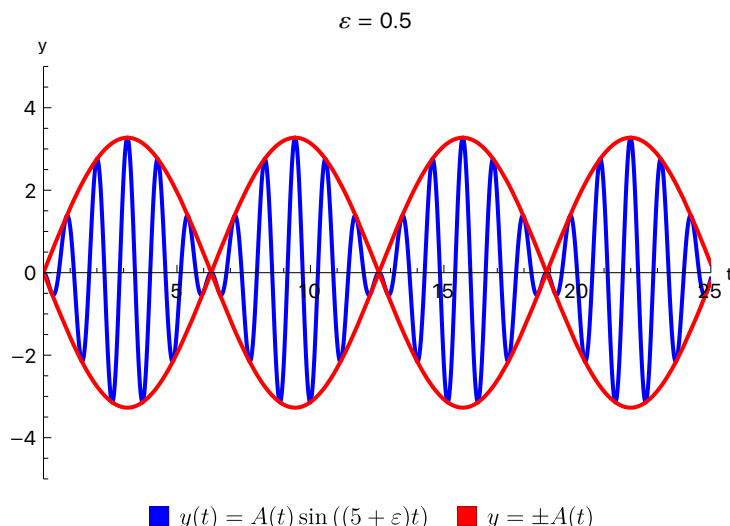


Figure 3.5.14 The graph of $y(t) = 3600e^{-at} (\cos(bt) + \frac{a}{b} \sin(bt)) - 3600 \cos(5t)$.

In music, when both waves are heard simultaneously, the sound pattern is known as *beats* and the sound system is said to behave *near resonance*.

Using a graphing utility, the reader is suggested to plot $y(t)$ for a smaller value of ε (e.g., $\varepsilon = 0.02$) and for a larger value (e.g., $\varepsilon = 5$) to observe how the motion changes when the frequency of the external force is close to the natural frequency and when it is far from it. ▲

In [Example 3.5.13](#), we could have used $F(t) = 180 \sin(5t + 2\epsilon t)$ with reference to [Example 3.5.7](#). The choice of the cosine function is only for convenience.

3.5.2 Electrical Circuits

In this subsection, we first study ordinary differential equations as initial value problems in electric circuits that consist of a voltage source of the electromotive force $e(t)$ driving electric charges and producing a current $i(t)$ (e.g., a battery for direct voltage or a generator for alternating voltage), a resistor (e.g., appliances) of resistance R , and an inductor of inductance L in series as depicted in [Figure 3.5.15](#) or [Figure 3.5.16](#). We then briefly discuss the analogy between the spring-mass system and the electric circuit.

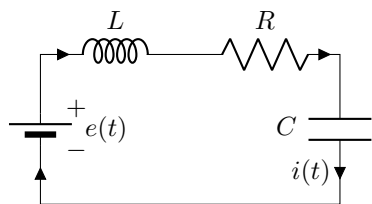


Figure 3.5.15 Direct voltage source

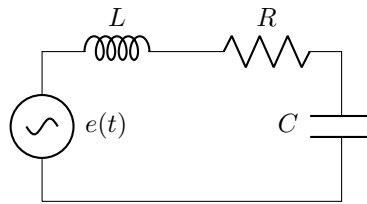


Figure 3.5.16 Alternating voltage source

Let $q(t)$ denote the electric charge on the capacitor at time t . According to Kirchhoff's second law, *the voltage impressed on a closed loop equals the sum of the voltage drops across the components in the loop*. The voltage drops across a resistor, inductor and capacitor are modeled as being proportional to $i(t)$, $i'(t)$, $q(t)$, respectively. These voltage drops are modeled as follows:

- i. the voltage drop across an inductor $= L i'(t)$,
- ii. the voltage drop across a resistor $= R i(t)$,
- iii. the voltage drop across a capacitor $= \frac{1}{C} q(t)$,

where the constants L , R , and C are called the inductance, resistance, and capacitance, respectively, and these quantities are typically measured in henry (h), ohm (Ω), and farad (f), respectively. The corresponding units for q , E , and I are coulomb (C), volt (V), and ampere (A).

We recall that the current $i(t)$ flowing through the capacitor is given by

$$i(t) = q'(t), \quad \text{so that} \quad i'(t) = q''(t).$$

and therefore Kirchhoff's law yields

$$Lq''(t) + Rq'(t) + \frac{1}{C}q(t) = e(t), \quad (3.5.25)$$

which is a second-order linear differential equation in $q(t)$ with constant coefficients.

For analogy, we can interchange the spring-mass system parameters in

$$my''(t) + by'(t) + ky(t) = F(t)$$

(see (3.5.5)) and series LRC circuit parameters in (3.5.25) as follows:

Table 3.5.17 Parameter Analogies

Spring-Mass		Series LRC
m	$\longleftrightarrow$	L
b	$\longleftrightarrow$	R
k	$\longleftrightarrow$	$\frac{1}{C}$
$\omega = \sqrt{k/m}$	$\longleftrightarrow$	$\omega = \sqrt{1/LC}$

In addition, the three significantly different cases of the spring-mass system apply to the LRC circuit as follows:

Table 3.5.18 Parameter Analogies

Damping Type	Spring-Mass	Series LRC
Underdamped	$b^2 < 4mk$	$R^2 < \frac{4L}{C}$
Critically damped	$b^2 = 4mk$	$R^2 = \frac{4L}{C}$
Overdamped	$b^2 > 4mk$	$R^2 > \frac{4L}{C}$

We now consider the parallel LRC circuit shown in Figure 3.5.19, where a current source supplies the input current $i(t)$. The diagram also shows that the current $i(t)$ splits into three branch currents $i_1(t)$, $i_2(t)$, and $i_3(t)$ flowing through a capacitor, resistor and inductor, respectively. We develop a differential equation for the impressed voltage $e(t)$ across the components produced from the source current $i(t)$.

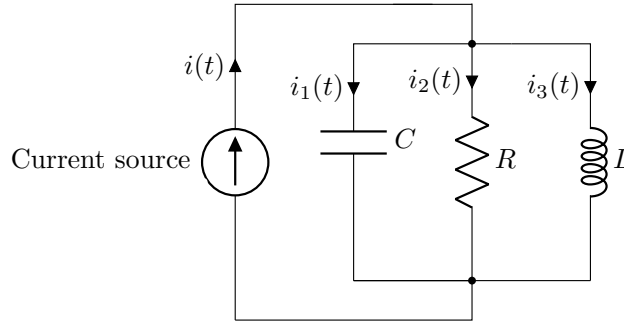


Figure 3.5.19 Parallel LRC circuit

By Kirchhoff's second law which states that *the current flowing to a point in a circuit must equal the current flowing away from the point*, we have

$$i_1(t) + i_2(t) + i_3(t) = i(t). \quad (3.5.26)$$

Since

$$i_1(t) = C e'(t), \quad i_2(t) = \frac{e(t)}{R}, \quad \text{and} \quad i_3(t) = \frac{1}{L} \int e(t) dt,$$

(3.5.26) reduces to

$$C e'(t) + \frac{e(t)}{R} + \frac{1}{L} \int e(t) dt = i(t).$$

Differentiating with respect to t yields

$$C e''(t) + \frac{1}{R} e'(t) + \frac{1}{L} e(t) = i'(t). \quad (3.5.27)$$

For analogy, we can interchange the spring-mass system parameters, series LRC circuit parameters, and parallel LRC circuit parameters in (3.5.27) as follows:

Table 3.5.20 Parameter analogies

Spring-Mass	Series LRC	Parallel LRC
m	L	C
b	R	$\frac{1}{R}$
k	$\frac{1}{C}$	$\frac{1}{L}$
$\omega = \sqrt{k/m}$	$\omega = \sqrt{1/LC}$	$\omega = \sqrt{1/LC}$

In addition, the three significantly different cases of the spring-mass system apply to the LRC circuits as follows:

Table 3.5.21 Damping parameter analogies

Damping Type	Spring-Mass	Series LRC	Parallel LRC
Underdamped	$b^2 < 4mk$	$R^2 < \frac{4L}{C}$	$R^2 > \frac{L}{4C}$
Critically damped	$b^2 = 4mk$	$R^2 = \frac{4L}{C}$	$R^2 = \frac{L}{4C}$
Overdamped	$b^2 > 4mk$	$R^2 > \frac{4L}{C}$	$R^2 < \frac{L}{4C}$

Example 3.5.22 Consider a series LRC -circuit with inductance $L = 1$ h, resistance $R = 12\ \Omega$, and capacitance $C = \frac{1}{40}$ f. The circuit is driven by an external voltage

$$e(t) = 10 \sin(2t) \quad t > 0,$$

measured in volts (V). Let $q(t)$ denote the charge measured in coulombs (C) on the capacitor, and let $i(t) = q'(t)$ be the current measured in amperes (A). Suppose the initial charge and current are

$$q(0) = 0, \quad q'(0) = 0.$$

- Derive the differential equation satisfied by $q(t)$ and solve it.
- Find the *transient part* and *steady-state* solution.
- Find the critical resistance at which the charge on the capacitor changes from oscillatory to nonoscillatory.

Solution. a. For a series LRC -circuit, Kirchhoff's voltage law gives

$$Lq''(t) + Rq'(t) + \frac{1}{C}q(t) = e(t).$$

With $L = 1$, $R = 12$, $C = \frac{1}{40}$, and $e(t) = 10 \sin(2t)$, we obtain the differential equation

$$q'' + 12q' + 40q = 10 \sin(2t).$$

The corresponding homogeneous equation is

$$q'' + 12q' + 40q = 0.$$

The roots of the characteristic equation

$$\lambda^2 + 12\lambda + 40 = 0$$

are

$$\frac{-12 \pm \sqrt{-16}}{2} = -6 \pm 2i.$$

Thus the general solution of the homogeneous equation is

$$q_h(t) = e^{-6t}(c_1 \cos(2t) + c_2 \sin(2t))$$

where c_1 and c_2 are arbitrary constants.

For a particular solution $q_p(t)$, suppose that A and B are constants to be determined, so that

$$q_p(t) = A \cos(2t) + B \sin(2t),$$

is a particular solution of the nonhomogeneous equation. Then

$$\begin{aligned} q_p'(t) &= -2A \sin(2t) + 2B \cos(2t), \text{ and} \\ q_p''(t) &= -4A \cos(2t) - 4B \sin(2t). \end{aligned}$$

Substituting into the equation $q'' + 12q' + 40q = 10 \sin(2t)$ yields

$$\begin{aligned} &(-4A \cos 2t - 4B \sin 2t) + 12(-2A \sin 2t + 2B \cos 2t) \\ &+ 40(A \cos 2t + B \sin 2t) \\ &= (36A + 24B) \cos(2t) + (-24A + 36B) \sin(2t) = 10 \sin(2t). \end{aligned}$$

Thus we must have

$$36A + 24B = 0, \quad -24A + 36B = 10.$$

Solving for A and B gives

$$A = -\frac{5}{39} \text{ and } B = \frac{5}{26},$$

and so

$$q_p(t) = -\frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t).$$

The general solution is

$$\begin{aligned} q(t) = q_h(t) + q_p(t) &= e^{-6t} (c_1 \cos(2t) + c_2 \sin(2t)) \\ &\quad - \frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t), \end{aligned}$$

where c_1 and c_2 are arbitrary constants. Using $q(0) = 0$ gives

$$c_1 - \frac{5}{39} = 0, \quad \text{which implies} \quad c_1 = \frac{5}{39}.$$

To use $q'(0) = 0$, we first compute $q'(t)$:

$$\begin{aligned} q'(t) &= e^{-6t} \left[-6(c_1 \cos(2t) \right. \\ &\quad \left. + c_2 \sin(2t)) + (-2c_1 \sin(2t) + 2c_2 \cos(2t)) \right] \\ &\quad + \frac{10}{39} \sin(2t) + \frac{5}{13} \cos(2t). \end{aligned}$$

Then, using $q'(0) = 0$, we get

$$-6c_1 + 2c_2 + \frac{5}{13} = 0.$$

Since $c_1 = \frac{5}{39}$, we have

$$-6 \cdot \frac{5}{39} + 2c_2 + \frac{5}{13} = 0,$$

which gives

$$c_2 = \frac{5}{26}.$$

Thus, the solution for $q(t)$ is

$$q(t) = e^{-6t} \left(\frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t) \right) - \frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t).$$

b. The **transient part** of the solution is the part that *decays* to zero as $t \rightarrow \infty$ and depends on the initial conditions. Since

$$\lim_{t \rightarrow \infty} e^{-6t} \left(\frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t) \right) = 0,$$

the transient part of the solution for $q(t)$ is

$$e^{-6t} \left(\frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t) \right)$$

The **steady-state solution** is the part of the solution $q(t)$ that *remains* after the transient part has decayed to zero. Therefore the steady-state solution for $q(t)$ is given by

$$\lim_{t \rightarrow \infty} q(t) = -\frac{5}{39} \cos(2t) + \frac{5}{26} \sin(2t).$$

- c. The series *LRC*-circuit is critically damped for

$$R = 2\sqrt{\frac{L}{C}} = 4\sqrt{10} \, \Omega.$$



3.5.3 Exercises

Subsection 3.5.1 Spring-Mass System.

1. An 8-kg mass attached to the lower end of a spring that is suspended vertically from a ceiling stretches the spring 5 cm from its natural length until the mass comes to rest. The mass is then pulled downward causing an additional stretch of 3 cm of the spring and then it is released. Assume that there is no air resistance and use the acceleration due to gravity $g = 980 \text{ cm/s}^2$. Let $y(t)$ be the displacement (in centimeters) of the mass from its equilibrium position measured positive in the downward direction.
 - i. Determine a differential equation in $y(t)$ with initial conditions that describe the motion of the mass.
 - ii. Solve for the initial value problem in [Item 3.5.3.1.i](#).
 - iii. Find the amplitude, period and circular frequency of the oscillatory motion.
2. Suppose a vertical spring with spring constant 8 N/m is suspended from a ceiling, and a 2 kg mass is attached to its lower end. The mass slides along a vertical wall that exerts a damping force proportional to its velocity, with damping constant 3 Ns/m. Let $y(t)$ denote the displacement (in meters) of the mass from its equilibrium position, where positive values of $y(t)$ correspond to positions below the equilibrium position.
 - i. Derive a differential equation governing the motion of the system in terms of $y(t)$.
 - ii. Find the general solution of the differential equation obtained in [Item 3.5.3.2.i](#).
 - iii. Determine whether the system is underdamped, critically damped, or overdamped.
 - iv. If the system is not critically damped, find the value of the damping constant that would make the system critically damped.
3. A 10-kg mass suspended from the lower end of a vertically hanging spring stretches the spring 9.8 centimeters. At time $t = 0$ when the mass is at rest in the equilibrium position, the mass-spring system is acted by the force $F(t) = 360 \cos(8t)$, measured positive (in newtons)

in the downward direction. Suppose that time t is measured in seconds. Let $y(t)$ denote the displacement (in centimeters) of the mass from its equilibrium position, measured positive in the downward direction.

- i. Determine the spring constant k .
 - ii. Develop an initial value problem for $y(t)$.
 - iii. Solve the initial value problem developed in [Item 3.5.3.3.ii](#).
 - iv. Using a graphing utility, plot the solution $y(t)$ and determine whether the motion exhibits a “beat”.
 - v. Find the maximum positive displacement of the mass from its equilibrium position.
4. A 8-pound weight attached to a spring of natural length 7 feet stretches the spring to a total length of 8.6 feet. The entire system is placed in a medium that exerts a damping force numerically equal to 4 times the instantaneous velocity.
- i. Write down an initial value problem describing the motion.
 - ii. Determine the equation of motion if the mass is released from rest at a point $\frac{1}{2}$ ft below the equilibrium position with a downward velocity of 1 ft/s.
 - iii. Express the equation of motion in terms of a single sine function.
 - iv. Find the times at which the mass passes through the equilibrium position moving downward.
 - v. Plot the equation of motion.

Subsection 3.5.2 Electrical Circuits.

5. A series LRC -circuit contains an inductor with inductance $L = 1$ h, a resistor with resistance $R = 10 \Omega$, and a capacitor with capacitance $C = 0.02$ f. The circuit is driven by a constant external voltage $e(t) = 5$ V. Let $q(t)$ denote the charge (in coulombs) on the capacitor, and let $i(t) = q'(t)$ be the current (in amperes). Given the initial conditions $q(0) = 4$ C and $i(0) = q'(0) = 0$ A,
- i. determine the charge $q(t)$ for $t > 0$; and
 - ii. find the transient current and steady-state current.
6. Consider a series LRC -circuit with inductance $L = 1$ h, capacitance $C = 16$ f, resistance R , and no external voltage source. Let $q(t)$ denote the charge (in coulombs) on the capacitor at time t (in seconds).
- i. Find $q(t)$ when $R = 4 \Omega$, $q(0) = 5$ C, and $q'(0) = 0$ C/s.
 - ii. Find the first time when the charge on the capacitor is equal to zero.
 - iii. Find the critical resistance at which the charge on the capacitor changes from oscillatory to nonoscillatory behavior.
7. Consider a parallel LRC -circuit with inductance $L = \frac{4}{5}$ h, capacitance $C = \frac{1}{16}$ f, resistance R , and a constant external current source. Let $q(t)$ denote the charge (in coulombs) on the capacitor at time t (in

seconds).

- i. Find $q(t)$ when $R = 4 \Omega$, $q(0) = 5 \text{ C}$, and $q'(0) = 0 \text{ C/s}$.
- ii. Find the first time when the charge on the capacitor is equal to zero.
- iii. Find the critical resistance at which the charge on the capacitor changes from oscillatory to nonoscillatory behavior.

3.6 Change of Variables

Objectives

- Understand the change of variables techniques.

We begin this section by exploring how a change of the dependent variable can be used to reduce a second order linear differential equation to a form can be solved using methods developed earlier. We then examine how a change of the independent variable can be used to develop solution techniques applicable to a variety of differential equations.

3.6.1 Change of the Dependent Variable

Let $u(x)$ be a function to be chosen later and let $w(x)$ be a function such that $y(x) = u(x)w(x)$ is a solution of

$$y'' + a_1(x)y' + a_0(x)y = f(x) \quad (3.6.1)$$

on some interval I , where a_1, a_0, f are continuous. Upon differentiation, we get

$$\begin{aligned} y' &= u'w + uw' \\ y'' &= u''w + 2u'w' + uw''. \end{aligned}$$

Substituting these into the differential equation, we get

$$(u'' + a_1u' + a_0u)w + (2u' + a_1u)w' + uw'' = f. \quad (3.6.2)$$

Our goal is to find w by choosing u in a way that simplifies (3.6.2). A choice of u can be made so that it satisfies one of the following two conditions:

Condition A: $u'' + a_1u' + a_0u = 0$, i.e., u is a solution of

$$y'' + a_1(x)y' + a_0(x)y = 0. \quad (3.6.3)$$

Condition B: $2u' + a_1(x)u = 0$ and a_1 is differentiable.

Suppose first that u satisfies **Condition A**. Then (3.6.2) becomes

$$(2u' + a_1u)w' + uw'' = f, \quad (3.6.4)$$

which, being a linear equation in w' , can be solved first for w' and then for w . Thus, we have obtained a particular solution $y_p(x) = u(x)w(x)$ of (3.6.1).

To find the general solution of (3.6.1), we need one another solution $v(x)$ (3.6.3), such that $u(x)$ and $v(x)$ are linearly independent. Let $v(x) = u(x)z(x)$

for some function $z(x)$ such that v satisfies (3.6.3). Running the same calculation as above with z in place of w , but with $f = 0$, we get

$$(2u' + a_1u)z' + uz'' = 0, \quad (3.6.5)$$

which solves for a nonzero function $z(x)$. Consequently, the general solution to (3.6.1) is given by

$$\begin{aligned} y &= c_1u(x) + c_2v(x) + u(x)w(x) \\ &= c_1u(x) + c_2u(x)z(x) + u(x)w(x) \\ &= u(x)(c_1 + c_2z(x) + w(x)), \end{aligned}$$

where c_1 and c_2 are arbitrary constants.

Let us work through an example to show how the above procedure works.

Example 3.6.1 Find the general solution of

$$(x-1)y'' - xy' + y = 1$$

using the fact that $u(x) = e^x$ is a solution to the associate homogeneous equation.

Solution. First, we note that $a_1(x) = -\frac{x}{x-1}$ and $f(x) = \frac{1}{x-1}$. We compute w and z satisfying (3.6.4) and (3.6.5), respectively. Using (3.6.4), we get

$$\left(2e^x - \frac{x}{x-1}e^x\right)w' + e^xw'' = \frac{1}{x-1}$$

which gives

$$w'' + \left(1 - \frac{1}{x-1}\right)w' = \frac{e^{-x}}{x-1}.$$

The integrating factor is

$$\frac{e^x}{x-1}.$$

Then

$$\left(\frac{e^x}{x-1}w'\right)' = \frac{1}{(x-1)^2},$$

which yields

$$\frac{e^x}{x-1}w' = -\frac{1}{x-1} + A,$$

so that

$$w' = -e^{-x} + A(x-1)e^{-x}.$$

Integrating yields

$$w = e^{-x} - Axe^{-x} + B,$$

where A and B are arbitrary constants. We simply take $w(x) = e^{-x}$, so that $y_p(x) = u(x)w(x) = e^xe^{-x} = 1$. Moreover, it is clear that $z(x) = -Axe^{-x} + B$ solves (3.6.5) for all A and B . Since we only need one nonzero function $z(x)$, we can simply take $z(x) = xe^{-x}$. Therefore $v(x) = e^xz(x) = x$. Hence the general solution of the given nonhomogeneous equation is

$$y = c_1e^x + c_2x + 1,$$

where c_1 and c_2 are arbitrary constants. ▲

Suppose now that u satisfies **Condition B**. Then $2u' + a_1(x)u = 0$ and so

$$u' = -\frac{a_1}{2}u \quad \text{so that} \quad u(x) = e^{-\frac{1}{2} \int a_1(x) dx}.$$

Moreover,

$$2u'' = -a_1u' - a_1'u = \left(\frac{a_1^2}{2} - a_1'\right)u. \quad (3.6.6)$$

Then

$$\begin{aligned} u'' + a_1u' + a_0u &= \left(\frac{a_1^2}{4} - \frac{1}{2}a_1'\right)u - \frac{a_1^2}{2}u + a_0u \\ &= \left(a_0 - \frac{a_1^2}{4} - \frac{a_1'}{2}\right)u. \end{aligned}$$

Substituting these into (3.6.2) gives

$$w'' + \left(a_0 - \frac{a_1^2}{4} - \frac{a_1'}{2}\right)w = g(x), \quad (3.6.7)$$

where $g(x) = \frac{f(x)}{u(x)}$.

The differential equation (3.6.7) is said to be the **normal form** of (3.6.1). The coefficient of w in (3.6.7) is called the **invariant** of (3.6.1) and is denoted by $I_y(x)$. That is,

$$I_y(x) = a_0(x) - \frac{a_1^2(x)}{4} - \frac{a_1'(x)}{2}.$$

We observe that (3.6.7) turns into a differential equation with constant coefficients if and only if $I_y(x)$ is constant. Consequently, (3.6.7) can be readily solved in this case. When $I_w(x)$ is a constant function,

$$y = u(x)w(x) = w(x) e^{-\frac{1}{2} \int a_1(x) dx}$$

is the general solution of (3.6.1) if and only if w is the general solution of (3.6.7).

We discuss some examples below to show how the above procedure works.

Example 3.6.2 Find the general solution of the differential equation

$$y'' + 4y' + 4y = \sin x$$

by using its normal form.

Solution. Comparing the differential equation with (3.6.1), we have $a_0 = a_1 = 4$. So the invariant of the differential equation is

$$I_y(x) = 4 - \frac{4^2}{4} - 0 = 0.$$

With the change of variable $y = u(x)w(x)$ with $u(x) = e^{-\frac{1}{2} \int 4 dx} = e^{-2x}$, the normal form of the equation is

$$w''(x) = \frac{\sin x}{u(x)} = e^{2x} \sin x.$$

Upon integration, we get

$$w(x) = c_1 + c_2x - \frac{1}{25}e^{2x}(4 \cos x - 3 \sin x).$$

Therefore the general solution is

$$y = e^{-2x}w(x) = c_1e^{-2x} + c_2xe^{-2x} - \frac{1}{25}(4 \cos x - 3 \sin x),$$

which is the same result we would have obtained by the method of undermined coefficients. ▲

The above procedure is particularly helpful for solving linear equations with variable coefficients, as the following examples illustrate.

Example 3.6.3 Find the general solution of

$$x^2 y'' - 2xy' + (k^2 x^2 + 2)y = x^3 \sin x,$$

where $k \neq 1$ is a positive constant, by using its normal form.

Solution. Comparing the given equation with (3.6.1), we have

$$a_0(x) = k^2 + \frac{2}{x^2} \quad \text{and} \quad a_1(x) = -\frac{2}{x}.$$

So the invariant of the differential equation is

$$I_y(x) = a_0(x) - \frac{a_1^2(x)}{4} - \frac{a_1'(x)}{2} = k^2 + \frac{2}{x^2} - \frac{1}{x^2} - \frac{1}{x^2} = k^2.$$

Since

$$u(x) = e^{-\frac{1}{2} \int (-\frac{2}{x}) dx} = e^{\ln|x|} = |x|,$$

we take $u(x) = x$. With the change of variable $y = u(x)w(x)$, the differential equation is transformed into its normal form

$$w'' + k^2 w = \frac{x \sin x}{x} = \sin x,$$

whose general solution, by the method of undetermined coefficients or variation of parameters, is

$$w = c_1 \cos(kx) + c_2 \sin(kx) + \frac{\sin x}{k^2 - 1},$$

where c_1 and c_2 are arbitrary constants. Hence the general solution of the given differential equation is

$$y = xw(x) = c_1 x \cos(kx) + c_2 x \sin(kx) + \frac{x \sin x}{k^2 - 1},$$

where c_1 and c_2 are arbitrary constants. ▲

3.6.2 Change of the Independent Variable

By changing the independent variable in the equation

$$y'' + a_1(x)y' + a_0(x)y = f(x) \tag{3.6.8}$$

from x to t via a one-to-one function $t = h(x)$ which is differentiable with the differentiable inverse $x = h^{-1}(t)$, we may be able to transform (3.6.8) into a differential equation which may be solved using the methods that are previously discussed. To see how such a change of the independent variable works, we note

$$\frac{dt}{dx} = h'(x)$$

and compute, by the chain rule, the derivatives

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt} h'(x) \tag{3.6.9}$$

and

$$\frac{d^2 y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dt} h'(x) \right) \frac{dt}{dx} = \frac{d^2 y}{dt^2} (h'(x))^2 + h''(x) \frac{dy}{dt}. \tag{3.6.10}$$

Substituting these into (3.6.8) gives

$$\frac{d^2 y}{dt^2} (h'(x))^2 + (h''(x) + a_1(x)h'(x)) \frac{dy}{dt} + a_0(x)y = f(x), \quad (3.6.11)$$

where $x = h^{-1}(t)$. The substitution $t = h(x)$ is beneficial only if it transforms (3.6.11) into an equation solvable by a known method.

To illustrate the effectiveness of changing the independent variable, we present two examples. The first involves a second order **Cauchy-Euler** equation, and the second is the **Legendre equation of order n** . Here, the term *order* refers to something different from the order of the differential equation itself. This distinction in the use of the same terminology is important to keep in mind.

Example 3.6.4 Find the general solution of the Cauchy-Euler equation

$$x^2 y'' + 3xy' + y = x \quad (3.6.12)$$

by solving the transformed equation that is obtained from the change of variable $t = \ln x$.

Solution. Using $t = h(x) = \ln x$ in (3.6.9) and (3.6.10), we find

$$\frac{dy}{dx} = \frac{dy}{dt} h'(x) = \frac{1}{x} \frac{dy}{dt}$$

and

$$\frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} (h'(x))^2 + h''(x) \frac{dy}{dt} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right).$$

Substituting these into (3.6.12) gives

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + y = e^t.$$

The general solution of this equation is

$$y = c_1 e^{-t} + c_2 t e^{-t} + \frac{1}{4} e^t,$$

where c_1 and c_2 are arbitrary constants. Since $t = \ln x$, it follows that the general solution of (3.6.12) is

$$y = \frac{c_1}{x} + \frac{c_2 e^x}{x} + \frac{x}{4},$$

where c_1 and c_2 are arbitrary constants. ▲

Example 3.6.5 Legendre Equation. Transform the Legendre equation

$$y'' + (\cot x) y' + n(n+1)y = 0 \quad (3.6.13)$$

of order n (a real number) using the change of variable $t = \cos x$.

Solution. Using $t = h(x) = \cos x$ in (3.6.9) and (3.6.10), we find

$$\frac{dy}{dx} = \frac{dy}{dt} h'(x) = -\sin x \frac{dy}{dt}$$

and

$$\frac{d^2 y}{dx^2} = (h'(x))^2 \frac{d^2 y}{dt^2} + h''(x) \frac{dy}{dt} = \sin^2 x \frac{d^2 y}{dt^2} - \cos x \frac{dy}{dt}.$$

Substituting these into (3.6.13) gives

$$\sin^2 x \frac{d^2 y}{dt^2} - \cos x \frac{dy}{dt} + \cot x (-\sin x) \frac{dy}{dt} + n(n+1)y = 0,$$

which simplifies to

$$\sin^2 x \frac{d^2 y}{dt^2} - 2 \cos x \frac{dy}{dt} + n(n+1)y = 0.$$

Using $t = \cos x$ gives the standard form of the Legendre equation

$$(1-t^2) \frac{d^2 y}{dt^2} - 2t \frac{dy}{dt} + n(n+1)y = 0. \quad (3.6.14)$$



Remark 3.6.6

- i. Since the invariant

$$I_y(t) = \frac{n(n+1)}{1-t^2} - \frac{1}{4} \left(\frac{-2t}{1-t^2} \right)^2 - \frac{1}{2} \left(\frac{-2t}{1-t^2} \right) = \frac{n(n+1)}{1-t^2} + \frac{1}{(1-t^2)^2}$$

of (3.6.12) is not a constant, the transformed equation in w by the change of dependent variable $y(t) = u(t)w(t)$ with

$$u(t) = e^{-\frac{1}{2} \int \left(\frac{-2t}{1-t^2} \right) dt}$$

does not have constant coefficients. Likewise, the invariant $I_y(x)$ of (3.6.13) for the change of variable $y(x) = u(x)w(x)$ with

$$u(x) = e^{-\frac{1}{2} \int \cot x dx}$$

is

$$I_y(x) = n(n+1) - \frac{1}{4} \cot^2 x + \frac{1}{2} \csc x \cot x$$

which is not a constant, and therefore the transformed equation in w does not have constant coefficients. A standard approach for solving the Legendre equation is the method of power series solutions, which will be discussed in Section 4.1.

- ii. The Legendre equation (3.6.13) lies in an important class known as **Sturm-Liouville problems**, which are parameter dependent problems of the form

$$(a_1(x)y')' + (\lambda w(x) - a_0(x))y = 0, \quad (3.6.15)$$

λ being a parameter (called an eigenvalue) and p, q, r known functions of x . In fact, (3.6.13) can be written as

$$((1-x^2)y')' + n(n+1)y = 0,$$

which is of the form (3.6.15) with $a_1(x) = 1-x^2$, $w(x) = 1$, $a_0(x) = 0$, and $\lambda = n(n+1)$. A detailed discussion of Sturm-Liouville problems, however, lies beyond the scope of this book.

We now discuss a useful change of dependent variable for the equation

$$y'' + a_1(x)y' + a_0(x)y = f(x) \quad (3.6.16)$$

given by

$$t = h(x) = \int \sqrt{a_0(x)} dx.$$

It is clear that

$$h'(x) = \sqrt{a_0(x)} \quad \text{and} \quad h''(x) = \frac{a'_0(x)}{2\sqrt{a_0(x)}}.$$

Using this change of variable in (3.6.11), we get

$$a_0(x) \frac{d^2 y}{dt^2} + \left(\frac{a'_0(x) + 2a_0(x)a_1(x)}{2\sqrt{a_0(x)}} \right) \frac{dy}{dt} + a_0(x)y = 0,$$

where $x = h^{-1}(t)$. The above equation can be rewritten as

$$\frac{d^2 y}{dt^2} + \left(\frac{a'_0(x) + 2a_0(x)a_1(x)}{2(a_0(x))^{3/2}} \right) \frac{dy}{dt} + y = 0. \quad (3.6.17)$$

The equation (3.6.17) can be readily solved when

$$\frac{a'_0(x) + 2a_0(x)a_1(x)}{2(a_0(x))^{3/2}} = \text{a constant}.$$

The arguments made above make sense only for those $a_0(x)$ in (3.6.16) for which these considerations are meaningful.

3.6.3 Exercises

- Find the general solution of each differential equation below. A solution to the associated homogeneous equation is provided.

- i. $x^2 y'' + xy' - 4y = x; \quad u(x) = x^2$

- ii. $x^3 y'' + x^2 y' - 4xy = 1 + x; \quad u(x) = x^2$

- iii. $x^4 y'' + x^3 y' - 4x^2 y = 1; \quad u(x) = x^2$

- iv. $(x^2 - 1)y'' - xy' + y = 1; \quad u(x) = x$

- v. $(\tan^2 x)y'' - (2 \tan x)y' + (\sec^2 x + 1)y = \sin^2 x \tan x; \quad u(x) = \sin x$

- Using the techniques discussed in this section, find the general solution of each differential equation below.

- i. $y'' + 6y' + 9y = e^{-3x}$

- ii. $y'' + 6y' + 9y = xe^{-3x}$

- iii. $y'' + 6y' + 9y = x^2 e^{-3x}$

- Find the general solution of

$$xy'' - 2(x-1)y' + 2(x-1)y = e^x$$

by using its normal form.

- Find the normal form of each differential equation below.

- i. $y'' - 2xy' + 2ny = 0$, n being a constant

- ii. $xy'' + (1-c)y' + y = 0$, c being a constant

- iii. $y'' + xy' + (x^2 - n^2)y = 0$, n being a constant
5. Find the general solution of each equation below by using the change of variable $t = h(x) = \int \sqrt{a_0(x)} dx$ that results in (3.6.11).

i. $y'' + (\tan x)y' + (\cos^2 x)y = 0$

ii. $x^4 y'' + x^2(2x - 3)y' + 2u = 0$

iii. $2xy'' + (5x^2 - 2)y' + 2x^3y = 0$

Chapter 4

Method of Series Solutions

Intro goes here.

4.1 Power Series Solutions

Text of section. Testing

Chapter 5

The Laplace Transform and Applications

In the previous chapters, we developed methods for solving initial value problems for ordinary differential equations with continuous forcing functions. In this chapter, we will extend our methods to solve initial value problems with discontinuous forcing functions and impulsive forcing functions modeled by the Dirac delta function.

5.1 The Laplace Transform

Consider a forced oscillator discussed in [Section 3.5](#) and described by

$$y''(t) + \omega^2 y(t) = f(t), \quad y(0) = 0, \quad y'(0) = 0, \quad (5.1.1)$$

where $f(t)$ is an impulsive force. At $t = 0$, the mass is at rest at its equilibrium position. At time $t = a$, the mass experiences a sharp, brief downward impact, such as a hammer strike, modeled by an the impulsive force $f(t)$. An impulsive force of this nature is commonly modeled by using an idealized mathematical object, called the **Dirac delta** “function” denoted by $\delta(t - a)$, which is characterized by the following properties:

- i. $\delta(t - a) = 0$ for all $t \neq a$; and
- ii. $\int_{-\infty}^{\infty} \delta(t - a) g(t) dt = g(a)$ for any function g continuous on an open interval containing a .

We note that when $g(t) = 1$ in (ii), we obtain $\int_{-\infty}^{\infty} \delta(t - a) dt = 1$. Based on the properties (i) and (ii) characterizing $\delta(t - a)$, it is important to recognize that $\delta(t - a)$ is not a function in the usual sense; rather it is a typical example of *generalized functions* or *distributions* which were first developed rigorously by Sergei Sobolev in 1936 and later in 1950s by Laurent Schwartz who is known to have given the most definitive and systematic development of the theory of distributions. For an elegant introduction to distribution theory with application, the reader is referred to Zemanian¹.

The Dirac delta function is particularly useful for modeling phenomena involving instantaneous or localized effects, such as a lightning strike, a point

¹A.H. Zemanian, *Distribution theory and transform analysis*, second edition, Dover, New York, 1987; MR0918977

heat explosion, or the sudden burst of dye at a specific location. Because of the properties (i) and (ii), the Dirac delta function $\delta(t - a)$ is said to be concentrated at $t = a$.

Mathematically, $\delta(t - a)$ can be interpreted as the limit of a sequence of approximating impulse functions. One such approximation is given by

$$d_{a,\varepsilon}(t) = \begin{cases} \frac{1}{\varepsilon} & \text{if } a \leq t \leq a + \varepsilon \\ 0 & \text{otherwise,} \end{cases}$$

which represents a pulse of height $1/\varepsilon$ and width $\varepsilon > 0$, centered at $t = a$.

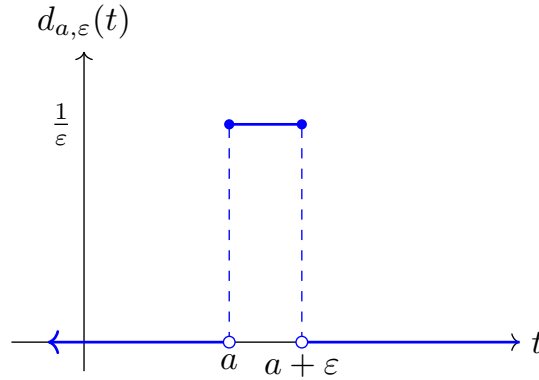


Figure 5.1.1 The graph of $d_{a,\varepsilon}(t)$.

As $\varepsilon \rightarrow 0$, this pulse becomes increasingly narrow and tall, concentrating its effect at $t = a$, i.e.,

$$\lim_{\varepsilon \rightarrow 0} d_{a,\varepsilon}(t) = \begin{cases} \infty & \text{if } t = a \\ 0 & \text{otherwise,} \end{cases} \quad (5.1.2)$$

while maintaining a total area of 1—mirroring the property (ii) of $\delta(t - a)$, i.e.,

$$\int_{-\infty}^{\infty} d_{a,\varepsilon}(t) dt = \frac{1}{\varepsilon} \varepsilon = 1,$$

so that

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} d_{a,\varepsilon}(t) dt = 1. \quad (5.1.3)$$

Furthermore, if g is continuous from right at $t = a$, then

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} d_{a,\varepsilon}(t) g(t) dt = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_a^{a+\varepsilon} g(t) dt = \lim_{\varepsilon \rightarrow 0} g(a + \varepsilon) = g(a) \quad (5.1.4)$$

by using l'Hôpital's rule and the continuity of g at $t = a$ from right.

By formally interchanging the limit and the integral in both (5.1.3) and (5.1.4), an operation which needs justification in integration theories, we obtain

$$\int_{-\infty}^{\infty} \lim_{\varepsilon \rightarrow 0} d_{a,\varepsilon}(t) dt = 1 \quad (5.1.5)$$

and

$$\int_{-\infty}^{\infty} \lim_{\varepsilon \rightarrow 0} d_{a,\varepsilon}(t) g(t) dt = g(a). \quad (5.1.6)$$

We observe in (5.1.2) and (5.1.4) that $\lim_{\varepsilon \rightarrow 0} d_{a,\varepsilon}(t)$ exhibits the properties (i) and (ii) that characterize $\delta(t-a)$, and therefore $\delta(t-a)$ can be interpreted as the limit of a sequence of functions that approximate an impulsive behavior. We recognize that $\lim_{\varepsilon \rightarrow 0} d_{a,\varepsilon}(t)$ also does not define a function on $(-\infty, \infty)$ in the classical sense because its value at $t = a$ is ∞ . Instead, the limit is understood as a generalized function, or a distribution, because of the way it operates primarily through its action on other functions within integrals, as described by the property ii., which is motivated by (5.1.6).

By using the property ii., we find that for all s in $(-\infty, \infty)$, we have

$$\int_{-\infty}^{\infty} \delta(t-a) e^{-st} dt = e^{-as} \quad (5.1.7)$$

for all real numbers s . From this point onward, we will focus on the case where $a \geq 0$ and $t \geq 0$. Under this assumption, (5.1.7) reduces to

$$\int_0^{\infty} \delta(t-a) e^{-st} dt = e^{-as} \quad (5.1.8)$$

for all real values of s . This integral is known as the *Laplace transform* (see Definition 5.1.2) of the Dirac delta function $\delta(t-a)$, denoted by

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}. \quad (5.1.9)$$

In particular, when $a = 0$ we have

$$\mathcal{L}\{\delta(t)\} = 1. \quad (5.1.10)$$

The main objective of this chapter is to develop a method for solving initial value problems such as (5.1.1) in which $f(t)$ is either discontinuous or involves the Dirac delta functions. The methods of undetermined coefficients and variation of parameters are not relevant to (5.1.1) with $f(t) = \delta(t-a)$ because the calculus of distributions is beyond the scope of this book. In particular, we do not have a meaning of y'' in this case.

Being motivated from (5.1.8) with the Dirac delta function $\delta(t-a)$ in the integrand, we first define the *Laplace transform* of classical functions defined on $[0, \infty)$.

Definition 5.1.2 (Laplace transform). Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function. The **Laplace transform** of f is a function $F(s)$, denoted commonly by $\mathcal{L}\{f(t)\}$, defined by

$$F(s) := \int_0^{\infty} f(t) e^{-st} dt$$

for those s for which the integral converges. ◆

The symbol $\mathcal{L}$ represents a transformation that maps an input function f of the variable t (usually time) to an output function F of the variable s (usually involving frequency when s is a complex number). This action of the transformation is depicted in the figure below.

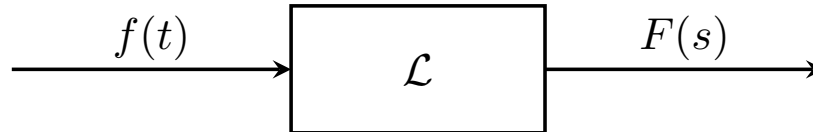


Figure 5.1.3 The Laplace transform $\mathcal{L}$ as an operator mapping $f(t)$ to $F(s)$.

The variable s in the definition of the Laplace transform can, in general, be a complex variable; however, in the current chapter, we will restrict s to a real variable.

The Laplace transform is a powerful tool for solving initial value problems for ordinary differential equations, especially those with nonhomogeneous terms that are piecewise continuous or involve the Dirac delta function. The key idea is to apply the Laplace transform to both sides of the differential equation, which converts the problem from the time domain (involving t) to the complex frequency domain (involving s). This transformation turns differentiation into algebraic manipulation, making it easier to solve for the transformed function $F(s)$. Once we have $F(s)$, we can then find the original function $f(t)$ by applying the inverse Laplace transform $\mathcal{L}^{-1}$ to $F(s)$. Before we begin using Laplace transforms to solve initial value problems, we compute the Laplace transforms of several common functions that appeared in the previous chapters, such as polynomial functions, exponential functions, trigonometric functions, and some combinations of these functions. The Laplace transform of the Dirac delta function is encountered in (5.1.8) and will be used in solving initial value problems with impulsive forcing.

By using Definition 5.1.2, we compute the Laplace transforms of some common functions in the examples below.

Example 5.1.4 Compute $\mathcal{L}\{e^{at}\}$ for a fixed real number a .

Solution. We have

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^\infty e^{at} e^{-st} dt = \lim_{h \rightarrow \infty} \int_0^h e^{(a-s)t} dt \\ &= \lim_{h \rightarrow \infty} \left[\frac{e^{(a-s)t}}{a-s} \right]_0^h \\ &= \lim_{h \rightarrow \infty} \frac{e^{(a-s)h}}{a-s} - \frac{1}{a-s} \\ &= \frac{1}{s-a}\end{aligned}$$

for $s > a$. Thus, we have

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \text{ for } s > a.$$



Example 5.1.5 Compute $\mathcal{L}\{1\}$.

Solution. We have

$$\begin{aligned}\mathcal{L}\{1\} &= \int_0^\infty e^{-st} dt \\ &= \lim_{h \rightarrow \infty} \left[\frac{e^{-st}}{-s} \right]_0^h \\ &= \lim_{h \rightarrow \infty} \left(\frac{e^{-sh}}{-s} - \frac{1}{-s} \right) \\ &= \frac{1}{s}\end{aligned}$$

for $s > 0$. Thus, we have

$$\boxed{\mathcal{L}\{1\} = \frac{1}{s} \text{ for } s > 0.}$$

▲

Example 5.1.6 Compute $\mathcal{L}\{t\}$.

Solution. We have

$$\begin{aligned} \mathcal{L}\{t\} &= \int_0^\infty te^{-st} dt \\ &= \lim_{h \rightarrow \infty} \int_0^h te^{-st} dt \\ &= \lim_{h \rightarrow \infty} \left[-\frac{te^{-st}}{s} + \frac{1}{s} \int_0^h e^{-st} dt \right] \\ &= \frac{1}{s^2} \end{aligned}$$

for $s > 0$. Thus, we have

$$\boxed{\mathcal{L}\{t\} = \frac{1}{s^2} \text{ for } s > 0.}$$

▲

Example 5.1.7 Compute $\mathcal{L}\{t^n\}$ for a nonnegative integer n .

Solution. We have

$$\begin{aligned} \mathcal{L}\{t^n\} &= \int_0^\infty t^n e^{-st} dt \\ &= \lim_{h \rightarrow \infty} \int_0^h t^n e^{-st} dt \\ &= \lim_{h \rightarrow \infty} \left[-\frac{t^n e^{-st}}{s} + \frac{n}{s} \int_0^h t^{n-1} e^{-st} dt \right] \\ &= \frac{n}{s} \lim_{h \rightarrow \infty} \int_0^h t^{n-1} e^{-st} dt \\ &= \frac{n}{s} \mathcal{L}\{t^{n-1}\}. \end{aligned}$$

Using this recursive relation, we obtain

$$\begin{aligned} \mathcal{L}\{t^n\} &= \frac{n}{s} \left(\frac{n-1}{s} \right) \mathcal{L}\{t^{n-2}\} \\ &= \frac{n}{s} \left(\frac{n-1}{s} \right) \left(\frac{n-2}{s} \right) \mathcal{L}\{t^{n-3}\} \\ &\vdots \\ &= \frac{n}{s} \left(\frac{n-1}{s} \right) \left(\frac{n-2}{s} \right) \cdots \frac{3}{s} \frac{2}{s} \mathcal{L}\{t\} \\ &= \frac{n}{s} \left(\frac{n-1}{s} \right) \left(\frac{n-2}{s} \right) \cdots \frac{3}{s} \frac{2}{s} \frac{1}{s^2} \\ &= \frac{n!}{s^{n+1}} \end{aligned}$$

for $s > 0$. Thus, we have

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad \text{for all } s > 0.$$

▲

Example 5.1.8 Compute $\mathcal{L}\{\sin(\omega t)\}$ for a positive real number ω .

Solution. By using integration by parts, we have

$$\int e^{-st} \sin(\omega t) dt = -\frac{e^{-st}}{s^2 + \omega^2} [s \sin(\omega t) + \omega \cos(\omega t)] + C,$$

where C is an arbitrary constant. Then

$$\int_0^\infty e^{-st} \sin(\omega t) dt = -\left[\frac{e^{-st}}{s^2 + \omega^2} (s \sin(\omega t) + \omega \cos(\omega t)) \right]_0^\infty.$$

Since $e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ and $s \sin(\omega t) + \omega \cos(\omega t)$ is bounded, we have

$$\lim_{t \rightarrow \infty} e^{-st} (s \sin(\omega t) + \omega \cos(\omega t)) = 0$$

for $s > 0$. The improper integral converges only for $s > 0$ when $\omega \neq 0$. Therefore, we have

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2} \quad \text{for all } s > 0.$$

▲

Example 5.1.9 Compute $\mathcal{L}\{\cos(\omega t)\}$ for a positive real number ω .

Solution. By using integration by parts, we have

$$\int e^{-st} \cos(\omega t) dt = -\frac{e^{-st}}{s^2 + \omega^2} [s \cos(\omega t) - \omega \sin(\omega t)] + C,$$

where C is an arbitrary constant. Then

$$\int_0^\infty e^{-st} \cos(\omega t) dt = -\left[\frac{e^{-st}}{s^2 + \omega^2} (s \cos(\omega t) - \omega \sin(\omega t)) \right]_0^\infty.$$

Since $e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ and $s \cos(\omega t) - \omega \sin(\omega t)$ is bounded, we have

$$\lim_{t \rightarrow \infty} e^{-st} (s \cos(\omega t) - \omega \sin(\omega t)) = 0$$

for $s > 0$. The improper integral converges only for $s > 0$ when $\omega \neq 0$. Therefore, we have

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2} \quad \text{for all } s > 0.$$

▲

Example 5.1.10 Shifting on the s -axis. Compute $\mathcal{L}\{e^{at}f(t)\}$ for each fixed real number a .

Solution. By using [Definition 5.1.2](#), we have

$$\mathcal{L}\{e^{at}f(t)\} = \int_0^\infty e^{at}f(t)e^{-st}dt = \int_0^\infty f(t)e^{-(s-a)t}dt = F(s-a).$$

Therefore, we have

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a).$$

A simple but useful example of a discontinuous function is the **unit step function** $\mathcal{U}$, also known as the **Heaviside** function (named in honor of the English mathematician, engineer, and physicist Oliver Heaviside (1850–1925)). It is defined on $(-\infty, \infty)$ by

$$\mathcal{U}(t) = \begin{cases} 0 & \text{if } t < 0, \\ 1 & \text{if } t \geq 0. \end{cases}$$

The unit step function with the jump discontinuity at $t = a$ is given by

$$\mathcal{U}(t-a) = \begin{cases} 0 & \text{if } t < a, \\ 1 & \text{if } t \geq a, \end{cases}.$$

The graph of the unit step function $\mathcal{U}(t-2)$ is shown in the figure below.

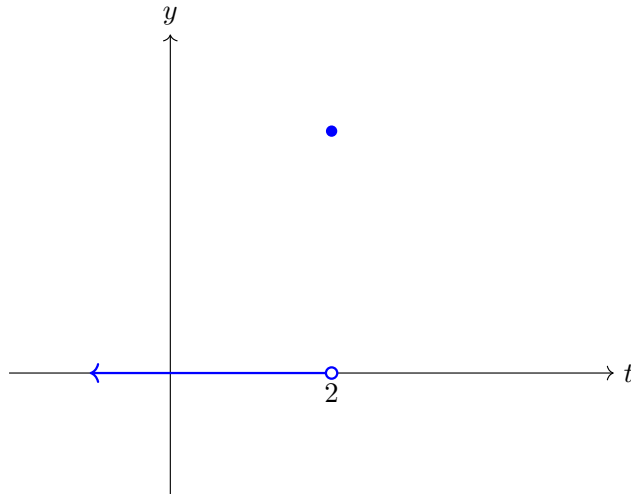


Figure 5.1.11 The graph of $\mathcal{U}(t-2)$

The unit step function is used to model situations where a quantity changes abruptly at a certain point in time, such as a switch being turned on or off, or a force being applied suddenly.

Example 5.1.12 Compute $\mathcal{L}\{\mathcal{U}(t-a)\}$ for a nonnegative real number a .

Solution. By using [Definition 5.1.2](#), we have

$$\begin{aligned} \mathcal{L}\{\mathcal{U}(t-a)\} &= \int_0^{\infty} e^{-st}\mathcal{U}(t-a)dt \\ &= \int_a^{\infty} e^{-st}dt \\ &= \left[\frac{e^{-st}}{-s} \right]_a^{\infty} \\ &= \frac{e^{-as}}{s} \end{aligned}$$

for $s > 0$. Thus, we have

$$\mathcal{L}\{\mathcal{U}(t-a)\} = \frac{e^{-as}}{s} \text{ for } s > 0.$$

▲
To model situations in applications where an external force or source begins acting only after time $t = a$, we analyze the effect of shifting the source function along the t -axis.

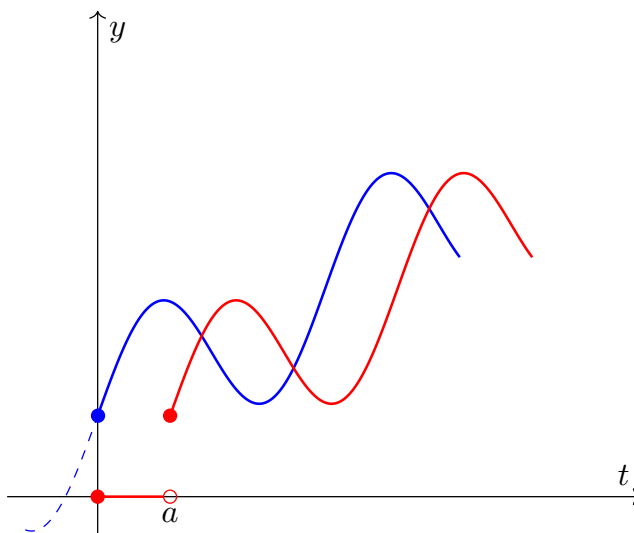
Suppose the function $g(t)$ is effective only at and after the time $t = a$. In that case, we can write

$$g(t) = \begin{cases} f(t-a) & \text{if } t \geq a, \\ 0 & \text{if } t < a, \end{cases}$$

which is obtained by shifting the graph of $f(t)$ to the right by a units and taking the zero function to the left of $t = a$. This function can be expressed in terms of the unit step function as

$$g(t) = \mathcal{U}(t-a)f(t-a),$$

whose graph is shown in the figure below, where the blue curve represents $f(t)$ and the red curve represents $g(t)$.



$$\begin{array}{ll} \text{---} & f(t) \\ \text{---} & g(t) = \mathcal{U}(t-a)f(t-a) \end{array}$$

Figure 5.1.13 Shifting on the t -axis

Example 5.1.14 Laplace transform of a shifted function. Compute $\mathcal{L}\{\mathcal{U}(t-a)f(t-a)\}$ for a nonnegative real number a .

Solution. By [Definition 5.1.2](#), we have

$$\begin{aligned} \mathcal{L}\{\mathcal{U}(t-a)f(t-a)\} &= \int_0^{\infty} e^{-st}\mathcal{U}(t-a)f(t-a) dt \\ &= \int_a^{\infty} e^{-st}f(t-a) dt \end{aligned}$$

$$\begin{aligned}
 &= e^{-as} \int_0^\infty e^{-s\tau} f(\tau) d\tau \quad (\text{letting } \tau = t - a) \\
 &= e^{-as} \mathcal{L}\{f(t)\} \\
 &= e^{-as} F(s).
 \end{aligned}$$

Thus, we have

$$\boxed{\mathcal{L}\{\mathcal{U}(t-a)f(t-a)\} = e^{-as}F(s).}$$

The linearity of the Riemann integration yields the linearity of the Laplace transform operator $\mathcal{L}$, as stated in the following theorem. ▲

Theorem 5.1.15 Linearity of the Laplace transform. *Let f and g be functions defined on $[0, \infty)$ such that their Laplace transforms exist for $s \geq s_0$ for some real number s_0 . Then, for any real numbers α and β , we have*

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha \mathcal{L}\{f(t)\} + \beta \mathcal{L}\{g(t)\}.$$

Proof. By using the definition of the Laplace transform, we have

$$\begin{aligned}
 \mathcal{L}\{\alpha f(t) + \beta g(t)\} &= \int_0^\infty e^{-st}(\alpha f(t) + \beta g(t))dt \\
 &= \alpha \int_0^\infty e^{-st}f(t)dt + \beta \int_0^\infty e^{-st}g(t)dt \\
 &= \alpha \mathcal{L}\{f(t)\} + \beta \mathcal{L}\{g(t)\}.
 \end{aligned}$$

■

5.1.1 When does the Laplace transform exist?

An immediate question one might ask is: what sort of functions $f(t)$ actually have Laplace transforms according to [Definition 5.1.2](#)? To answer this in the context of the Riemann integration, we note that $F(s) = \mathcal{L}\{f(t)\}$ is the limit of the definite integrals

$$\int_0^b e^{-st} f(t) dt$$

as $b \rightarrow \infty$ whenever it exists, and therefore $f(t)$ cannot have a vertical asymptote at any point $t = b$. To avoid any chances for vertical asymptotes, we take $f(t)$ to be **piecewise continuous** on $[0, \infty)$. A piecewise continuous function on $[0, \infty)$ is one which can have at most a finite number of jump discontinuities on each closed subinterval $[a, b]$ of $[0, \infty)$. An example of a piecewise continuous function is depicted in the figure below.

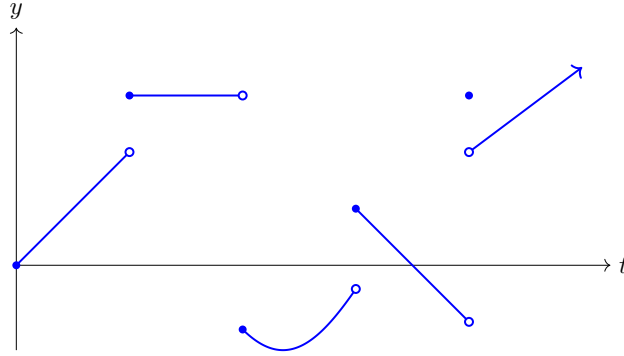


Figure 5.1.16 A piecewise continuous function

However, the condition that f be piecewise continuous on $[0, \infty)$ would not be sufficient for the existence of its Laplace transform. For instance, the function $f(t) = e^{t^2}$ does not have its Laplace transform. In fact, for each fixed number c , there exists t_0 such that $t^2 > ct$ for all $t \geq t_0$, and this implies that

$$\int_{t_0}^b e^{-st} e^{t^2} dt = \int_{t_0}^b e^{t^2 - st} dt > \int_{t_0}^b e^{(c-s)t} dt.$$

This results in

$$\int_0^\infty e^{-st} e^{t^2} dt \geq \int_{t_0}^\infty e^{-st} e^{t^2} dt \geq \int_{t_0}^\infty e^{(c-s)t} dt = \infty$$

for all $s \leq c$. This holds for every number c , and therefore $f(t) = e^{t^2}$ does not have its Laplace transform. From this example, we observe that e^{-st} must control the growth property of $f(t)$ as $t \rightarrow \infty$ for the existence of its Laplace transform. In particular, in addition to the piecewise continuity, if $f(t)$ is of an **exponential order** as $t \rightarrow \infty$, i.e., if there exist constants α , $M > 0$, and $t_0 > 0$ such that

$$|f(t)| \leq M e^{\alpha t} \quad \text{for all } t \geq t_0,$$

then, for all $b \geq t_0$, we find

$$\left| \int_0^b e^{-st} f(t) dt \right| \leq \int_0^b e^{-st} |f(t)| dt \leq \int_0^b e^{-st} M e^{\alpha t} dt = M \int_0^b e^{-(s-\alpha)t} dt.$$

Consequently,

$$F(s) = \int_0^\infty e^{-st} f(t) dt$$

exists for $s > \alpha$. We formalize all these ideas together in the following existence theorem.

Theorem 5.1.17 *Let f be a piecewise continuous function on $[0, \infty)$ and of an exponential order as $t \rightarrow \infty$. Then $F(s) = \mathcal{L}\{f(t)\}$ exists.*

Proof. By the piecewise continuity of f on $[0, \infty)$, we have

$$\int_0^b e^{-st} f(t) dt = \sum_{i=1}^n \int_{t_{i-1}}^{t_i} e^{-st} f(t) dt$$

for some finite number of points $0 = t_0 < t_1 < \cdots < t_n = s_0$. By the exponen-

tial order of f , we have

$$\left| \int_{s_0}^b e^{-st} f(t) dt \right| \leq M \int_{s_0}^b e^{-(s-\alpha)t} dt$$

for all $b \geq s_0$. Thus, the limit

$$\lim_{b \rightarrow \infty} \int_{s_0}^b e^{-st} f(t) dt$$

exists for all $s > \alpha$. This implies that

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \lim_{b \rightarrow \infty} \int_0^b e^{-st} f(t) dt \\ &= \sum_{i=1}^n \int_{t_{i-1}}^{t_i} e^{-st} f(t) dt + \lim_{b \rightarrow \infty} \int_{s_0}^b e^{-st} f(t) dt \end{aligned}$$

exists for all $s > \alpha$. ■

We are also interested in whether two different functions can have the same Laplace transform. Trivial examples of functions that have the same Laplace transform are $g(t) = \mathcal{U}(t - a)$ with $a > 0$ and $h(t)$ defined by

$$h(t) = \begin{cases} 1 & \text{if } t > a, \\ 0 & \text{if } t \leq a. \end{cases}$$

Note that $g(a) = 1$ and $h(a) = 0$, and $g(t) = h(t)$ for all $t \neq a$. Since g and h differ only at one single point in $[0, \infty)$, it follows, by the definition of the Laplace transform, that

$$G(s) = H(s) = \frac{e^{-s}}{s}$$

for $s > 0$. This implies that $\mathcal{L}$ is not a one-to-one function from functions $f(t)$ of the variable t to functions $F(s)$ of the variable s .

5.1.2 Inverse Laplace transform

As mentioned at the end of Section [Section 5.1](#), the forced oscillator (5.1.1) cannot be solved using the methods from previous chapters when f is a Dirac delta function. Even if f is only piecewise continuous, the earlier methods can become quite tedious to apply. The Laplace transform method provides a more efficient approach, and this method involves taking the Laplace transform of a linear differential equation and converting it into an algebraic equation. After solving the algebraic equation in the s -domain for the Laplace transform of the solution function, we then apply the one-to-one property of the Laplace transform to return to the solution in the t -domain. Evidently, for the method to work, we need to know when $\mathcal{L}$ is one-to-one, so that the inverse $\mathcal{L}^{-1}$ makes sense.

For continuous functions, the following theorem, whose proof is beyond the scope of this text, establishes the invertibility of the Laplace transform.

Theorem 5.1.18 Invertibility of the Laplace transform. *Let f and g be two piecewise continuous functions defined on $[0, \infty)$ such that their Laplace transforms exist for $s > s_0$ for some real number s_0 . If*

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{g(t)\},$$

then $f(t) = g(t)$ for all $t \geq 0$ except possibly at points where either f or g is

discontinuous.

By [Theorem 5.1.18](#), for continuous functions f and g satisfying $F(s) = G(s)$ for all $s > s_0$, where s_0 is a positive number, we have $f(t) = g(t)$ for all t in $[0, \infty)$. Consequently, $\mathcal{L}$ is **one-to-one on the class of continuous functions whose Laplace transforms exist**. Thus, for every continuous function f on $[0, \infty)$ with the Laplace transform of $F(s)$, we have

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

for all $t \geq 0$. Here, $\mathcal{L}^{-1}\{F(s)\}$ is called the **inverse Laplace transform** $F(s)$.

We recall from [Example 5.1.12](#) that

$$\mathcal{L}\{\mathcal{U}(t-a)\} = \frac{e^{-as}}{s}$$

for $a \geq 0$. Therefore, when we speak of

$$\mathcal{L}^{-1}\left\{\frac{e^{-as}}{s}\right\},$$

we mean a function of t that agrees with $\mathcal{U}(t-a)$ for $t \neq a$, and it can have any arbitrary value at $t = a$. In practice, we choose the function that best fits the problem at hand, but in this book, we will use

$$\mathcal{L}^{-1}\left\{\frac{e^{-as}}{s}\right\} = \mathcal{U}(t-a)$$

unless this choice is clearly inappropriate. It is understood that any other function that differs from $\mathcal{U}(t-a)$ only at $t = a$ may also be used. These considerations will also be made at the points of jump discontinuity when we take the inverse Laplace transform of the Laplace transform of a piecewise continuous functions of exponential order as $t \rightarrow \infty$.

In view of [Theorem 5.1.18](#), $\mathcal{L}^{-1}\{F(s)\}$ may represent multiple functions when $f(t)$ has jump discontinuities on $[0, \infty)$. For most functions discussed in this chapter, it won't be much of an issue since we will be working with either continuous functions or, at worse, functions that have finite number of discontinuity, with the exception of a Dirac delta distributions which has its Laplace transform encoded in the definition itself as

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}.$$

Since $\delta(t-a)$ is discontinuous only at $t = a$, we will adopt

$$\mathcal{L}^{-1}\{e^{-as}\} = \delta(t-a).$$

More rigorous arguments about such ideas pertaining to Dirac delta functions exist in an area called “distribution theory,” which is beyond the scope of this book.

It follows that the inverse Laplace transform operator $\mathcal{L}^{-1}$ is linear, and we formalize this linearity property of $\mathcal{L}^{-1}$ in the following theorem.

Theorem 5.1.19 *The inverse Laplace transform operator $\mathcal{L}^{-1}$ is linear, that is,*

$$\mathcal{L}^{-1}\{c_1 F_1(s) + c_2 F_2(s)\} = c_1 \mathcal{L}^{-1}\{F_1(s)\} + c_2 \mathcal{L}^{-1}\{F_2(s)\},$$

where c_1 and c_2 are constants and F_1 and F_2 are Laplace transforms of functions that are either piecewise continuous and of exponential order at infinity or delta functions.

Proof. By the linearity of the Laplace transform operator $\mathcal{L}$, we have

$$\begin{aligned} & \mathcal{L}\{c_1\mathcal{L}^{-1}\{F_1(s)\} + c_2\mathcal{L}^{-1}\{F_2(s)\}\} \\ &= c_1\mathcal{L}\{\mathcal{L}^{-1}\{F_1(s)\}\} + c_2\mathcal{L}\{\mathcal{L}^{-1}\{F_2(s)\}\} \\ &= c_1F_1(s) + c_2F_2(s). \end{aligned}$$

By the one-to-one property of the Laplace transform, we conclude that

$$\mathcal{L}^{-1}\{c_1F_1(s) + c_2F_2(s)\} = c_1\mathcal{L}^{-1}\{F_1(s)\} + c_2\mathcal{L}^{-1}\{F_2(s)\}.$$

■

5.1.3 Exercises

- Find the Laplace transforms of the following functions on $[0, \infty)$ by using formulas derived in [Section 5.1](#).

- $f(t) = 4 + t^5 + \sin(5t) - 7\cos(3t)$
- $f(t) = e^{-11t} + e^t \sin(3t) + e^{t-3} \mathcal{U}(t-3)$
- $f(t) = e^{\alpha t} (A \cos(\beta t) + B \sin(\beta t))$ (A , B , α and β are fixed real numbers.)
- $f(t) = e^{7t} \delta(t-3)$
- $f(t) = e^{21t} \delta(t-3)$
- $f(t) = \sum_{k=0}^{\infty} \delta(t-7k)$ (Assume that the integration and summation can be interchanged.)

- Find the Laplace transforms of the following piecewise continuous functions either by using the definition or by first expressing the functions as a linear combination of unit step functions and then using formulas derived in [Section 5.1](#).

- $f(t) = \begin{cases} 3 & \text{if } 0 \leq t < 2 \\ 0 & \text{if } t \geq 2 \end{cases}$
- $f(t) = \begin{cases} 0 & \text{if } 0 \leq t < 2 \\ 3 & \text{if } t \geq 2 \end{cases}$
- $f(t) = \begin{cases} e^{2t} & \text{if } 0 \leq t < 2 \\ 0 & \text{if } t \geq 2 \end{cases}$
- $f(t) = \begin{cases} e^{2t} \sin t & \text{if } 0 \leq t < \pi \\ 0 & \text{if } t \geq \pi \end{cases}$
- $f(t) = \begin{cases} 0 & \text{if } 0 \leq t < 2 \\ 2 & \text{if } 2 \leq t < 4 \\ 0 & \text{if } t \geq 4 < 6 \\ 4 & \text{if } 6 \leq t \end{cases}$
- $f(t) = \begin{cases} e^t & \text{if } 0 \leq t < \pi \\ \sin t & \text{if } t \geq \pi \end{cases}$

- Find the Laplace transform of each of the following functions by using formulas derived in [Section 5.1](#).

- $(t-4)^2$
- e^{3t-7}
- $\sin^2(2t)$
- $\cosh(t)$
- $\sinh(t)$

- Find the inverse Laplace transform of each of the following Laplace transforms.

- | | |
|-------------------------------|---------------------------------|
| i. $\frac{1}{s(s+1)}$ | v. $\frac{4}{s^2 - 2s + 5}$ |
| ii. $\frac{s}{(s-1)^3}$ | vi. $\frac{e^{-s}}{s+1}$ |
| iii. $\frac{1}{(s+2)(s-1)}$ | vii. $\frac{2}{(s^2+4)(s^2-4)}$ |
| iv. $\frac{2s}{(s-1)^2(s+1)}$ | viii. 5 |
| | ix. e^{-7s} |

5.2 Applications to Initial Value Problems

As point out earlier in [Section 5.1](#), we will need to convert a linear initial value problem into an algebraic equation using the Laplace transform. After solving the equation in the s -domain for the Laplace transform of the solution function, we then apply the inverse Laplace transform operator $\mathcal{L}^{-1}$ to return to the t -domain and obtain the solution of the initial value problem. For these steps, we will need the Laplace transform of derivatives.

Definition 5.2.1 A function $y(t)$ defined on $[0, \infty)$ is said to be of an exponential order as $t \rightarrow \infty$ if there exists constants α , $M > 0$ and $t_0 \geq 0$ such that

$$|y(t)| \leq Me^{\alpha t} \quad \text{for all } t \geq t_0.$$

The number α is referred to as an **exponential order** of y . ◆

Then, for $s > \alpha$, we have

$$|y(t)e^{-st}| \leq Me^{-(s-\alpha)t} \quad \text{for all } t \geq t_0,$$

and consequently,

$$\lim_{t \rightarrow \infty} y(t)e^{-st} = 0 \quad \text{for } s > \alpha.$$

5.2.1 Laplace Transforms of Derivatives

Suppose that y' is piecewise continuous on $[0, \infty)$ and of an exponential order as $t \rightarrow \infty$. Assume that $\mathcal{L}\{y(t)\}$ exists. Then, using integration by parts, we have

$$\begin{aligned} \mathcal{L}\{y'(t)\} &= \int_0^\infty e^{-st} y'(t) dt \\ &= [e^{-st} y(t)]_0^\infty + s \int_0^\infty e^{-st} y(t) dt \\ &= -y(0) + s\mathcal{L}\{y(t)\} \\ &= sY(s) - y(0) \end{aligned}$$

for $s > \alpha$. Thus,

$$\boxed{\mathcal{L}\{y'(t)\} = sY(s) - y(0).} \tag{5.2.1}$$

We extend the above idea to higher order derivatives. For instance, suppose that $y(t)$ and $y'(t)$ are continuous and of exponential order. If $y''(t)$ exists and is piecewise continuous on $[0, \infty)$, then applying the formula [\(5.2.1\)](#) first to $z'(t)$ with $z(t) = y'(t)$ and then to $y'(t)$, we obtain

$$\mathcal{L}\{y''(t)\} = \mathcal{L}\{z'(t)\}$$

$$\begin{aligned}
 &= sZ(s) - z(0) \\
 &= s\mathcal{L}\{y'(t)\} - y'(0) \\
 &= s(sY(s) - y(0)) - y'(0) \\
 &= s^2Y(s) - sy(0) - y'(0),
 \end{aligned}$$

i.e.,

$$\boxed{\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)}. \quad (5.2.2)$$

Applying these ideas repeatedly to higher order derivatives, we obtain the following theorem whose proof is omitted.

Theorem 5.2.2 (Laplace transform of derivatives). *Let $n \geq 1$ be a positive integer. If $y, y', \dots, y^{(n-1)}$ are continuous on $[0, \infty)$ and of exponential order as $t \rightarrow \infty$ and if $y^{(n)}(t)$ is piecewise continuous on $[0, \infty)$ and of exponential order as $t \rightarrow \infty$, then*

$$\mathcal{L}\{y^{(n)}(t)\} = s^n Y(s) - s^{n-1}y(0) - s^{n-2}y'(0) - \dots - y^{(n-1)}(0),$$

where $Y(s) = \mathcal{L}\{y(t)\}$.

[Theorem 5.2.2](#) presents how $y(0), y'(0), \dots, y^{(n-1)}(0)$ naturally arise, reflecting the initial conditions of an n -th order linear initial value problem and indicating the potential use of the Laplace transform to convert a linear differential equation with constant coefficients to an algebraic equation with no derivatives of the Laplace transform of the solution variable. Specifically, consider the linear initial value problem of the form:

$$\begin{cases} a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = f(t), \\ y(0) = y_0, \dots, y^{(n-1)}(0) = y_{n-1}. \end{cases} \quad (5.2.3)$$

Applying [Theorem 5.2.2](#) in conjunction with the linearity of $\mathcal{L}$, the initial value problem in [\(5.2.3\)](#) gives rise to the following algebraic equation in the s -domain:

$$\begin{aligned}
 &a_n (s^n Y(s) - s^{n-1}y(0) - s^{n-2}y'(0) - \dots - y^{(n-1)}(0)) \\
 &+ a_{n-1} (s^{n-1}Y(s) - s^{n-2}y(0) - s^{n-3}y'(0) - \dots - y^{(n-2)}(0)) \\
 &\vdots \\
 &+ a_1 (sY(s) - y(0)) + a_0 Y(s) = F(s),
 \end{aligned} \quad (5.2.4)$$

which takes the form

$$p(s)Y(s) = q(s) + F(s), \quad (5.2.5)$$

where $p(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$ and $q(s)$ is a polynomial in s obtained from all the remaining terms on the left side of the equation [\(5.2.5\)](#). It then follows that

$$Y(s) = \frac{q(s)}{p(s)} + \frac{F(s)}{p(s)}. \quad (5.2.6)$$

The solution $y(t)$ to the initial value problem [\(5.2.3\)](#) is given by

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{q(s)}{p(s)}\right\} + \mathcal{L}^{-1}\left\{\frac{F(s)}{p(s)}\right\}. \quad (5.2.7)$$

To find

$$\mathcal{L}^{-1}\left\{\frac{q(s)}{p(s)}\right\} \quad \text{and} \quad \mathcal{L}^{-1}\left\{\frac{F(s)}{p(s)}\right\},$$

we typically use partial fraction decomposition of $q(s)/p(s)$ and $F(s)/p(s)$ and then use appropriate inverse Laplace transforms term by term. For more details on the techniques of partial fraction decomposition, see [Section 6.2](#).

In principle, one should verify that the equation (5.2.7) satisfies the initial value problem (5.2.3) by verifying that $y(t)$ solves the differential equation and satisfies the initial conditions, especially important when the differential equation is defined also for $t < 0$ since the Laplace transform only makes sense to functions on $[0, \infty)$. However, this verification is unnecessary when $f(t)$ is continuous on an open interval containing $t = 0$, as the initial value problem (5.2.3) is guaranteed to have a unique local solution.

In the following examples, we apply the Laplace transform method to solve initial value problems.

Example 5.2.3 Solve the initial value problem using the Laplace transform:

$$y' - y = 2 \cos(3t), \quad y(0) = 1.$$

Answer. $y(t) = \frac{6}{5}e^t - \frac{1}{5}\cos(3t) + \frac{3}{5}\sin(3t)$ on $(-\infty, \infty)$.

Solution. Applying the Laplace transform to both sides of the equation, we have

$$\begin{aligned} \mathcal{L}\{y'\} - \mathcal{L}\{y\} &= 2\mathcal{L}\{\cos(3t)\} \\ sY(s) - y(0) - Y(s) &= \frac{2s}{s^2 + 9} \\ (s - 1)Y(s) &= 1 + \frac{2s}{s^2 + 9} \\ Y(s) &= \frac{1}{s - 1} + \frac{2s}{(s - 1)(s^2 + 9)}. \end{aligned}$$

To find the inverse Laplace transform of $2s/((s - 1)(s^2 + 9))$, we use partial fraction decomposition to write

$$\frac{2s}{(s - 1)(s^2 + 9)} = \frac{A}{s - 1} + \frac{Bs + C}{s^2 + 9}.$$

Solving for A, B, C , we have $A = 1/5, B = -1/5, C = 9/5$. Thus,

$$\begin{aligned} Y(s) &= \frac{1}{s - 1} + \frac{1/5}{s - 1} + \frac{-1/5s + 9/5}{s^2 + 9} \\ &= \frac{6/5}{s - 1} + \frac{-1/5s + 9/5}{s^2 + 9}. \end{aligned}$$

Applying the inverse Laplace transform term by term, we have

$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\left\{\frac{6/5}{s - 1}\right\} + \mathcal{L}^{-1}\left\{\frac{-1/5s + 9/5}{s^2 + 9}\right\} \\ &= \frac{6}{5}e^t - \frac{1}{5}\cos(3t) + \frac{3}{5}\sin(3t). \end{aligned}$$

▲

Example 5.2.4 Solve the initial value problem using the Laplace transform:

$$y'' - 5y' + 4y = t, \quad y(0) = 1, y'(0) = 0.$$

Answer. $y(t) = e^t - \frac{5}{16}e^{4t} + \frac{1}{4}t + \frac{5}{16}$ on $(-\infty, \infty)$.

Solution. Applying the Laplace transform to both sides of the equation, we have

$$\begin{aligned}\mathcal{L}\{y''\} - 5\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} &= \mathcal{L}\{t\} \\ (s^2Y(s) - sy(0) - y'(0)) - 5(sY(s) - y(0)) + 4Y(s) &= \frac{1}{s^2} \\ (s^2 - 5s + 4)Y(s) &= \frac{1}{s^2} + s - 5 \\ Y(s) &= \frac{1}{s^2(s-1)(s-4)} + \frac{s-5}{(s-1)(s-4)} = \frac{s^3 - 5s^2 + 1}{s^2(s-1)(s-4)}.\end{aligned}$$

Decompose into partial fractions:

$$\frac{s^3 - 5s^2 + 1}{s^2(s-1)(s-4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1} + \frac{D}{s-4}.$$

Solving for the constants gives

$$A = \frac{5}{16}, \quad B = \frac{1}{4}, \quad C = 1, \quad D = -\frac{5}{16}.$$

Hence

$$Y(s) = \frac{5}{16s} + \frac{1}{4s^2} + \frac{1}{s-1} - \frac{5}{16(s-4)}.$$

Now apply the inverse Laplace transform:

Therefore,

$$\begin{aligned}y(t) &= \mathcal{L}^{-1}\left\{\frac{5}{16s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{4s^2}\right\} \\ &\quad + \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \mathcal{L}^{-1}\left\{\frac{5}{16(s-4)}\right\} \\ &= \frac{5}{16} + \frac{1}{4}t + e^t - \frac{5}{16}e^{4t}.\end{aligned}$$

We can show that

$$y(t) = \frac{5}{16} + \frac{1}{4}t + e^t - \frac{5}{16}e^{4t} \quad \text{on } (-\infty, \infty)$$

is the solution of the initial value problem on by verifying that it satisfies the differential equation and the initial conditions. $\blacktriangle$

Example 5.2.5 Move this to a later section. Solve the initial value problem using the Laplace transform:

$$y'' + 4y' + 5y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 2.$$

Solution. Applying the Laplace transform to both sides of the equation, we have

$$\begin{aligned}\mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} &= \mathcal{L}\{e^{-t}\} \\ (s^2Y(s) - sy(0) - y'(0)) + 4(sY(s) - y(0)) + 5Y(s) &= \frac{1}{s+1} \\ (s^2 + 4s + 5)Y(s) &= \frac{1}{s+1} + 2 \\ Y(s) &= \frac{1}{(s+1)(s^2 + 4s + 5)} + \frac{2}{s^2 + 4s + 5}.\end{aligned}$$

To find the inverse Laplace transform of $1/((s+1)(s^2+4s+5))$, we use partial fraction decomposition to write

$$\frac{1}{(s+1)(s^2+4s+5)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+4s+5}.$$

Solving for A, B, C , we have $A = 1/5, B = -1/5, C = 1/5$. Thus,

$$\begin{aligned} Y(s) &= \frac{1/5}{s+1} + \frac{-1/5s+1/5}{s^2+4s+5} + \frac{2}{s^2+4s+5} \\ &= \frac{1/5}{s+1} + \frac{-1/5s+11/5}{s^2+4s+5}. \end{aligned}$$

Applying the inverse Laplace transform term by term, we have

$$\begin{aligned} y(t) &= \mathcal{L}^{-1} \left\{ \frac{1/5}{s+1} \right\} + \mathcal{L}^{-1} \left\{ \frac{-1/5s+11/5}{s^2+4s+5} \right\} \\ &= \frac{1}{5}e^{-t} - \frac{1}{5}e^{-2t}\cos(t) + \frac{9}{5}e^{-2t}\sin(t). \end{aligned}$$



The general principle for solving initial value problems by the Laplace transform method remains the same as demonstrated in the previous examples. The major effort to make is in computing the Laplace transforms of more complicated functions and finding inverse transforms of more complicated expressions. In the following sections, we continue exploring additional methods for computing Laplace transforms and solve initial value problems that involve more complicated functions.

5.2.2 Exercises

1. Solve the following initial value problems using the Laplace transform method:

- i. $y' - y = 1, \quad y(0) = 0$

- ii. $y' - 6y = e^{2t}, \quad y(0) = 1$

- iii. $y'' + 5y + 6y = e^{2t}, \quad y(0) = 0, \quad y'(0) = 0$

- iv. $y'' - 5y' + 6y = e^{2t}, \quad y(0) = 1, \quad y'(0) = 0$

- v. $y'' + 9y = e^{2t}, \quad y(0) = 0, \quad y'(0) = 1$

2. Solve the following initial value problems using the Laplace transform method:

- i. $y' + y = e^{3t}\sin(2t), \quad y(0) = 0$

- ii. $y'' - 4y' + 5y = 0, \quad y(0) = 0, \quad y'(0) = 1$

- iii. $y'' - y' = e^t \sin t, \quad y(0) = 0, \quad y'(0) = 0$

- iv. $y''' + 2y'' - y' - 2y = \sin t, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$

3. Solve the following initial value problems using the Laplace transform method:

- i. $y' + 4y = f(t)$, $y(0) = 0$, where $f(t) = \begin{cases} 1 & \text{if } 0 \leq t < 1 \\ -1 & \text{if } t \geq 1 \end{cases}$
- ii. $y' + 4y = f(t)$, $y(0) = 0$, where $f(t) = \begin{cases} \sin t & \text{if } 0 \leq t < 1 \\ 0 & \text{if } t \geq 1 \end{cases}$
- iii. $y'' - y' = \cos t \mathcal{U}(t - 2\pi)$ $y(0) = 1$, $y'(0) = 1$
- iv. $y'' + 5y' + 6y = \mathcal{U}(t - 2\pi)$, $y(0) = 0$, $y'(0) = 1$
- v. $y'' + 4y' + 3y = 1 - \mathcal{U}(t - 2) - \mathcal{U}(t - 4)$, $y(0) = 0$, $y'(0) = 0$

5.3 Derivatives and Integrals of Laplace Transforms

The evaluation of Laplace transforms and inverse Laplace transforms can often be carried out efficiently by applying differentiation or integration techniques to certain Laplace transforms.

5.3.1 Derivatives of Laplace Transforms

Let f be a piecewise continuous function on $[0, \infty)$ and of exponential order as $t \rightarrow \infty$. Using the Leibniz rule from [Theorem 6.1.2](#) in [Section 6.1](#) for differentiating under the integral sign, we have

$$\begin{aligned}
 F'(s) &= \frac{d}{ds} \int_0^\infty e^{-st} f(t) dt \\
 &= \int_0^\infty (-t) e^{-st} f(t) dt \\
 &= - \int_0^\infty e^{-st} [t f(t)] dt \\
 &= -\mathcal{L}\{t f(t)\}.
 \end{aligned}$$

Thus, we have

$$\mathcal{L}\{t f(t)\} = -F'(s).$$

In a similar fashion, we obtain

$$\mathcal{L}\{t^2 f(t)\} = (-1)^2 F''(s).$$

Repeating this procedure inductively we obtain

$$\boxed{\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)} \quad (5.3.1)$$

for all positive integers n .

Example 5.3.1 Find the Laplace transform $\mathcal{L}\{t \sin(\omega t)\}$.

Solution. We have

$$\begin{aligned}
 \mathcal{L}\{t \sin(\omega t)\} &= -\frac{d}{ds} \mathcal{L}\{\sin(\omega t)\} \\
 &= -\frac{d}{ds} \left(\frac{\omega}{s^2 + \omega^2} \right)
 \end{aligned}$$

$$= \frac{2\omega s}{(s^2 + \omega^2)^2}.$$

▲

Example 5.3.2 Find the Laplace transform $\mathcal{L}\{t^2 \sin(\omega t)\}$.

Solution. We have

$$\begin{aligned} \mathcal{L}\{t^2 \sin(\omega t)\} &= \frac{d^2}{ds^2} \mathcal{L}\{\sin(\omega t)\} \\ &= \frac{d^2}{ds^2} \left(\frac{\omega}{s^2 + \omega^2} \right) \\ &= \frac{2\omega(3s^2 - \omega^2)}{(s^2 + \omega^2)^3}. \end{aligned}$$

▲

5.3.2 Integrals of Laplace Transforms

Let us discuss about what happens when you integrate a Laplace transform. Let f be a piecewise continuous function of exponential order as $t \rightarrow \infty$. Suppose s_0 is a real number such that $F(s)$ exists for all $s > s_0$. To compute $\mathcal{L}\{f(t)/t\}$, let $g(t) = f(t)/t$ so that $tg(t) = f(t)$. Then, in view of (5.3.1), we obtain

$$G'(s) = -F(s).$$

Consequently, we have

$$G(s) = - \int_s^\infty G'(p) dp = \int_s^\infty F(p) dp.$$

Here, we have considered only those f for which the Laplace transform $G(s)$ of g satisfies the limiting property that $G(s) \rightarrow 0$ as $s \rightarrow \infty$. Thus, we have

$$\boxed{\mathcal{L}\left\{\frac{f(t)}{t}\right\}(s) = \int_s^\infty F(p) dp.} \quad (5.3.2)$$

Moreover, we can use (5.3.2) to determine the existence of the Laplace transform of $f(t)/t$. For instance, if the integral on the right-hand side of (5.3.2) diverges for all $s > s_0$, then the Laplace transform of $f(t)/t$ does not exist for any $s > s_0$.

From the definition of the Laplace transform, it follows that

$$F(0) = \int_0^\infty f(t) dt.$$

provided the improper integral on the right-hand side converges.

Example 5.3.3 Compute $\mathcal{L}\{\sin(\omega t)/t\}$ for positive values of ω .

Solution. We have

$$\begin{aligned} \mathcal{L}\left\{\frac{\sin(\omega t)}{t}\right\}(s) &= \int_s^\infty \frac{\omega}{p^2 + \omega^2} dp \\ &= \left[\tan^{-1}\left(\frac{p}{\omega}\right) \right]_s^\infty \\ &= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{\omega}\right). \end{aligned}$$

Note that the improper integral

$$\int_0^{\infty} \frac{\sin(\omega t)}{t} dt$$

converges and is equal to $\pi/2$. ▲

Example 5.3.4 Compute $\mathcal{L}\{\cos(\omega t)/t\}$ for positive values of ω .

Solution. We have

$$\mathcal{L}\{\cos(\omega t)\}(p) = \frac{p}{p^2 + \omega^2}.$$

Since

$$\begin{aligned} \int_s^{\infty} \frac{p}{p^2 + \omega^2} dp \\ &= \left[\frac{1}{2} \ln(p^2 + \omega^2) \right]_s^{\infty} \\ &= \infty, \end{aligned}$$

it follows that the improper integral

$$\int_0^{\infty} \frac{\cos(\omega t)}{t} dt$$

diverges. ▲

Example 5.3.5 Compute $\mathcal{L}\left\{\frac{e^t - 1}{t}\right\}$.

Solution. Since

$$\mathcal{L}\{e^t - 1\}(p) = \frac{1}{p - 1} - \frac{1}{p},$$

we have

$$\begin{aligned} \mathcal{L}\left\{\frac{e^t - 1}{t}\right\}(s) &= \int_s^{\infty} \left(\frac{1}{p - 1} - \frac{1}{p} \right) dp \\ &= \left[\ln\left(\frac{p - 1}{p}\right) \right]_s^{\infty} \\ &= \ln\left(\frac{s}{s - 1}\right). \end{aligned}$$

for $s > 1$. ▲

5.3.3 Exercises

1. Find the Laplace transform of each of the following functions:

i. te^{-7t}

v. $te^{-3t} \cos(2t)$

ii. $t^2 e^{-7t}$

vi. $t^2 e^{-3t} \cos(2t)$

iii. $t \cos(2t)$

iv. $t^2 \cos(2t)$

vii. $te^{3t} \sin(5t)$

2. Compute $\mathcal{L}\left\{\frac{e^{-t} - 1}{t}\right\}$

3. Determine whether $\frac{e^t - 2}{t}$ has its Laplace transform.

Hint. Use (5.3.2). to determine whether the Laplace transform of $(e^t -$

2)/ t exists.

4. Solve the following initial value problems using the Laplace transform method.

(a) $y' + y = t \cos t, \quad y(0) = 0$

(b) $y' - y = te^t \cos t, \quad y(0) = 0$

(c) $y'' + 4y = \cos(2t), \quad y(0) = 2, y'(0) = 1$

(d) $y'' + 4y = f(t), \quad y(0) = 2, y'(0) = 1, \text{ with } f(t) = \begin{cases} \cos(2t) & \text{if } t < 2\pi, \\ 0 & \text{if } t \geq 2\pi. \end{cases}$

5.4 Laplace Transforms of Periodic Functions

In [Section 5.1](#), we computed the Laplace transforms of the periodic functions $\sin(\omega t)$ and $\cos(\omega t)$ that are continuous. In applications, we also have to address periodic functions that are periodic but piecewise continuous on $[0, \infty)$.

5.4.1 Periodic Functions

Definition 5.4.1 A function f is said to be *periodic* on $(-\infty, \infty)$ if there exists a number $T > 0$ such that

$$f(t) = f(t + T) \quad \text{for all } t \in (-\infty, \infty).$$

The smallest positive number T for which this property holds is called the *fundamental period* (or simply the *period*) of f . ♦

Graphically, a periodic function f repeats the portion of its graph restricted to $[0, T]$ across subsequent intervals of the same length T .

A piecewise continuous periodic function $f(t)$ with a period T has the following property:

$$f(t) = f(t + T) = f(t + 2T) = \cdots = f(t + nT)$$

for all positive integers n . This property is visualized in [Figure 5.4.2](#) below.

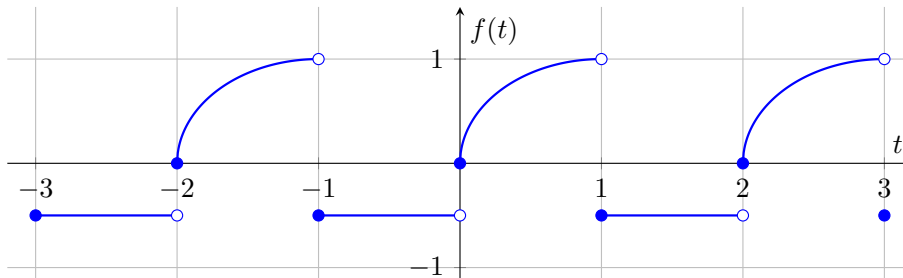


Figure 5.4.2 The graph of $f(t)$.

To compute $\mathcal{L}\{f(t)\}$, we express it as an infinite series of integrals as follows:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^{\infty} f(t)e^{-st} dt \\ &= \int_0^T f(t)e^{-st} dt + \int_T^{2T} f(t)e^{-st} dt + \int_{2T}^{3T} f(t)e^{-st} dt + \cdots \end{aligned}$$

$$\begin{aligned}
 &= \int_0^T f(t)e^{-st} dt + \int_0^T f(t+T)e^{-s(t+T)} dt \\
 &\quad + \int_0^T f(t+2T)e^{-s(t+2T)} dt + \cdots + \int_0^T f(t+nT)e^{-s(t+nT)} dt + \cdots \\
 &= \int_0^T f(t)e^{-st} dt \left(1 + e^{-sT} + e^{-2sT} + \cdots + e^{-nsT} + \cdots \right) \\
 &= \int_0^T f(t)e^{-st} dt \sum_{n=0}^{\infty} e^{-nsT}.
 \end{aligned}$$

Since the geometric series $\sum_{n=0}^{\infty} e^{-nsT}$ converges for $s > 0$ to $\frac{1}{1 - e^{-sT}}$, we obtain

$$\boxed{\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T f(t)e^{-st} dt.} \quad (5.4.1)$$

Example 5.4.3 Let us revisit $\mathcal{L}\{\sin(\omega t)\}$. ▲

Using (5.4.1) with $T = 2\pi/\omega$, we obtain

$$\mathcal{L}\{\sin(\omega t)\} = \frac{1}{1 - e^{-2\pi s/\omega}} \int_0^{2\pi/\omega} \sin(\omega t)e^{-st} dt.$$

We compute

$$\int_0^{2\pi/\omega} \sin(\omega t)e^{-st} dt = \frac{\omega}{s^2 + \omega^2} \left(1 - e^{-2\pi s/\omega} \right).$$

Hence

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2},$$

as expected.

The example above suggests that the direct computation of the Laplace transform of a periodic function may be more convenient.

The example below concerns a periodic function acting as the source term of an initial value problem.

Example 5.4.4 Consider the initial value problem $y' + y = f(t)$, $y(0) = 0$, where $f(t)$ is given by

$$f(t) = \begin{cases} 1 & \text{if } 0 \leq t \leq 1, \\ 0 & \text{if } 1 < t < 2, \end{cases}$$

and $f(t) = f(t+2)$ for all t in $(-\infty, \infty)$. ▲

The graph of f on $[0, \infty)$ is shown below.

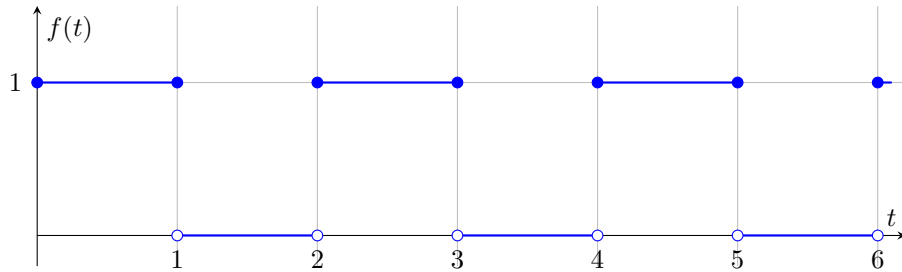


Figure 5.4.5 The graph of $f(t)$.

Then

$$\begin{aligned}
 F(s) &= \frac{1}{1 - e^{-2s}} \int_0^2 f(t) e^{-st} dt \\
 &= \frac{1}{1 - e^{-2s}} \int_0^1 e^{-st} dt \\
 &= \frac{1}{1 - e^{-2s}} \left(\frac{e^{-s} - 1}{-s} \right) \\
 &= \frac{1}{s(1 + e^{-s})}.
 \end{aligned}$$

Taking the Laplace transform of the differential equation $y' + y = f(t)$, we get

$$sY(s) - y(0) + Y(s) = F(s).$$

With $y(0) = 0$, we have

$$Y(s) = \frac{1}{s(s+1)(1+e^{-s})}.$$

Since

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

and

$$\frac{1}{1+e^{-s}} = \sum_{n=0}^{\infty} (-1)^n e^{-ns},$$

it follows that

$$Y(s) = \left(\frac{1}{s} - \frac{1}{s+1} \right) \sum_{n=0}^{\infty} (-1)^n e^{-ns}.$$

In view of the shifting on the t -axis property discussed in [Example 5.1.14](#), the solution $y(t)$ of the initial value problem is

$$y(t) = \sum_{n=0}^{\infty} (-1)^n \mathcal{U}(t-n) (1 - e^{-(t-n)}). \quad (5.4.2)$$

The graph of this solution is in [Figure 5.4.6](#) below. The solution $y(t)$ is a piecewise continuous periodic function with period 2, and it converges to the periodic function $f(t)$ as $t \rightarrow \infty$.

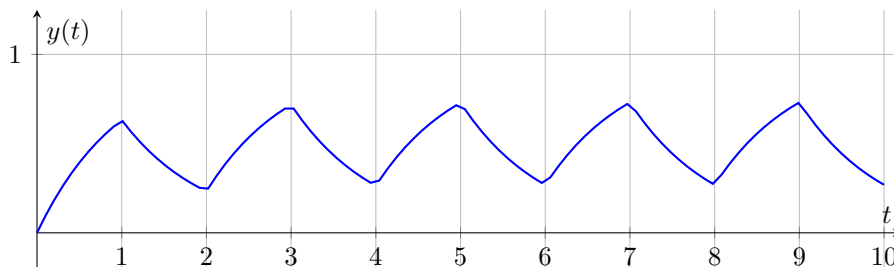


Figure 5.4.6 The graph of $y(t)$.

5.4.2 Exercises

1. Sketch the graph of three periods of each function below and compute its Laplace transform. The interval for the first period is provided.

- i. $f(t) = \cos t, \quad 0 \leq t < \pi$
 - ii. $f(t) = t, \quad 0 \leq t < 3$
 - iii. $f(t) = t - 2, \quad 0 \leq t < 4$
 - iv. $f(t) = \begin{cases} 1 & \text{if } 0 \leq t < 1 \\ -1 & \text{if } 1 \leq t < 2 \end{cases}$
 - v. $f(t) = \begin{cases} \sin t & \text{if } 0 \leq t < \pi \\ 0 & \text{if } \pi \leq t < 2\pi \end{cases}$
 - vi. $f(t) = \begin{cases} e^{-t} \sin t & \text{if } 0 \leq t < \pi \\ 0 & \text{if } \pi \leq t < 2\pi \end{cases}$
2. Solve the following initial value problems using the Laplace transform method. You will need $\frac{1}{1-t} = \sum_{n=0}^{\infty} (-1)^n t^n$ and $\frac{1}{1+t} = \sum_{n=0}^{\infty} (-1)^n t^n$ when $|t| < 1$.
- i. $y'' + 4y' + 3y = f(t), \quad y(0) = 0, y'(0) = 0$ where $f(t) = \begin{cases} 0 & \text{if } 0 \leq t \leq 1 \\ 1 & \text{if } 1 < t < 2 \end{cases}$ and $f(t) = f(t+2)$ for all t in $(-\infty, \infty)$.
 - ii. $y'' + 4y' + 4y = f(t), \quad y(0) = 0, y'(0) = 0$, where $f(t)$ is the periodic function in [Item 5.4.2.1.iv](#) of [Exercise 5.4.2.1](#).
 - iii. $y'' + 3y' + 2y = f(t), \quad y(0) = 0, y'(0) = 0$, where $f(t)$ is the periodic function in [Item 5.4.2.1.v](#) of [Exercise 5.4.2.1](#).
 - iv. $y'' + 4y = f(t), \quad y(0) = 0, y'(0) = 0$, where $f(t)$ is the periodic function in [Item 5.4.2.1.vi](#) of [Exercise 5.4.2.1](#).

5.5 A Convolution Theorem and its Applications

A crucial step in solving a linear initial value problem using Laplace transforms is determining the inverse Laplace transform of the solution's Laplace transform. This step can become challenging when the expression, whose inverse Laplace transform is desired, is complicated. In this section, we discuss a procedure with which we can determine the inverse Laplace transform of an expression that is a product of known Laplace transforms. For example, we recall that

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} = t \quad \text{and} \quad \mathcal{L}^{-1} \left\{ \frac{2}{s^3} \right\} = t^2.$$

However, if we were to compute

$$\mathcal{L}^{-1} \left\{ \left(\frac{1}{s^2} \right) \left(\frac{2}{s^3} \right) \right\},$$

a direct way is to see this problem as

$$\mathcal{L}^{-1} \left\{ \frac{2}{s^5} \right\} = \frac{2}{4!} t^4 = \frac{t^4}{12}.$$

This leads to a natural question: in what way can we combine t and t^2 to produce $t^4/12$? The operation for combining inverse Laplace transforms—specifically, to determine the inverse of a product—is known as *convolution*, which is formalized in the following definition.

Definition 5.5.1 Let f and g be functions on $[0, \infty)$. The **convolution** of f and g , denoted by $f * g$, is the function defined by

$$(f * g)(t) = \int_0^t f(u)g(t-u) du$$

whenever the integral exists for a given t in $[0, \infty)$. ◆

For computational purposes, $(f * g)(t)$ may also be written as $f(t) * g(t)$. It can be proved that $f * g = g * f$; see, for example, [Item 5.5.3.2.ii](#) in [Exercises 5.5.3](#). The convolution $1 * f$ is particularly useful, and it is given by

$$(1 * f)(t) = \int_0^t f(u) du.$$

See, also, [Item 5.5.3.2.v](#) in [Exercises 5.5.3](#).

Example 5.5.2 Let us return to the example discussed before [Definition 5.5.1](#) and calculate the convolution of $f(t) = t$ and $g(t) = t^2$. We have

$$\begin{aligned} (f * g)(t) &= \int_0^t u(t-u)^2 du \\ &= \int_0^t (t^2u - 2tu^2 + u^3) du \\ &= \left[t^2 \frac{u^2}{2} - 2t \frac{u^3}{3} + \frac{u^4}{4} \right]_0^t \\ &= \frac{t^4}{2} - \frac{2t^4}{3} + \frac{t^4}{4} \\ &= \frac{t^4}{12}, \end{aligned}$$

as expected. ▲

5.5.1 Convolution Theorem

We now state and prove the most important theorem of this section.

Theorem 5.5.3 Let f and g be piecewise continuous functions on $[0, \infty)$ and of the exponential order α as $t \rightarrow \infty$. Then

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s);$$

or equivalently,

$$\mathcal{L}^{-1}\{F(s)G(s)\} = \mathcal{L}^{-1}\{F(s)\} * \mathcal{L}^{-1}\{G(s)\} = (f * g)(t)$$

for $s > \alpha$.

To see how this theorem holds, we need to use a method for changing the order of an iterated integral. We include a proof of the Convolution Theorem below for the sake of completeness; however, not to be distracted from the proof, the reader may choose to skip it for now and proceed directly to apply the theorem.

Proof of Theorem 5.5.3. We recall that

$$F(s) = \int_0^\infty e^{-s\eta} f(\eta) d\eta \quad \text{and} \quad G(s) = \int_0^\infty e^{-su} g(u) du.$$

Then

$$\begin{aligned} F(s)G(s) &= \int_0^\infty e^{-s\eta} f(\eta) d\eta \int_0^\infty e^{-su} g(u) du \\ &= \int_0^\infty g(u) \left(\int_0^\infty e^{-s(\eta+u)} f(\eta) d\eta \right) du \\ &= \int_0^\infty g(u) \left(\int_u^\infty e^{-st} f(t-u) dt \right) du, \end{aligned}$$

where $t = \eta + u$. By changing the order of integration, we obtain

$$\begin{aligned} F(s)G(s) &= \int_0^\infty e^{-st} \left(\int_0^t g(u) f(t-u) du \right) dt \\ &= \int_0^\infty e^{-st} (g * f)(t) dt \\ &= \mathcal{L}\{(f * g)(t)\}. \end{aligned}$$

■

Some fundamental properties of the convolution operation are discussed in [Exercises 5.5.3](#); see the problem [Item 5.5.3.2.ii](#).

5.5.2 Application of the Convolution Theorem to Initial Value Problems

The following theorem concerns an interesting application of [Theorem 5.5.3](#) to linear initial value problems that involve the Dirac delta functions.

Theorem 5.5.4 *Let $y_\delta(t)$ denote the solution of the initial value problem*

$$ay'' + by' + cy = \delta(t), \quad y(0) = 0, y'(0) = 0. \quad (5.5.1)$$

The the solution $y(t)$ of the initial value problem

$$ay'' + by' + cy = g(t), \quad y(0) = 0, y'(0) = 0 \quad (5.5.2)$$

where a, b, c are constants with $a \neq 0$, and g is a piecewise continuous function that is of exponential order as $t \rightarrow \infty$, is given by

$$y(t) = (y_\delta * g)(t).$$

We first remark that y' and y'' in [\(5.5.1\)](#) are to be interpreted differently (as *distributional derivatives*) because of the presence of $\delta(t)$ on the right side of the equation. The fact beyond the scope of this book is that the Laplace transforms of distributional derivatives coincide with the Laplace transforms of the classical derivatives.

Proof of Theorem 5.5.4. We first find the solution of the initial value problem [\(5.5.1\)](#). Applying the Laplace transform *athcalL* to the initial value problem [\(5.5.1\)](#) and using $y(0) = y'(0) = 0$ yields

$$(as^2 + bs + c)Y_\delta(s) = 1,$$

that is,

$$Y_\delta(s) = \frac{1}{as^2 + bs + c}.$$

Then, by taking the inverse Laplace transform, we have

$$y_\delta(t) = \mathcal{L}^{-1} \left\{ \frac{1}{as^2 + bs + c} \right\}.$$

Next, we take the Laplace transform of both sides of the equation in (5.5.2) to get

$$Y(s) = \frac{G(s)}{as^2 + bs + c} = G(s)Y_\delta(s).$$

Finally, by applying the convolution theorem [Theorem 5.5.3](#), we conclude that

$$y(t) = (y_\delta * g)(t)$$

solves the initial value problem (5.5.2). ■

Example 5.5.5 Consider the initial value problem $y'' + y = e^t$, $y(0) = y'(0) = 0$. To solve this problem, we first find the solution of the initial value problem $y'' + y = \delta(t)$, $y(0) = y'(0) = 0$. Applying the Laplace transform to this problem yields

$$(s^2 + 1)Y_\delta(s) = 1,$$

that is,

$$Y_\delta(s) = \frac{1}{s^2 + 1}.$$

Then, by taking the inverse Laplace transform, we have

$$y_\delta(t) = \sin t.$$

Finally, by applying [Theorem 5.5.4](#) with $g(t) = e^t$, we conclude that

$$y(t) = y_\delta(t) * e^t = \sin t * e^t = -\frac{1}{2}(-e^t + \sin t + \cos t)$$

on $(-\infty, \infty)$. ▲

5.5.3 Exercises

1. For the functions f and g on $[0, \infty)$ given below, compute $f * g$, $\mathcal{L}\{(f * g)(t)\}$ and verify that $\mathcal{L}\{(f * g)(t)\} = \mathcal{L}\{f(t)\} \mathcal{L}\{g(t)\}$. In the following, k and ω are any real numbers.

- i. $f(t) = g(t) = 1$

- ii. $f(t) = g(t) = e^{kt}$

- iii. $f(t) = \sin t$, $g(t) = e^{kt}$

- iv. $f(t) = g(t) = \sin(\omega t)$

- v. $f(t) = \sin(\omega t)$, $g(t) = \cos(\omega t)$

2. Let f, g, h be functions defined on the interval $[0, \infty)$, and assume that all the convolutions below are well-defined. Use the definition of the convolution to prove the following properties.

- i. $f * g = g * f$.

- ii. $f * (g * h) = (f * g) * h$

- iii. $f * (g + h) = f * g + f * h$

- iv. $f * (kg) = k(f * g)$ for any constant k
- v. $(1 * f)(t) = \int_0^t f(u) du.$
3. Use the convolution theorem [Theorem 5.5.3](#) to find the Laplace transform of each of the following functions.
- i. $1 * t * t^2$
- ii. $1 * t^2 * \sin t$
- iii. $1 * \cos t * \sin t$
- iv. $1 * \sin t * \mathcal{U}(t - 2)$
- v. $1 * \mathcal{U}(t - 2) * \mathcal{U}(t - 5)$
- vi. $\int_0^t e^u du$
- vii. $\int_0^t \sin u du$
- viii. $\int_0^t u \sin u du$
- ix. $\int_0^t \sin u \cos(t - u) du$
- x. $t \int_0^t \sin u \cos(t - u) du$
4. Use the convolution theorem [Theorem 5.5.3](#) to find each of the following inverse Laplace transforms.
- i. $\mathcal{L}^{-1} \left\{ \frac{1}{(s+1)(s-1)} \right\}$
- ii. $\mathcal{L}^{-1} \left\{ \frac{1}{s^3(s-1)} \right\}$
- iii. $\mathcal{L}^{-1} \left\{ \frac{2}{(s^2+4)^2} \right\}$
- iv. $\mathcal{L}^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\}$
- v. $\mathcal{L}^{-1} \left\{ \frac{s^2}{(s^2-4)^2} \right\}$
- vi. $\mathcal{L}^{-1} \left\{ \frac{1}{s^2(s^2+4)^2} \right\}$
5. Solve the following integral or integro-differential equations using the Laplace transform.
- i. $y(t) + \int_0^t y(u) du = 1$
- ii. $y(t) = te^t + \int_0^t u y(t - u) du$
- iii. $y'(t) = 1 - \cos t - \int_0^t y(u) du, \quad y(0) = 0$
- iv. $y'(t) + 4y(t) + 4 \int_0^t y(u) du = 1, \quad y(0) = 0$
6. Use [Theorem 5.5.4](#) to solve the following initial value problems.
- i. $y' - 3y = \sin t, \quad y(0) = 0$
- ii. $y'' + y = \sin t, \quad y(0) = 0, \quad y'(0) = 0$
- iii. $y'' + 4y' + 3y = e^t, \quad y(0) = 0, \quad y'(0) = 0$
- iv. $y'' + 4y' + 13y = \cos t, \quad y(0) = 0, \quad y'(0) = 0$
7. Let f be a continuous function defined on the interval $[0, \infty)$, and assume that all the convolutions below are well-defined. Show the following properties of the convolution operation.
- i. $f * \delta = f.$ (This means that δ is an identity element for the convolu-

tion operation.)

- ii. $f(t) * \delta(t - a) = \mathcal{U}(t - a)f(t - a)$, where $a \geq 0$ and is a fixed real number.
8. Let f, g be functions with continuous derivatives on the interval $[0, \infty)$ and of exponential order as $t \rightarrow \infty$. Assume further that all the convolutions below are well-defined. Use the convolution theorem [Theorem 5.5.3](#) to prove the following properties.

- i. $1 * f'(t) = f(t) - f(0)$.
- ii. $(f * g)'(t) = g(0)f(t) + (g' * f)(t) = f(0)g(t) + (g * f')(t)$.

The above properties are also valid for any differentiable functions f and g , not necessarily of exponential order as $t \rightarrow \infty$, in light of the Leibniz rule¹ for differentiating an integral (see [Theorem 6.1.2](#)).

9. By using the Leibniz rule ([5.5.3](#)), convert each of the following integral or integro-differential equations into an equivalent initial value problem and then solve it by using your favorite method. Compare your answers to those obtained in [Exercise 5.5.3.5](#).

- i. $y(t) + \int_0^t y(u) du = 1$
- ii. $y(t) = te^t + \int_0^t u y(t - u) du$
- iii. $y'(t) = 1 - \cos t - \int_0^t y(u) du, \quad y(0) = 0$
- iv. $y'(t) + 4y(t) + 4 \int_0^t y(u) du = 1, \quad y(0) = 0$

10. Solve the following initial value problems using the Laplace transform. Also, determine whether the solutions are continuous at the point(s) of concentration of the involved Dirac delta functions(s).

- i. $y' - 3y = \delta(t - 1), \quad y(0) = 0$
- ii. $y' + 4y = \delta(t - 2), \quad y(0) = 1$
- iii. $y'' + 4y = \delta(t - 2\pi), \quad y(0) = 0, \quad y'(0) = 0$
- iv. $y'' + 4y' + 3y = \delta(t - 2\pi), \quad y(0) = 0, \quad y'(0) = 0$
- v. $y'' + 4y' + 13y = \delta(t - 2\pi), \quad y(0) = 0, \quad y'(0) = 0$
- vi. $y'' + 4y' + 3y = \delta(t - 2\pi) + \delta(t - 3\pi), \quad y(0) = 0, \quad y'(0) = 0$
- vii. $y'' + 4y' + 13y = \delta(t - 2\pi) + \delta(t - 3\pi), \quad y(0) = 0, \quad y'(0) = 0$

11. A 32 lb mass stretches a vertically hanging spring 4 feet to equilibrium. The mass is then pulled down 1 foot and released from rest. Assume no

¹

$$\begin{aligned} \frac{d}{dt} \int_{a(t)}^{b(t)} f(u, t) du &= f(b(t), t) \frac{db(t)}{dt} - f(a(t), t) \frac{da(t)}{dt} \\ &\quad + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(u, t) du \end{aligned} \quad (5.5.3)$$

damping or external forces. Let $y(t)$ denote the displacement of the mass from equilibrium at time t , measured positive in the downward direction. (Use $g = 32 \text{ ft/s}^2$.)

- i. Set up an initial value problem for $y(t)$.
- ii. Determine a Dirac delta function that models the impulsive force on the mass exerted by striking it with a hammer to stop it at the first time it passes through equilibrium, keeping it at the equilibrium position thereafter.
- iii. Repeat [Item 5.5.3.11.ii](#) to stop the mass at the second time it passes through equilibrium.

5.6 Systems of Linear Differential Equations

This chapter focuses on the applications of the Laplace transform to solve systems of linear ordinary differential equations. Using the Laplace transform, we convert a system of differential equations into a system of algebraic equations involving Laplace transforms of the unknown functions (dependent variables), solve this algebraic system for the Laplace transforms, and then apply the inverse Laplace transform to recover the functions from their Laplace transforms. These functions then become solutions of the differential equations in the system.

To fix the ideas, we consider below a first-order linear system of ordinary differential equations in the three variables $x(t), y(t), z(t)$:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y + a_{13}z + f_1(t), \\ \frac{dy}{dt} = a_{21}x + a_{22}y + a_{23}z + f_2(t), \\ \frac{dz}{dt} = a_{31}x + a_{32}y + a_{33}z + f_3(t), \end{cases} \quad (5.6.1)$$

where

- $x(t), y(t), z(t)$ are unknown functions of time t ,
- the coefficients a_{ij} are fixed real numbers, and
- $f_i(t)$ are given functions of t , possibly zero.

The system (5.6.1) is said to be **decoupled** if $a_{ij} = 0$ for all i, j with $i \neq j$ and **coupled** if for each $i = 1, 2, 3$ there is $j = 1, 2, 3$ with $i \neq j$ and $a_{ij} \neq 0$. The linear system (5.6.1) is said to be **partially decoupled** if it has a structure that allows solving one or more of the equations in isolation, and then using those solutions to simplify or solve the remaining equations.

5.6.1 The Laplace Transform Method

To solve the linear system (5.6.1) with the initial conditions

$$x(0) = x_0, \quad y(0) = y_0, \quad z(0) = z_0,$$

we apply the Laplace transform $\mathcal{L}$ to each equation. The system is then transformed into

$$\begin{cases} sX(s) - x_0 = a_{11}X(s) + a_{12}Y(s) + a_{13}Z(s) + F_1(s), \\ sY(s) - y_0 = a_{21}X(s) + a_{22}Y(s) + a_{23}Z(s) + F_2(s), \\ sZ(s) - z_0 = a_{31}X(s) + a_{32}Y(s) + a_{33}Z(s) + F_3(s), \end{cases}$$

where $X(s), Y(s), Z(s)$ are the Laplace transforms of $x(t), y(t), z(t)$, and $F_i(s) = \mathcal{L}\{f_i(t)\}, i = 1, 2, 3$.

We solve the above system for $X(s), Y(s), Z(s)$ using algebraic techniques, such as Cramer's rule, Gaussian elimination, etc. and then apply the inverse Laplace transform to obtain solutions $x(t), y(t), z(t)$.

We proceed to discuss four main examples; namely, radioactive decay series, mixing solutions, coupled spring-mass systems, and electrical circuits.

5.6.2 Radioactive Decay Series

Consider a three-stage radioactive decay series of a radioactive element represented in a schematic diagram below:

$$X \longrightarrow Y \longrightarrow Z$$

where

- X is a parent radioactive isotope,
- Y is a radioactive daughter element, and
- Z is a stable (non-decaying) end element.

Let

$$\begin{aligned} x(t) &= \text{the amount of isotope } X \text{ at time } t, \\ y(t) &= \text{the amount of isotope } Y \text{ at time } t, \\ z(t) &= \text{the amount of isotope } Z \text{ at time } t, \\ a &= \text{the decay constant of } X, \text{ and} \\ b &= \text{the decay constant of } Y. \end{aligned}$$

We assume that the disintegration rate of X at time t is proportional to its amount at that time, and the same holds true for Y . Then the system of differential equations modeling the decay series is

$$\begin{cases} \frac{dx}{dt} = -ax \\ \frac{dy}{dt} = ax - by \\ \frac{dz}{dt} = by \end{cases} \quad (5.6.2)$$

subject to the initial conditions:

$$x(0) = x_0, \quad y(0) = 0, \quad z(0) = 0.$$

The linear system (5.6.2) describes how the parent isotope X decays into Y , which in turn decays into a stable isotope Z . We can adopt this scheme to model real-world decay chains such as

$$\text{Uranium-238} \rightarrow \text{Thorium-234} \rightarrow \cdots \rightarrow \text{Lead-206}.$$

The linear system (5.6.2) is an example of a *partially decoupled* system. We observe that the solutions of the differential equation in the system (5.6.2) can be obtained in a sequence of steps: first solve the first equation for $x(t)$, substitute this $x(t)$ into the second equation to find $y(t)$ using the integrating factor method, and then substitute $y(t)$ into the third equation to solve for $z(t)$. Details are omitted, but the solutions can be found to be

$$\begin{cases} x(t) &= x_0 e^{-at}, \\ y(t) &= \frac{ax_0}{b-a} (e^{-at} - e^{-bt}), \\ z(t) &= x_0 \left[1 + \frac{ae^{-bt} - be^{-at}}{b-a} \right]. \end{cases}$$

Let us obtain these solutions by using the Laplace transform method.

Step 1: Laplace Transforms. Since

$$\mathcal{L}\left\{\frac{dx}{dt}\right\} = -a\mathcal{L}\{x(t)\},$$

we have

$$sX(s) - x_0 = -aX(s),$$

and therefore

$$X(s) = \frac{x_0}{s+a}.$$

Similarly,

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} = aX(s) - bY(s),$$

which yields

$$sY(s) = aX(s) - bY(s),$$

and therefore

$$Y(s) = \frac{aX(s)}{s+b} = \frac{ax_0}{(s+a)(s+b)}.$$

Also,

$$\mathcal{L}\left\{\frac{dz}{dt}\right\} = bY(s)$$

gives

$$Z(s) = \frac{b}{s}Y(s) = \frac{abx_0}{s(s+a)(s+b)}$$

Step 2: Inverse Laplace Transforms. We have

$$x(t) = \mathcal{L}^{-1}\left\{\frac{x_0}{s+a}\right\} = x_0 e^{-at}.$$

Since

$$Y(s) = \frac{ax_0}{(s+a)(s+b)} = \frac{ax_0}{b-a} \left(\frac{1}{s+a} - \frac{1}{s+b} \right),$$

we have

$$y(t) = \frac{ax_0}{b-a} (e^{-at} - e^{-bt}).$$

Finally, since

$$Z(s) = \frac{abx_0}{s(s+a)(s+b)},$$

we have

$$\begin{aligned} z(t) &= \int_0^t by(\tau) d\tau \\ &= \frac{abx_0}{b-a} \left(\frac{1-e^{-at}}{a} - \frac{1-e^{-bt}}{b} \right) \\ &= x_0 \left[1 + \frac{ae^{-bt} - be^{-at}}{b-a} \right] \end{aligned}$$

Thus, the solutions are given by

$$\begin{cases} x(t) &= x_0 e^{-at}, \\ y(t) &= \frac{ax_0}{b-a} (e^{-at} - e^{-bt}), \\ z(t) &= x_0 \left[1 + \frac{ae^{-bt} - be^{-at}}{b-a} \right], \end{cases}$$

as expected.

5.6.3 Mixing Tanks

Consider two connected mixing tanks as shown in the figure below.

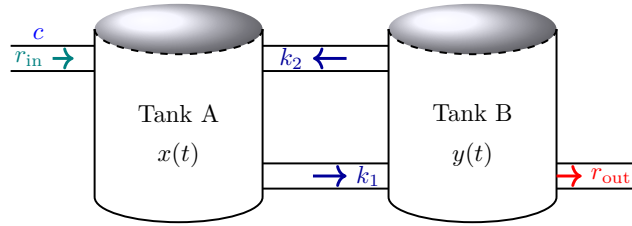


Figure 5.6.1 Two connected mixing tanks.

$$\begin{cases} x'(t) = r_{\text{in}} c + \frac{k_2 y(t)}{V_2 + t(k_1 - k_2 - r_{\text{out}})} - \frac{k_1 x(t)}{V_1 + t(r_{\text{in}} + k_2 - k_1)}, \\ y'(t) = \frac{k_1 x(t)}{V_1 + t(r_{\text{in}} + k_2 - k_1)} - \frac{(k_2 + r_{\text{out}})y(t)}{V_2 + t(k_1 - k_2 - r_{\text{out}})}. \end{cases}$$

When $r_{\text{in}} = k_1 - k_2 = r_{\text{out}}$, the system becomes

$$\begin{cases} x'(t) = r_{\text{in}} c + \frac{k_2 y(t)}{V_2} - \frac{k_1 x(t)}{V_1}, \\ y'(t) = \frac{k_1 x(t)}{V_1} - \frac{(k_2 + r_{\text{out}})y(t)}{V_2}. \end{cases}$$

5.6.4 Coupled Spring-Mass Systems

Suppose that two masses m_1 and m_2 are connected to two springs A and B hanging vertically from a rigid support to form a coupled spring-mass system as shown in the figure below.

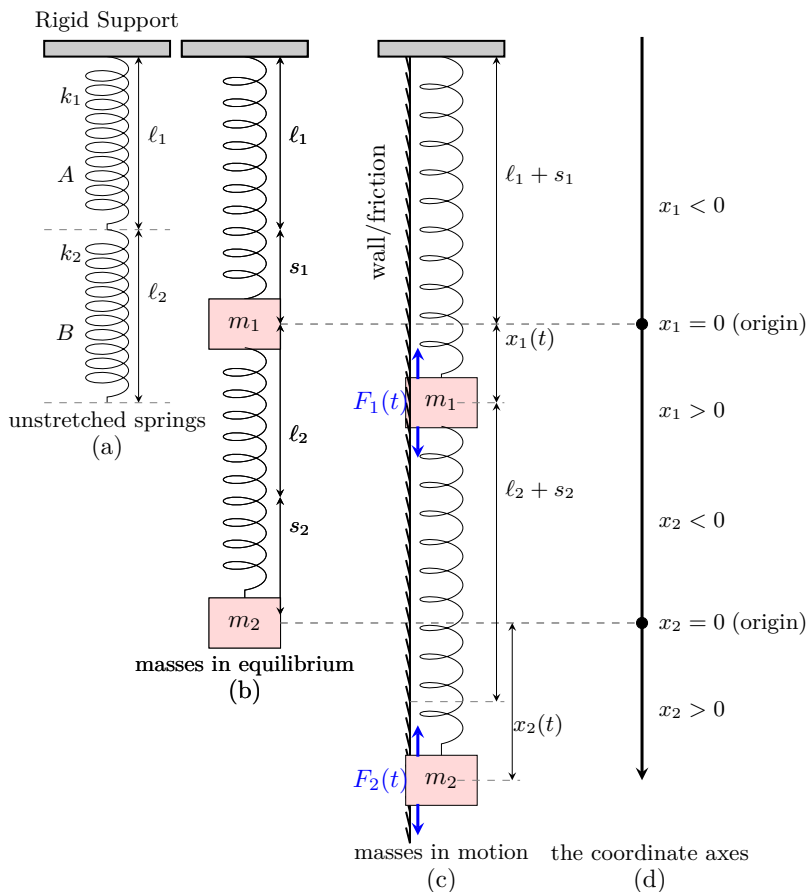


Figure 5.6.2 A coupled mass-spring system showing unstretched, equilibrium, motion, and coordinate-axis.

Let k_1 and k_2 be the constants of proportionality in Hooke's law for the springs A and B, respectively. When both masses m_1 and m_2 are in equilibrium, we have

$$\begin{cases} m_1 g &= k_1 s_1 - k_2 s_2, \\ m_2 g &= k_2 s_2. \end{cases} \quad (5.6.3)$$

Adding together the two relations in the above display yields

$$(m_1 + m_2)g = k_1 s_1, \quad (5.6.4)$$

which reflects the intuition that the presence of the spring B could be thought of considering the mass $m_1 + m_2$ in place of m_1 and removing the spring B entirely. We recall that the stretch s_1 of the spring A depends on both masses m_1 and m_2 , while the stretch s_2 of the spring B depends only on the mass m_2 when the spring constants are computed using the gravitational force.

Let $x_1(t)$ and $x_2(t)$ denote the vertical displacements of the masses from their equilibrium positions at time t . We take the positive direction of the displacements to be vertically downward. When the masses are in motion, the spring B is subject to both a stretch and a compression, so that its net stretch of the spring B is given by $s_2 + (x_2(t) - x_1(t))$ when the stretch of the spring A is $s_1 + x_1(t)$.

The restoring force of the spring A acting on the mass m_1 at time t is $-k_1(s_1 + x_1(t))$, and the restoring force of the spring B acting on the mass m_2 at time t is $-k_2(s_2 + x_2(t) - x_1(t))$. In addition to the restoring force of the

spring A , the mass m_1 also experiences the force of $k_2(s_2 + x_2(t) - x_1(t))$ at time t . Let b_1 and b_2 denote the damping coefficients for the masses m_1 and m_2 , respectively. Also, let $F_1(t)$ and $F_2(t)$ denote the external forces acting on the masses m_1 and m_2 , respectively. We assume that the spring weights are negligible compared to the weights of the masses. Suppressing t in $x_1(t)$ and $x_2(t)$ and using Newton's law, we see that the motion of the two masses in the spring-mass system is described by

$$\begin{aligned} m_1 x_1'' &= m_1 g - k_1(s_1 + x_1) + k_2(s_2 + x_2 - x_1) - b_1 x_1' + F_1(t) \\ &= m_1 g - k_1 s_1 + k_2 s_2 - (k_1 + k_2)x_1 + k_2 x_2 - b_1 x_1' + F_1(t) \end{aligned}$$

and

$$\begin{aligned} m_2 x_2'' &= m_2 g - k_2(s_2 + x_2 - x_1) - b_2 x_2' + F_2(t) \\ &= m_2 g - k_2 s_2 + k_2 x_1 - k_2 x_2 - b_2 x_2' + F_2(t). \end{aligned}$$

Using (5.6.3), we obtain

$$\begin{cases} m_1 x_1'' + b_1 x_1' + (k_1 + k_2)x_1 - k_2 x_2 = F_1(t), \\ m_2 x_2'' + b_2 x_2' + k_2 x_2 - k_2 x_1 = F_2(t). \end{cases} \quad (5.6.5)$$

The above system of second-order coupled linear differential equations describes the motion of the two masses in the coupled spring-mass system, which, together with the initial conditions $x_1(t_0) = a$, $x_1'(t_0) = b$, $x_2(t_0) = c$, $x_2'(t_0) = d$, forms an initial value problem.

In the next example, we solve the initial value problem (5.6.5) for $m_1 = m_2 = 1$, $k_1 = 3$, $k_2 = 2$, $b_1 = b_2 = 0$, $F_1(t) = F_2(t) = 0$, and $x_1(0) = 0$, $x_1'(0) = -1$, $x_2(0) = 0$, $x_2'(0) = 1$.

Example 5.6.3 Solve the second-order coupled system of linear differential equations with initial conditions

$$\begin{cases} x_1'' = -5x_1 + 2x_2, \\ x_2'' = 2x_1 - 2x_2, \\ x_1(0) = 0, \quad x_1'(0) = -1, \\ x_2(0) = 0, \quad x_2'(0) = 1. \end{cases} \quad (5.6.6)$$

Taking the Laplace transform yields

$$\begin{cases} s^2 X_1(s) - s x_1(0) - x_1'(0) = -5X_1(s) + 2X_2(s), \\ s^2 X_2(s) - s x_2(0) - x_2'(0) = 2X_1(s) - 2X_2(s). \end{cases}$$

Using the given initial conditions yields

$$\begin{cases} (s^2 + 5)X_1(s) + 1 = 2X_2(s), \\ (s^2 + 2)X_2(s) - 1 = 2X_1(s). \end{cases}$$

Solving for $X_1(s)$ and $X_2(s)$ gives

$$X_1(s) = \frac{-s^2}{(s^2 + 1)(s^2 + 6)} \quad \text{and} \quad X_2(s) = \frac{s^2 + 3}{(s^2 + 1)(s^2 + 6)}.$$

Partial fraction decomposition of $X_1(s)$ and $X_2(s)$ are

$$X_1(s) = \frac{1/5}{s^2 + 1} - \frac{6/5}{s^2 + 6} \quad \text{and} \quad X_2(s) = \frac{2/5}{s^2 + 1} + \frac{3/5}{s^2 + 6}.$$

Taking the inverse Laplace transform yields

$$\begin{cases} x_1(t) &= \frac{1}{5} \sin t - \frac{6}{5\sqrt{6}} \sin(\sqrt{6}t), \\ x_2(t) &= \frac{2}{5} \sin t + \frac{3}{5\sqrt{6}} \sin(\sqrt{6}t), \end{cases}$$

which render the solutions of the given initial value problem (5.6.6).

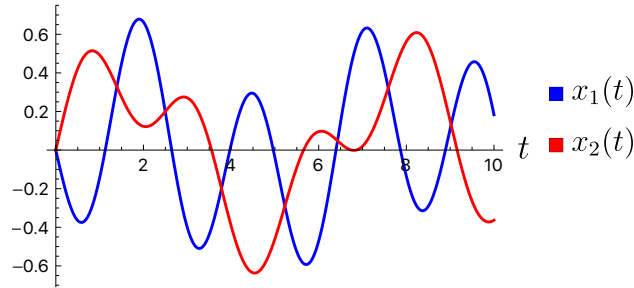


Figure 5.6.4 Graphs of the solutions $x_1(t)$ and $x_2(t)$



5.6.5 Electrical Network Systems

We discussed series and parallel LRC circuits in Chapter 3. In Figure 5.6.5 below, we have an LRC circuit with a voltage source $e(t)$. We develop a system of linear differential equations to find the charge $q_1(t)$ on the capacitor and currents $i(t)$ and $i_2(t)$ flowing through the resistor and inductor, respectively. Let $i_1(t)$ be the current flowing through the capacitor.

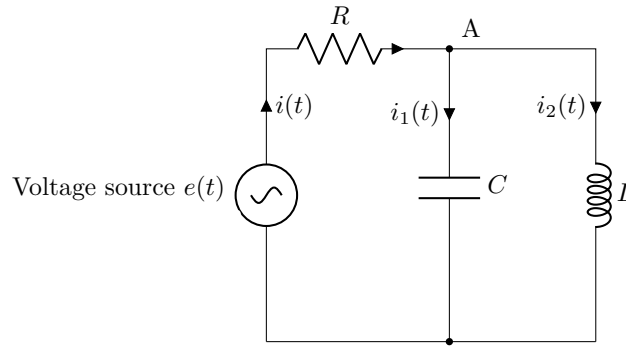


Figure 5.6.5 An LRC circuit with a voltage source.

We observe that the circuit contains three distinct closed loops:

- **Loop 1:** consisting of the battery, resistor, and capacitor;
- **Loop 2:** consisting of the battery, resistor, and inductor; and
- **Loop 3:** formed by the capacitor and inductor alone.

The system of corresponding equations for the loops are:

$$\begin{cases} Ri(t) + \frac{1}{C}q_1(t) = e(t) & \text{(for Loop 1)} \\ Li_2'(t) + Ri(t) = e(t) & \text{(for Loop 2)} \\ Li_2'(t) - \frac{1}{C}q_1(t) = 0 & \text{(for Loop 3)} \end{cases}$$

We observe that each equation in the system can be obtained from the remaining two. For example, the equation for Loop 2 can be obtained by adding the equations for Loop 1 and Loop 3. For the discussion that follows, we omit the equation for Loop 2.

Furthermore, by Kirchhoff's second law, the current split at the junction A is given by

$$i(t) = i_1(t) + i_2(t),$$

which, in view of $i_1(t) = q_1'(t)$, becomes

$$q_1'(t) + i_2(t) - i(t) = 0.$$

Thus, a linear system in $i(t)$, $q_1(t)$ and $i_2(t)$ is

$$\begin{cases} Ri(t) + \frac{1}{C}q_1(t) = e(t) & \text{(for Loop 1)} \\ L i_2'(t) - \frac{1}{C}q_1(t) = 0 & \text{(for Loop 3)} \\ q_1'(t) + i_2(t) - i(t) = 0 & \text{(the current split at the junction A)} \end{cases} \quad (5.6.7)$$

We now proceed to solve the system (5.6.7). Differentiating the third equation in (5.6.7) with respect to t yields

$$i'(t) = q_1''(t) + i_2'(t). \quad (5.6.8)$$

Solving for $i_2'(t)$ from the second equation in (5.6.7) and substituting it into (5.6.8) gives

$$i'(t) = q_1''(t) + \frac{1}{LC}q_1(t).$$

Substituting this expression for $i'(t)$ into the equation obtained by differentiating the first equation in (5.6.7) with respect to t , we get

$$LRCq_1''(t) + Lq_1'(t) + Rq_1(t) = LCe'(t), \quad (5.6.9)$$

which can be solved for $q_1(t)$. Once $q_1(t)$ is determined, we find $i_1(t)$ by using $i_1(t) = q_1'(t)$. We then find $i(t)$ from the first equation in (5.6.7) by using $q_1(t)$ there. Finally, we find $i_2(t) = i(t) - i_1(t)$.

5.6.6 Exercises

1. Solve the initial value problem

$$\begin{cases} x_1'' = -5x_1 + 2x_2, \\ x_2'' = 2x_1 - 2x_2, \\ x_1(0) = 0, \quad x_1'(0) = -1, \\ x_2(0) = 0, \quad x_2'(0) = 1. \end{cases}$$

by using the Laplace transform.

Chapter 6

Supplementary Topics

Intro goes here yes.

6.1 Differentiation Under the Integral

Something about differentiating under the integral goes here yes. Testing

Theorem 6.1.1 Leibniz Rule. *Let $f(x, y)$ be a function on a rectangle $R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$ such that both f and $\frac{\partial f}{\partial y}$ are continuous on R . Then*

$$\frac{d}{dy} \int_a^b f(x, y) dx = \int_a^b \frac{\partial f}{\partial y}(x, y) dx.$$

Theorem 6.1.2 Leibniz Rule for Improper Integrals. *Let $f(x, y)$ be a function on a rectangle $R = \{(x, y) : a \leq x < \infty, c \leq y \leq d\}$ such that, for each y in I , both f and $\frac{\partial f}{\partial y}$ have improper integrals on $[a, \infty)$ and such that, for each x in $[a, \infty)$, $f(x, y)$ is differentiable in y for each y in I . Suppose, further, that there exists a function g on $[a, \infty)$ such that*

$$\left| \frac{\partial f}{\partial y}(x, y) \right| \leq g(x)$$

for all $x \in [a, \infty)$ and for all y in I and such that

$$\int_a^\infty g(x) dx$$

is convergent. Then

$$\frac{d}{dy} \int_a^\infty f(x, y) dx = \int_a^\infty \frac{\partial f}{\partial y}(x, y) dx.$$

Theorem 6.1.3 *Let $f(x, y)$ be a function on a rectangle $R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$ such that both f and $\frac{\partial f}{\partial y}$ are continuous on R . Let a and b be differentiable functions on $[c, d]$ such that $a \leq a(y), b(y) \leq b$ for all y in $[c, d]$. Then*

$$\frac{d}{dy} \int_{a(y)}^{b(y)} f(x, y) dx = b'(y) f(b(y), y) - a'(y) f(a(y), y) + \int_{a(y)}^{b(y)} \frac{\partial f}{\partial y}(x, y) dx.$$

6.2 Partial Fraction Decomposition

Let P and Q be general polynomials in x , and let m and n be the degrees of P and Q , respectively. Then the **rational** function

$$R(x) = \frac{P(x)}{Q(x)}$$

is said to be **proper** if $m < n$ and **improper** if $m \geq n$. When $R(x)$ is improper, we perform polynomial division to obtain

$$R(x) = q(x) + \frac{r(x)}{Q(x)},$$

where $q(x)$ is the **quotient** polynomial of degree $m - n$, and $r(x)$ is called the **remainder** polynomial and it is of degree less than n . For example, if $P(x) = x^3, Q(x) = x^2 - 1$, then

$$\frac{x^3 + 2}{x^2 - 1} = x + \frac{x + 2}{x^2 - 1},$$

so that $q(x) = x$ and $r(x) = x + 2$.

After performing polynomial division, the term $q(x)$ is already a polynomial and requires no decomposition. The term $\frac{r(x)}{Q(x)}$ is a proper rational function and the only part of $R(x)$ that is usually decomposed into partial fractions. Therefore, in what follows, we assume that $R(x)$ itself is proper. The method of partial fraction decomposition then depends entirely on the factorization of the denominator $Q(x)$. We now examine all possible cases that may occur.

6.2.1 Linear Factors $x - a$

If the denominator contains distinct linear factors,

$$Q(x) = (x - a_1)(x - a_2) \cdots (x - a_n),$$

then the decomposition has the form

$$\frac{P(x)}{Q(x)} = \frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \cdots + \frac{A_n}{x - a_n},$$

where $A_1, \dots, A_n$ are constants to be determined. Let us illustrate this with an example below.

Example 6.2.1 Perform partial fraction decomposition of

$$\frac{3x + 5}{(x - 1)(x + 2)}.$$

Suppose there are constants A and B such that

$$\frac{3x + 5}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}.$$

Multiplying through by $(x - 1)(x + 2)$ gives

$$3x + 5 = A(x + 2) + B(x - 1)$$

which must be an identity in x . For convenience, we do allow $x = 1$ or $x = -2$ in this identity. We find $A = 2$, $B = 1$, so

$$\frac{3x+5}{(x-1)(x+2)} = \frac{2}{x-1} + \frac{1}{x+2}.$$



6.2.2 Repeated Linear Factors $(x-a)^n$

If

$$Q(x) = (x-a)^n,$$

then the decomposition includes every power of the linear factor:

$$\frac{P(x)}{(x-a)^n} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_n}{(x-a)^n},$$

$A_1, \dots, A_n$ are constants to be determined.

Example 6.2.2 Perform the partial fraction decomposition of

$$\frac{x+1}{(x-1)^2}.$$

Suppose there exist constants A and B such that

$$\frac{x+1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}.$$

Multiplying through by $(x-1)^2$ gives

$$x+1 = A(x-1) + B.$$

We find $A = 1$ and $B = 2$. Then

$$\frac{x+1}{(x-1)^2} = \frac{1}{x-1} + \frac{2}{(x-1)^2}.$$



6.2.3 Irreducible Quadratic Factors x^2+bx+c

Suppose that $Q(x)$ contains a factor x^2+bx+c that is irreducible over the real numbers. The partial fraction decomposition of $\frac{P(x)}{Q(x)}$ contains a term of the form

$$\frac{Ax+B}{x^2+bx+c},$$

where A and B are constants to be determined. We illustrate this in the next example.

Example 6.2.3 To perform the partial fraction decomposition of

$$\frac{x}{(x^2+1)(x-2)},$$

we find A, B and C such that

$$\frac{x}{(x^2+1)(x-2)} = \frac{Ax+B}{x^2+1} + \frac{C}{x-2}.$$

Notice the term $\frac{C}{x-2}$ corresponding to the linear factor $x-2$ of the denominator. Multiplying through by $(x^2+1)(x-2)$ yields

$$x = (Ax + B)(x - 2) + C(x^2 + 1).$$

Comparing coefficients gives

$$A = \frac{1}{3}, \quad B = \frac{2}{3}, \quad C = -\frac{1}{3},$$

so

$$\frac{x}{(x^2+1)(x-2)} = \frac{\frac{1}{3}x + \frac{2}{3}}{x^2+1} - \frac{1}{3(x-2)}.$$



6.2.4 Repeated Irreducible Quadratic Factors

If $Q(x) = (x^2 + bx + c)^k$, where the quadratic $x^2 + bx + c$ is irreducible and repeated with multiplicity k , then the decomposition $\frac{P(x)}{Q(x)}$ is of the form

$$\frac{P(x)}{(x^2 + bx + c)^k} = \frac{A_1x + B_1}{x^2 + bx + c} + \frac{A_2x + B_2}{(x^2 + bx + c)^2} + \cdots + \frac{A_kx + B_k}{(x^2 + bx + c)^k},$$

with the constants $A_1, \dots, A_k$ and $B_1, \dots, B_k$ are to be determined. We illustrate this in the next example.

Example 6.2.4 To perform the partial fraction decomposition of

$$\frac{3x + 5}{(x^2 + 1)^2},$$

we find constants A, B, C, D such that

$$\frac{3x + 5}{(x^2 + 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}.$$

Multiplying both sides by $(x^2 + 1)^2$ gives

$$3x + 5 = (Ax + B)(x^2 + 1) + (Cx + D).$$

Expanding and matching coefficients yields

$$A = 0, \quad B = 3, \quad C = -3, \quad D = 5,$$

and hence

$$\frac{3x + 5}{(x^2 + 1)^2} = \frac{3}{x^2 + 1} + \frac{-3x + 5}{(x^2 + 1)^2}.$$



6.2.5 Mixed Factors

If the denominator $Q(x)$ factors as¹

$$Q(x) = \prod_{i=1}^{\ell_1} (x - a_i)^{k_i} \prod_{j=1}^{\ell_2} (x^2 + b_jx + c_j)^{m_j},$$

¹The symbol $\prod_{i=1}^{\ell} p_i$ denotes product $p_1 p_2 \dots p_{\ell}$.

then the full partial fraction decomposition of $\frac{P(x)}{Q(x)}$ contains terms

- $\frac{A}{x-a}, \frac{A}{(x-a)^2}, \text{ etc.}, \text{ and}$
- $\frac{Ax+B}{x^2+bx+c}, \frac{Ax+B}{(x^2+bx+c)^2}, \text{ etc.}$

Let us illustrate this in the next example

Example 6.2.5 To perform the partial fraction decomposition of

$$\frac{2x^2 + 3x + 1}{(x-1)(x^2+1)},$$

we find constants A, B, C such that

$$\frac{2x^2 + 3x + 1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}.$$

Multiplying both sides by $(x-1)(x^2+1)$ gives

$$2x^2 + 3x + 1 = A(x^2+1) + (Bx+C)(x-1).$$

Expanding and comparing coefficients yields

$$A = 1, \quad B = 1, \quad C = 2,$$

so

$$\frac{2x^2 + 3x + 1}{(x-1)(x^2+1)} = \frac{1}{x-1} + \frac{x+2}{x^2+1}.$$

▲

6.2.6 Exercises

1. Decompose each rational function into partial fractions.

(a) $\frac{5x+3}{(x-1)(x+4)}$

(b) $\frac{7}{(x+2)^2}$

(c) $\frac{3x^2+2x+1}{x(x^2+4)}$

(d) $\frac{x^2+5}{(x^2+1)^2}$

(e) $\frac{4x+9}{(x-1)^3}$

(f) $\frac{2x^2-1}{(x^2+1)(x-3)}$

(g) $\frac{6x}{(x^2+2x+5)(x-1)}$

Answers.

1. $\frac{2}{x-1} + \frac{3}{x+4}$

2. $\frac{7}{(x+2)^2}$

3. $\frac{1}{x} + \frac{x+2}{x^2+4}$

4. $\frac{5}{x^2+1} - \frac{x}{(x^2+1)^2}$

5. $\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3}$ (determine A, B, C)

6. $\frac{1}{x-3} + \frac{x+3}{x^2+1}$

7. Divide first: $x+1+\frac{2}{x^2-1}$ and then decompose to get $x+1+\frac{1}{x-1}-\frac{1}{x+1}$.

8. $\frac{2}{x-1} + \frac{-2x+1}{x^2+2x+5}$

6.3 Formulas

6.3.1 Common Laplace Transforms

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
1. 1	$\frac{1}{s}, s > 0$
2. t	$\frac{1}{s^2}, s > 0$
3. t^n , n being a natural number	$\frac{n!}{s^{n+1}}, s > 0$
4. $t^{-1/2}$	$\frac{\sqrt{\pi}}{\sqrt{s}}, s > 0$
5. $t^{1/2}$	$\frac{\sqrt{\pi}}{2s^{3/2}}, s > 0$
6. t^α , $\alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}, s > 0$
7. $\sin(kt)$	$\frac{k}{s^2 + k^2}, s > 0$
8. $\cos(kt)$	$\frac{s}{s^2 + k^2}, s > 0$
9. $\sin^2(kt)$	$\frac{2k^2}{s(s^2 + 4k^2)}, s > 0$
10. $\cos^2(kt)$	$\frac{s^2 + 2k^2}{s(s^2 + 4k^2)}, s > 0$
11. e^{at}	$\frac{1}{s-a}, s > a$
12. $\sinh(kt)$	$\frac{k}{s^2 - k^2}, s > k $
13. $\cosh(kt)$	$\frac{s}{s^2 - k^2}, s > k $
14. $\sinh^2(kt)$	$\frac{2k^2}{s(s^2 - 4k^2)}, s > k $
15. $\cosh^2(kt)$	$\frac{s^2 - 2k^2}{s(s^2 - 4k^2)}, s > k $
16. te^{at}	$\frac{1}{(s-a)^2}, s > a$
17. $t^n e^{at}$, n being a natural number	$\frac{n!}{(s-a)^{n+1}}, s > a$
18. $e^{at} \sin(kt)$	$\frac{k}{(s-a)^2 + k^2}, s > a$
19. $e^{at} \cos(kt)$	$\frac{s-a}{(s-a)^2 + k^2}, s > a$
20. $e^{at} \sinh(kt)$	$\frac{k}{(s-a)^2 - k^2}, s > a + k $
21. $e^{at} \cosh(kt)$	$\frac{s-a}{(s-a)^2 - k^2}, s > a + k $
22. $t \sin(kt)$	$\frac{2ks}{(s^2 + k^2)^2}, s > 0$
23. $t \cos(kt)$	$\frac{s^2 - k^2}{(s^2 + k^2)^2}, s > 0$
24. $\sin(kt) + kt \cos(kt)$	$\frac{2ks^2}{(s^2 + k^2)^2}, s > 0$
25. $\sin(kt) - kt \cos(kt)$	$\frac{2k^3}{(s^2 + k^2)^2}, s > 0$
26. $t \sinh(kt)$	$\frac{2ks}{(s^2 - k^2)^2}, s > k $
27. $t \cosh(kt)$	$\frac{s^2 + k^2}{(s^2 - k^2)^2}, s > k $
28. $\frac{e^{at} - e^{bt}}{a-b}$	$\frac{1}{(s-a)(s-b)}, s > \max\{a, b\}$
29. $\frac{ae^{at} - be^{bt}}{a-b}$	$\frac{s}{(s-a)(s-b)}, s > \max\{a, b\}$
30. $1 - \cos(kt)$	$\frac{k^2}{s(s^2 + k^2)}, s > 0$
31. $kt - \sin(kt)$	$\frac{k^3}{s^2(s^2 + k^2)}, s > 0$
32. $\frac{a \sin(bt) - b \sin(at)}{ab(a^2 - b^2)}$	$\frac{1}{(s^2 + a^2)(s^2 + b^2)}, s > 0$

6.3.2 Derivatives

We use notations $\frac{d}{dx}f(x)$ and $f'(x)$ to synonymously denote the derivative of $f(x)$ with respect to x .

Differentiation Rules

$$\text{Constant: } \frac{d}{dx}c = 0$$

$$\text{Constant Multiple: } \frac{d}{dx}(cf(x)) = cf'(x)$$

$$\text{Sum: } \frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$$

$$\text{Product: } \frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + g'(x)f(x)$$

$$\text{Quotient: } \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

$$\text{Chain: } \frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

$$\text{Power: } \frac{d}{dx}x^n = nx^{n-1}$$

$$\text{Power: } \frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x)$$

$$\frac{d}{dx}\sin x = \cos x$$

$$\frac{d}{dx}\cos x = -\sin x$$

$$\frac{d}{dx}\tan x = \sec^2 x$$

$$\frac{d}{dx}\cot x = -\csc^2 x$$

$$\frac{d}{dx}\sec x = \sec x \tan x$$

$$\frac{d}{dx}\csc x = -\csc x \cot x$$

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\cos^{-1}x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$$

$$\frac{d}{dx}\cot^{-1}x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}\sec^{-1}x = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}\csc^{-1}x = -\frac{1}{|x|\sqrt{x^2-1}}$$

6.4 Worksheets

6.4.1 WS-1

This is a test worksheet.

6.4.2 WS-2

This is a test worksheet.

Appendix A

Selected Hints

3 · Second and Higher Order Equations and Applications

3.3 · Second Order Nonhomogeneous Linear Differential Equations

3.3.4 · Exercises

Subsection **3.3.1 Variation of Parameters.**

3.3.4.7. Use the following rule for differentiating under the integral sign:
if

$$F(x) = \int_{x_0}^x g(x, t) dt,$$

then

$$F'(x) = g(x, x) + \int_{x_0}^x \frac{\partial g}{\partial x}(x, t) dt.$$

5 · The Laplace Transform and Applications

5.3 · Derivatives and Integrals of Laplace Transforms

5.3.3 · Exercises

5.3.3.3. Use (5.3.2). to determine whether the Laplace transform of $(e^t - 2)/t$ exists.

Appendix B

Selected Solutions

Appendix C

List of Symbols

Symbol	Description	Page
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Appendix D

List of Theorems

Chapter 3 Second and Higher Order Equations and Applications

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Colophon

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